Optimal Dynamic Selection Under Costly Evaluation: Evidence from Graduate Admissions

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Motivation

- **Economic Problem:** Selecting best options under costly evaluation
  - Determining each option's true quality costs decision-maker time and effort
  - Low-cost screening of all options $\Rightarrow$ high-cost evaluation of surviving options
  - **Examples:** Admissions and hiring, clinical trials, matching platforms
- **Why Graduate Admissions:** Benefits graduate programs and future applicants
  - Graduate programs invest time and money into students' development
  - If programs were undervaluing applicants with certain characteristics, future such applicants may receive formerly-withheld opportunities

Research Questions

1. Which factors in applications affect admission and success in graduate school?
   - Literature measures success with program completion and job placement ranking
2. Are admissions committees optimizing, or are they making mistakes?
   - Goal is to accept applicants with highest success probabilities who will matriculate
   - Predictors of admission only (success only) may be overvalued (undervalued)
3. If they are making mistakes, how can they make better admissions decisions?
   - In turn, by changing decision rules, how much better of a cohort can they admit?

★ This paper's methods generalize to admissions, hiring, and other settings

Methodology and Main Results

- Build structural models of admissions committee's decision problem that support:
  - Statistical tests of whether committee's decision rule aligns with desired objective
  - Counterfactual simulations quantifying gains from deviating to model's decision rule
- Estimate model parameters with admissions data from a major research university
  - Do estimated success probabilities align with those implied by committee's decisions?
  - Tests reject optimality and counterfactual deviation gains statistically significant when committee misvalues applicant characteristics, but not when it optimizes

Literature Review

- **Graduate Admissions:** Regress acceptance and success on applicant charateristics
  - **Original Literature:** Attiyeh & Attiyeh (1997); Krueger & Wu (2000); Grove & Wu (2007); Athey, Katz, Krueger, Levitt & Poterba (2007)
  - **Study of 2002 Cohort:** Stock, Finegan & Siegfried (2006–2015)
  - **New Predictors/Methods:** Bai, Esche, MacLeod & Shi (2022)
- **Optimal Dynamic Selection Under Costly Evaluation:**
  - **Sequential Search:** McCall (1970); Mortensen (1970); Weitzman (1979)
  - **Rational Inattention:** Sims (2003); Matějka & McKay (2015); Caplin & Dean (2015)

★ **This Paper:** Builds and estimates dynamic selection models of admissions, which test optimality by relating acceptance effects to success effects

Synthetic Data Protocol (IRB-Approved)

1. **Administrative office receives data from university departments**
   - Algorithmically extracts variables from textual files, such as recommendation letters
2. **Synthetic data (Patki et al. 2016) used for model development**
   - Estimate joint distribution of variables, and simulate synthetic dataset from it
   - Rows do not correspond to real subjects, preventing identity disclosure
   - Cells with < 5 comparable observations binned to prevent attribute disclosure
3. **Administrative office executes final estimation code on actual data**
   - Only summary statistics and parameter estimates are returned to researcher

★ **Contribution:** Privacy-protecting framework for researchers studying sensitive data

▸▸ Synthetic Data Example

Outline

1. Reduced-Form Analysis
2. Structural Models
3. Structural Estimation
4. Optimality Tests

Data Description

- **Sample:** Applicants to research university's economics, government, history, linguistics, and psychology PhD programs from 2015 to 2023
- **Dependent Variables:** Measures of admission and success
  - **Admission:** Indicators for all stages of admissions process, including waitlist
  - **Success:** Completion and placement category/quality for **all** applicants, with academia-equivalent rankings for non-academic placements (Krueger & Wu 2000)
- **Independent Variables:** Objective vs. subjective
  - **Objective:** Demographics and performance measures, such as GRE scores
  - **Subjective:** Textual processing of recommendation letters, application essays, CVs, and supplemental forms using dictionaries to construct variables

▸▸ Variables ▸▸ Inference vs. Prediction ▸▸ Sample Selection

Reduced-Form Summary

- **Data:** All applicants to all 5 graduate programs from 2020 to 2023
  - Currently have results for acceptance and objective variables only
  - Implementing synthetic data protocol to complete the empirical work
- **Method:** Logistic regressions with lasso for variable selection
- **Results:** Economics admissions cares most about quantitative GRE and undergraduate GPA, and is most data-driven based on goodness of fit
  - Need success data to see if other programs are making mistakes
  - Prolific recommenders increase acceptance probability $\Rightarrow$ network effects

Summary Statistics · Summary Statistics (Matriculation)

Post-Lasso AMEs: Predicting Acceptance

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
GRE Verbal Percentile	0.0012***	0.0013**	0.0032***		0.0022**
GRE Quantitative Percentile	0.0119***	0.0006*	0.0019***	0.0009	
GRE Analytic Percentile	0.0014***	0.0009**		0.0009	
Undergraduate GPA	0.1710***	0.0267	0.0985**	0.0499	
TOEFL/IELTS	0.0299	0.0254	0.0758*	0.0474	
Elite U.S. University	0.0635***	0.0144	0.0452*		0.0346
Elite U.S. Liberal Arts College	0.0806***	0.0247*	0.0771**		0.0604
Elite Foreign University	0.0365*		0.0432	0.0535	
Graduate Degree			0.0210		
Work Experience	0.0293**	0.0157	0.0383		0.0722**
#Professor Recommenders			0.0436**	0.0259*	
#Prolific Recommenders	0.0209***	0.0115**	0.0160	0.0184**	0.0195
#Math Courses	0.0131***				
Advanced Math	0.0180				
Pseudo-R2	0.2671	0.1500	0.1847	0.1722	0.1844

Full Post-Lasso AMEs   Logistic Regression AMEs   Logistic Regression Results (Matriculation)

Acceptance vs. Success AMEs

Program = Economics	Accept	Success
GRE Verbal Percentile	0.0012***	
GRE Quantitative Percentile	0.0119***	
GRE Analytic Percentile	0.0014***	
Undergraduate GPA	0.1710***	
TOEFL/IELTS	0.0299	
Elite U.S. University	0.0635***	
Elite U.S. Liberal Arts College	0.0806***	
Elite Foreign University	0.0365*	
Work Experience	0.0293**	
#Prolific Recommenders	0.0209***	
#Math Courses	0.0131***	
Advanced Math	0.0180	

- Admissions literature regresses acceptance and success on variables in applications, and qualitatively compares the effects
- While such comparisons are of interest, they are insufficient to test optimality
- Need model of admissions to relate acceptance effects to success effects

Insufficiency of Comparing AMEs

Outline

Why Structural Models?

- **Reason 1:** Test whether committee's decision rule aligns with desired objective
  - Per literature, committee should accept applicants with highest success probabilities
  - Committees also accept applicants with higher matriculation probabilities
- **Reason 2:** Find counterfactual decision rule in which committee optimizes
  - Quantify gains from deviating from actual to counterfactual decision rule

Model Framework

- **Portfolio Choice Problem:** Accept subset of applicants with highest sum of success probabilities subject to capacity constraint on number of acceptances

- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}}, \text{ such that: } \sum_{n=1}^{N} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_A,$$

where x = objective variables, z = subjective variables, $y \in \{\text{Success}, \text{Failure}\}$, N = number of applicants, and N_A = maximum number of acceptances

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- BEMS (2022) prove that portfolio choice problem reduces to cutoff rule problem
  - Accept applicant if and only if their success probability exceeds cutoff probability
  - Decisions across applicants now independent, so no need to consider all subsets

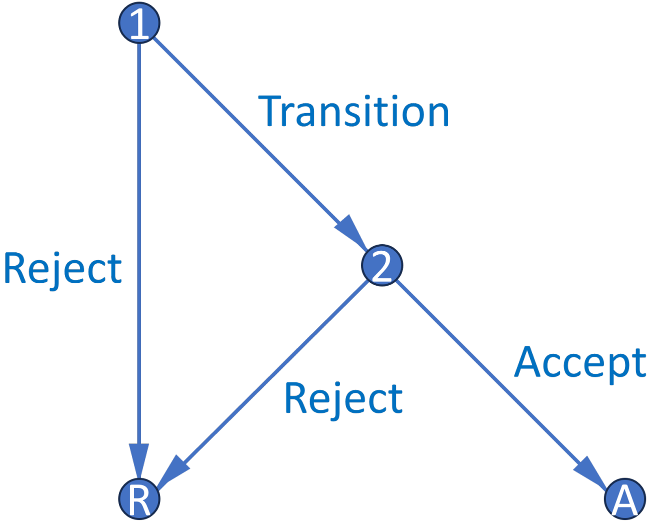
⏩ Portfolio Choice to Cutoff Rule

Cutoff Rule Problem

- Start with two-round cutoff rule problem that accounts for costly evaluation:
  - **Round 1:** See only objective variables (e.g. GRE). Can transition application file to Round 2 to see subjective variables (e.g. recommendation letters), or can reject
  - **Round 2:** If chose to transition application file in Round 1, can accept or reject
  - Assume for now committee transitions and accepts optimal number of files
- Along with the baseline dynamic model, there are two additional models
  - Static model for high-dimensional variables and comparing cutoffs across rounds
  - Waitlist model adds a third round and accounts for matriculation probabilities

▸▸ Models of Admissions

Decision Graph



Dynamic Model

- **States:** $r =$ round, $x =$ objective variables, $z =$ subjective variables, $t =$ year, $\epsilon_r =$ Type 1 extreme value preference shock with scale parameter σ
- **Controls:** $d_1 \in \{\text{Transition}, \text{Reject}\}$, $d_2 \in \{\text{Accept}, \text{Reject}\}$
- **Outcomes:** $y \in \{\text{Success}, \text{Failure}\}$, $y_r^* =$ marginal outcome

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Round 2:

$$v_2(x, z, t, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, t, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, t, d_2) = \underbrace{P(y = S|x, z)\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_2^* = S|t)\mathbb{1}(d_2 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Round 2 Portfolio Choice
Dynamic Parameterization

Dynamic Model

Round 1:

$$v_1(x, t, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, d_1) = \underbrace{\mathbb{E}[v_2(x, z, t, \epsilon_2)|x, t]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, t, \epsilon_2)|x, t] = \int_{\widetilde{z}} \sigma \log \left(e^{P(y=S|x,\widetilde{z})/\sigma} + e^{P(y_2^*=S|t)/\sigma} \right) dF(\widetilde{z}|x)$$

Dynamic Model

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Conditional Choice Probabilities:

$$P(d_r|x,(z),t) = \frac{\exp(u_r(x,(z),t,d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(x,(z),t,\widetilde{d_r})/\sigma)}$$

Round 1 Portfolio Choice
Dynamic Parameterization

Static Model

- **Goals:** Handle high-dimensional z, and make Round 1 cutoff a probability

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Round 2:

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$$v_1(x, t, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, d_1) + \epsilon_1(d_1), \text{ such that:}$$

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- $P(y = S|x)$ estimates inconsistent, but fitted values account for x's effect on z

Round 2 Portfolio Choice | Round 1 Portfolio Choice | Static Parameterization

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants
- For each applicant $n \in \{1, \ldots, N\}$, make decision $d_n \in \{\text{Accept}, \text{Reject}\}$ given:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M,$$

where $(x, z) =$ variables, $y \in \{\text{Success}, \text{Failure}\}$, $m \in \{\text{Matriculate}, \text{Decline}\}$, $N =$ number of applicants, and $N_M =$ maximum number of matriculants

Portfolio Choice with Matriculation

- Accept subset of applicants with highest sum of success times matriculation probabilities subject to capacity constraint on expected number of matriculants

$$\max_{(d_1,\dots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} \underbrace{P(y_n = S|x_n, z_n)}_{\text{Success probability}} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}},$$

$$\text{such that: } \sum_{n=1}^{N} \underbrace{P(m_n = M|x_n, z_n)}_{\text{Matriculation probability}} \underbrace{\mathbb{1}(d_n = A)}_{\text{Acceptance}} \leq N_M$$

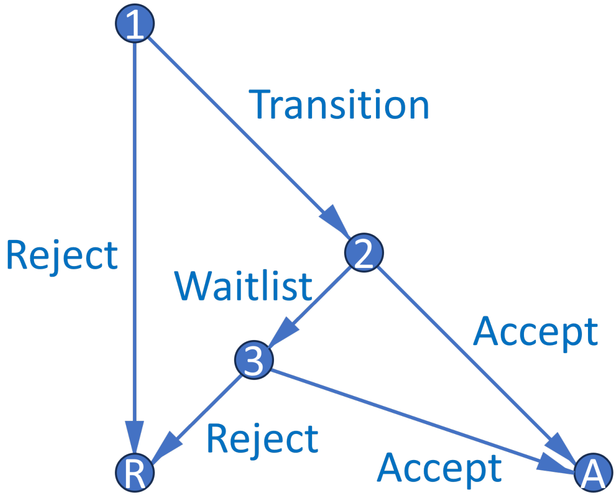
Theorem 1 (Proof)

For large N, this problem has a cutoff rule solution on **success** *probabilities.*

- Likely matriculants more valuable, but also more expensive $\Rightarrow$ effects cancel
- **Upshot:** Matriculation probabilities only matter if there are dynamic effects

Cutoff Rule with Matriculation

Waitlist Decision Graph



- Three-round model features waitlist and accounts for applicants' decisions
- Dynamic mechanism incentivizes giving early acceptances to likely matriculants

Waitlist Model, Round 3

- Respecify Round 3 cutoff $P(y_3^* = S|t)$, where $t =$ year, as $P(y_3^* = S|\overline{m_{2A}})$
  - $\overline{m_{2A}} =$ average early accepted applicant's matriculation probability
    - Enthusiasm variables can affect matriculation independently of success
  - $P(y_3^* = S|\overline{m_{2A}})$ increases in $\overline{m_{2A}}$ because larger $\overline{m_{2A}}$ means fewer openings

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$$v_3(x, z, \overline{m_{2A}}, \epsilon_3) = \max_{d_3 \in \{A,R\}} u_3(x, z, \overline{m_{2A}}, d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x, z, \overline{m_{2A}}, d_3) = \underbrace{P(y = S|x, z)\mathbb{1}(d_3 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{P(y_3^* = S|\overline{m_{2A}})\mathbb{1}(d_3 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Waitlist Model, Round 2

- What information available at start of Round 2 determines $\overline{m_{2A}}$?
  - $\ddot{m}_2$ is defined for all transitioned applicants as $\overline{m_2}$ with m_2 excluded, where $m_2 =$ applicant's own matriculation probability. Within a year, high m_2 implies low $\ddot{m}_2$
  - $\ddot{m}_2$ appears when applicant is waitlisted $\Rightarrow$ represents $\overline{m_2}$ of applicants committee *can* accept (vs. $\overline{m_{2A}}$, which represents $\overline{m_2}$ of applicants committee *does* accept)

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$$v_2(x, z, \ddot{m}_2, \epsilon_2) = \max_{d_2 \in \{A,W\}} u_2(x, z, \ddot{m}_2, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x, z, \ddot{m}_2, d_2) = \underbrace{P(y = S|x, z)\mathbb{1}(d_2 = A)}_{\text{If accept, get } P(\text{Success})} + \underbrace{\beta\mathbb{E}[v_3(x, z, \overline{m_{2A}}, \epsilon_3)|x, z, \ddot{m}_2]\mathbb{1}(d_2 = W)}_{\text{If waitlist, get discounted } \mathbb{E}[\text{Social Surplus}]}, \text{ such that:}$$

$$\mathbb{E}[v_3(x, z, \overline{m_{2A}}, \epsilon_3)|x, z, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log\left(\sum_{\widetilde{d}_3 \in \{A,R\}} e^{u_3(x,z,\widetilde{\overline{m_{2A}}},\widetilde{d}_3)/\sigma}\right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

Waitlist Model, Round 1

Waitlist Dynamics

Theorem 2 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- Compared to low m_2 applicant, high m_2 applicant has low $\ddot{m}_2$, and therefore low $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, $\mathbb{E}[P(\text{Cutoff})_3]$, and $\mathbb{E}[v_3]$. This increases relative payoff of accepting them
- Why not accept low m_2 applicant and fill spot with waitlisted high m_2 applicant?
  1. m_2 declines between Rounds 2 and 3 and has farther to fall for high m_2 applicant
  2. In general, matriculation probabilities can change unpredictably between rounds

★ **Upshot:** Accepting applicants who want to matriculate benefits the program

Waitlist Notes Symmetric Waitlist Model

Outline

Maximum Likelihood Estimation

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}} \underbrace{\left[P_{y_{1,n}}(\theta_{S_1}) P_{y_{2,n}}(\theta_{S_2})\right]^{\mathbb{1}(y_n=S)}}_{\text{Success probabilities}}$$

$$\underbrace{\left[(1 - P_{y_{1,n}}(\theta_{S_1}))(1 - P_{y_{2,n}}(\theta_{S_2}))\right]^{\mathbb{1}(y_n=F)}}_{\text{Success probabilities}} \prod_{n=N_T+1}^{N} \underbrace{(1 - P_{d_{1,n}}(\theta_T)) P_{y_{1,n}}(\theta_{S_1})^{\mathbb{1}(y_n=S)} (1 - P_{y_{1,n}}(\theta_{S_1}))^{\mathbb{1}(y_n=F)}}_{\text{Conditional choice probabilities}}$$

- In unrestricted likelihood $\mathcal{L}^U$, parameterize CCPs with flexible logit specifications
- In restricted likelihood $\mathcal{L}^R$, use model CCPs, which depend on structural parameters
- Do success probabilities implied by outcomes align with those implied by CCPs?

Dynamic Likelihood Waitlist Likelihood

Cutoff Probability Identification

Theorem 3 (Proof)

The maximum likelihood cutoff probability estimate is such that the number of acceptances N_A equals the expected N_A. Also, for any cutoff probability and N_A, the likelihood is highest when the committee accepts the best N_A applicants.

- Identify each year's cutoffs by how selective committee is that year
- While selectivity identifies cutoffs, sorting improves restricted likelihood
  - If committee engages in sorting, likelihood-ratio test less likely to reject
  - $\text{LR} = 2\left[\log(\hat{\mathcal{L}}^U) - \log(\hat{\mathcal{L}}^R)\right] \sim \chi^2_Q$, such that $Q = \dim(\theta^U) - \dim(\theta^R)$

Empirical Considerations

Unrestricted Static AMEs (Optimal)

	Transition	Accept	Success 1	Success 2
GRE Quantitative Percentile	0.0208***	0.0177***	0.0213***	0.0177***
	(0.0019)	(0.0033)	(0.0019)	(0.0039)
Gender	0.0000	0.0347	0.0008	-0.0086
	(0.0298)	(0.0448)	(0.0281)	(0.0481)
Demographic 1	-0.0736*	-0.0776	-0.0549	-0.0583
	(0.0416)	(0.0662)	(0.0392)	(0.0709)
Demographic 2	0.0091	-0.0215	0.0467	0.1152*
	(0.0447)	(0.0693)	(0.0413)	(0.0678)
Advanced Math		0.1304***		0.1185**
		(0.0463)		(0.0513)
Committee Score		0.0174		0.0050
		(0.0113)		(0.0119)
Year Transition More	0.1726***	-0.2345***		
	(0.0276)	(0.0378)		

- Data simulated from reduced-form-calibrated data-generating parameters
- Committee's decision rule aligns with maximizing success probabilities

Summary Statistics (Optimal) | Full Unrestricted Static Estimates (Optimal) | Unrestricted Dynamic Estimates (Optimal)

Unrestricted Static AMEs (Suboptimal)

	Transition	Accept	Success 1	Success 2
GRE Quantitative Percentile	0.0193***	0.0077**	0.0213***	0.0173***
	(0.0020)	(0.0036)	(0.0019)	(0.0039)
Gender	0.0376	0.0646	0.0008	-0.0123
	(0.0297)	(0.0442)	(0.0281)	(0.0476)
Demographic 1	-0.1017**	-0.1036	-0.0549	-0.0487
	(0.0412)	(0.0639)	(0.0392)	(0.0701)
Demographic 2	0.0071	-0.0133	0.0467	0.1161*
	(0.0441)	(0.0676)	(0.0413)	(0.0661)
Advanced Math		0.0797*		0.1337***
		(0.0482)		(0.0504)
Committee Score		0.0425***		0.0012
		(0.0120)		(0.0121)
Year Transition More	0.1886***	-0.2317***		
	(0.0273)	(0.0397)		

- Data simulated from reduced-form-calibrated data-generating parameters
- *Demographic 1* and *Advanced Math* undervalued, *Committee Score* overvalued

Summary Statistics (Suboptimal) Full Unrestricted Static Estimates (Suboptimal) Unrestricted Dynamic Estimates (Suboptimal)

Restricted Static Estimates (Optimal)

	$P(\text{Cutoff})_1$	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Year Transition Fewer | Intercept	49.77%	36.60%	-10.0587***	-8.9338***	
	(38.43%, 61.13%)	(28.53%, 45.50%)	(2.2082)	(2.4363)	
GRE Quantitative Percentile			0.1090***	0.0882***	
			(0.0259)	(0.0275)	
Gender			0.0070	0.0220	
			(0.1080)	(0.1620)	
Demographic 1			-0.3346*	-0.2945	
			(0.1803)	(0.2626)	
Demographic 2			0.1444	0.2881	
			(0.1702)	(0.2608)	
Advanced Math				0.6055***	
				(0.1854)	
Committee Score				0.0657	
				(0.0419)	
Year Transition More	31.86%	64.56%			
	(27.27%, 36.83%)	(47.28%, 78.72%)			
Sigma					0.2149***
					(0.0597)
LR Statistic | DF | P-Value	8.4729	9	0.4873		

- Likelihood-ratio test does not reject model fitting the data
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 2 cutoff probability

Full Restricted Static Estimates (Optimal) Restricted Dynamic Estimates (Optimal)

Restricted Static Estimates (Suboptimal)

	$P(\text{Cutoff})_1$	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Year Transition Fewer | Intercept	53.84%	37.04%	-10.3327***	-7.2584***	
	(40.49%, 66.66%)	(27.83%, 47.29%)	(1.9333)	(2.2514)	
GRE Quantitative Percentile			0.1123***	0.0663**	
			(0.0227)	(0.0260)	
Gender			0.1010	0.0839	
			(0.1144)	(0.1650)	
Demographic 1			-0.4233**	-0.3725	
			(0.1783)	(0.2547)	
Demographic 2			0.1588	0.2935	
			(0.1773)	(0.2525)	
Advanced Math				0.5525***	
				(0.1957)	
Committee Score				0.1161***	
				(0.0447)	
Year Transition More	31.62%	66.37%			
	(26.62%, 37.07%)	(47.08%, 81.40%)			
Sigma					0.2473***
					(0.0680)
LR Statistic | DF | P-Value	19.2965	9	0.0228		

- Likelihood-ratio test rejects model fitting the data at 5% level
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 2 cutoff probability

Full Restricted Static Estimates (Suboptimal) Restricted Dynamic Estimates (Suboptimal)

Unrestricted Waitlist-Hybrid AMEs (Optimal)

	$\overline{m_{2A}}$	Transition	Accept 2	Accept 3	Success 1	Success 3	Matriculate
GRE Quantitative Percentile		0.0177***	0.0066**	0.0082**	0.0243***	0.0192***	-0.0095
		(0.0020)	(0.0029)	(0.0034)	(0.0020)	(0.0035)	(0.0149)
Gender		0.0124	-0.0510	-0.0173	-0.0145	-0.0111	0.3919**
		(0.0296)	(0.0363)	(0.0403)	(0.0306)	(0.0486)	(0.1629)
Demographic 1		-0.0601	0.0234	0.0611	-0.1469***	-0.0855	0.4612**
		(0.0407)	(0.0616)	(0.0654)	(0.0414)	(0.0718)	(0.1801)
Demographic 2		-0.0165	0.0475	0.0263	-0.0152	0.1131	0.7364***
		(0.0450)	(0.0605)	(0.0675)	(0.0450)	(0.0743)	(0.2082)
Advanced Math			0.0257	-0.0081		-0.0214	0.5676***
			(0.0367)	(0.0484)		(0.0520)	(0.1723)
Committee Score			-0.0017	0.0087		0.0293**	-0.1181***
			(0.0095)	(0.0118)		(0.0133)	(0.0236)
Year Transition/Matriculate More		0.1674***					0.4422***
		(0.0277)					(0.1208)
$\ddot{m}_2$	0.8325***		-0.0588				
	(0.2037)		(0.0841)				
$\overline{m_{2A}}$				-1.2509***			
				(0.4445)			

- Watlist-hybrid model: Round 1 is static, Rounds 2 and 3 are dynamic
- AME signs as expected for $\ddot{m}_2$ on $\overline{m_{2A}}$, and for $\overline{m_{2A}}$ on *Accept 3*

Waitlist Summary Statistics (Optimal) | Full Unrestricted W-H Estimates (Optimal) | Unrestricted W-D Estimates (Optimal)

Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

	$\overline{m_{2A}}$	Transition	Accept 2	Accept 3	Success 1	Success 3	Matriculate
GRE Quantitative Percentile		0.0218***	-0.0027	0.0039	0.0175***	0.0170***	0.0035
		(0.0019)	(0.0032)	(0.0035)	(0.0023)	(0.0040)	(0.0083)
Gender		0.0681**	0.0508	-0.0989**	-0.0446	-0.0030	-0.0525
		(0.0286)	(0.0387)	(0.0441)	(0.0314)	(0.0504)	(0.1051)
Demographic 1		-0.0430	-0.0202	-0.1463**	-0.0431	-0.0253	0.1897
		(0.0400)	(0.0589)	(0.0710)	(0.0427)	(0.0792)	(0.1567)
Demographic 2		0.0852**	0.0354	-0.1352**	-0.0320	-0.0452	0.0949
		(0.0431)	(0.0604)	(0.0662)	(0.0477)	(0.0803)	(0.1640)
Advanced Math			0.0623	0.0056		0.0310	0.1291
			(0.0414)	(0.0467)		(0.0543)	(0.1095)
Committee Score			0.0009	0.0167		0.0068	-0.1315***
			(0.0087)	(0.0102)		(0.0122)	(0.0272)
Year Transition/Matriculate More		0.1979***					0.1955*
		(0.0265)					(0.1053)
$\ddot{m}_2$	1.2940***		-0.5186				
	(0.2636)		(0.3836)				
$\overline{m_{2A}}$				-1.9986**			
				(0.9596)			

- Watlist-hybrid model: Round 1 is static, Rounds 2 and 3 are dynamic
- AME signs as expected for $\ddot{m}_2$ on $\overline{m_{2A}}$, and for $\overline{m_{2A}}$ on *Accept 3*

Waitlist Summary Statistics (Suboptimal) | Full Unrestricted W-H Estimates (Suboptimal) | Unrestricted W-D Estimates (Suboptimal)

Restricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m_{2A}}$	$P(\text{Cutoff})_1$	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	0.0760	38.21%	6.51%	-10.8672***	-9.9856***	1.5968		
	(0.1019)	(35.16%, 41.37%)	(0.67%, 41.91%)	(0.9634)	(1.6344)	(9.6640)		
GRE Quantitative Percentile				0.1217***	0.0996***	-0.0715		
				(0.0112)	(0.0187)	(0.1162)		
Gender				-0.0289	-0.1703	2.9518*		
				(0.1212)	(0.2000)	(1.5616)		
Demographic 1				-0.6107***	-0.2214	3.4735**		
				(0.1657)	(0.3133)	(1.6588)		
Demographic 2				-0.0829	0.4767	5.5471**		
				(0.1796)	(0.2952)	(2.2266)		
Advanced Math					-0.0150	4.2756***		
					(0.2301)	(1.6021)		
Committee Score					0.1114**	-0.8898***		
					(0.0557)	(0.2508)		
Year Transition/Matriculate More		27.53%	100.00%			3.3305***		
		(24.44%, 30.84%)	(29.39%, 100.00%)			(1.2386)		
$\ddot{m}_2$	0.8325***							
	(0.2037)							
Beta							0.9991***	
							(0.0023)	
Sigma								0.3013***
								(0.0253)
LR Statistic | DF | P-Value	13.3895	16	0.6441					

- Likelihood-ratio test does not reject model fitting the data
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 3 cutoff probability

▸▸ Full Restricted Waitlist-Hybrid Estimates (Optimal) ▸▸ Restricted Waitlist-Dynamic Estimates (Optimal)

Restricted Waitlist-Hybrid Estimates (Suboptimal)

	$\overline{m_{2A}}$	$P(\text{Cutoff})_1$	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Year Transition/Matriculate Fewer | Intercept	-0.0332	36.58%	24.35%	-9.0363***	-4.8172***	-1.3135		
	(0.0222)	(32.93%, 40.39%)	(14.18%, 38.53%)	(0.9974)	(1.6258)	(3.5812)		
GRE Quantitative Percentile				0.0969***	0.0482***	0.0064		
				(0.0118)	(0.0187)	(0.0410)		
Gender				0.0491	-0.1320	-0.4025		
				(0.1152)	(0.1781)	(0.5294)		
Demographic 1				-0.1924	-0.4306	1.9429*		
				(0.1572)	(0.2621)	(1.1547)		
Demographic 2				0.1492	-0.3491	1.2089		
				(0.1710)	(0.2696)	(1.2045)		
Advanced Math					0.2118	0.8972*		
					(0.1857)	(0.5374)		
Committee Score					0.0568	-0.6934***		
					(0.0439)	(0.1711)		
Year Transition/Matriculate More		25.84%	99.89%			1.7935***		
		(23.16%, 28.72%)	(17.09%, 100.00%)			(0.5537)		
$\ddot{m}_2$	1.0987***							
	(0.1004)							
Beta							1.0000***	
							(0.0000)	
Sigma								0.2437***
								(0.0405)
LR Statistic | DF | P-Value	57.1196	16	0.0000					

- Likelihood-ratio test rejects model fitting the data at 1% level
- In *Year Transition Fewer*, Round 1 cutoff probability > Round 3 cutoff probability

⏩ Full Restricted Waitlist-Hybrid Estimates (Suboptimal) ⏩ Restricted Waitlist-Dynamic Estimates (Suboptimal)

Outline

Likelihood-Ratio and Deviation Gains Tests

- **Likelihood-Ratio:** Determines whether model fits data generated by committee's decision rule, but not whether committee can do better by changing decision rule
- **Deviation Gains:** Suppose committee adopts model's decision rule. Will average success probability of acceptance/matriculation set increase, and if so, by how much?
  - Solve model with $\sigma = 0$ in CCPs, compute average yearly $P(\text{Success})$ of model's acceptance/matriculation set, and compare to that of actual decision rule's set
  - $\hat{\theta}_S^R$ may be biased to rationalize suboptimal behavior, so plug in $\hat{\theta}_S^U$
  - **Example:** If committee puts too little weight on *Advanced Math*, then its restricted success probability parameter estimate will be biased down

Optimality Tests Notes

Deviation Gains Interpretation

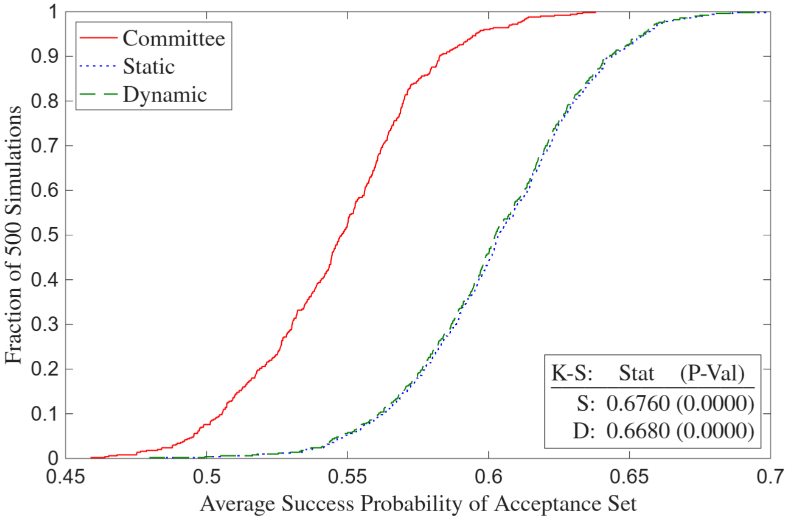
- Suppose there are meaningfully large deviation gains. Then committee may have incorrect beliefs about which types of applicants are most likely to succeed
  - Matriculation rate of model's acceptance set may be less than that of actual decision rule's acceptance set, making estimated gains an upper bound
  - In waitlist counterfactual, compare success probabilities of matriculation sets by simulating accepted applicants' matriculation decisions after Rounds 2 and 3

★ **Robustness:** Given too few observations for holdout set and infeasibility of large k-fold cross-validation due to runtime, implement randomization test

  - Compute average success probabilities with perturbations of $\hat{\theta}_S^U$

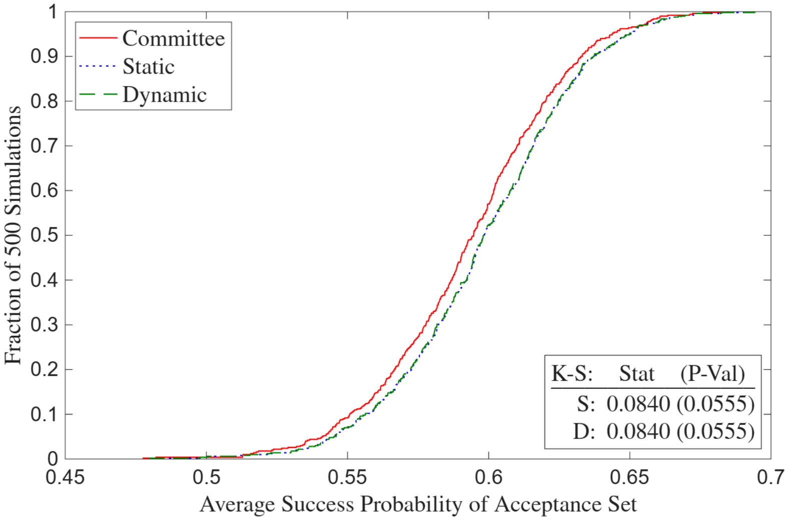
Deviation Gains Test

Deviation Gains (Suboptimal)



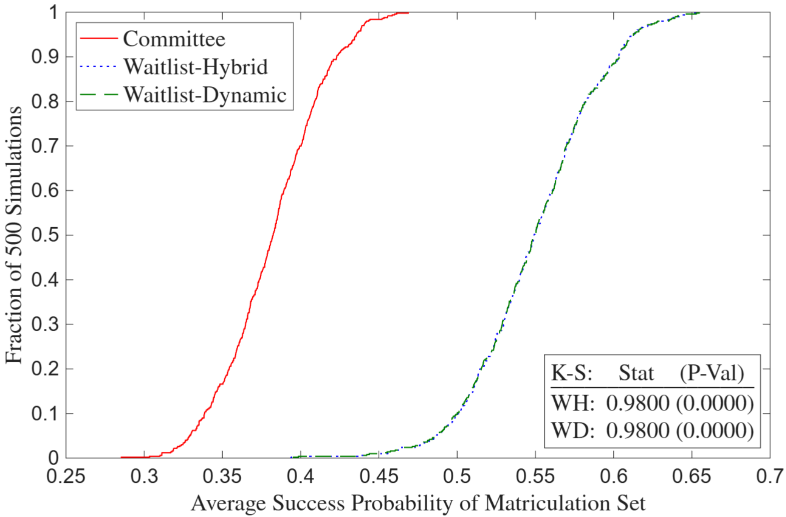
▸▸ Full Deviation Gains (Suboptimal) ▸▸ Deviation Gains, z Missing (Suboptimal)

Deviation Gains (Optimal)



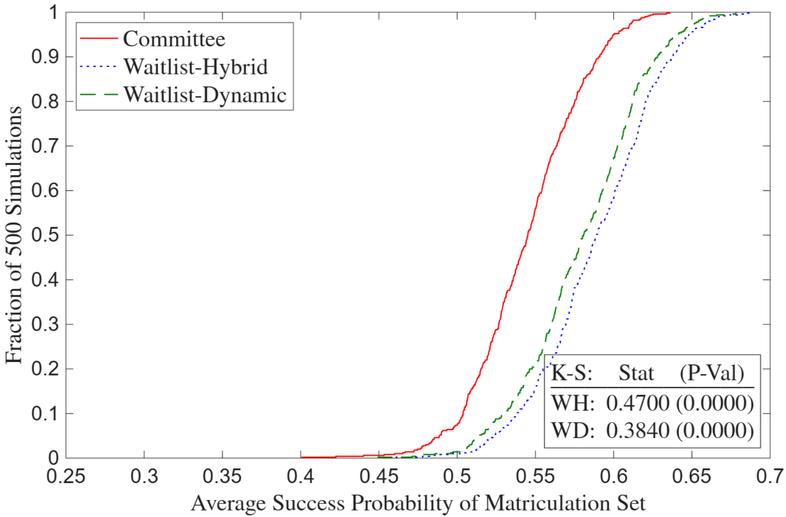
Full Deviation Gains (Optimal)
Deviation Gains, z Missing (Optimal)

Waitlist Deviation Gains (Suboptimal)



▸▸ Waitlist Deviation Gains, z Missing (Suboptimal)

Waitlist Deviation Gains (Optimal)



Waitlist Deviation Gains, z Missing (Optimal)

Conclusion

- **Economic Problem:** Selecting best options under costly evaluation
- **Application:** Graduate admissions using synthetic data protocol to protect privacy
- **Contribution:** Build models to test optimality and quantify deviation gains
  1. Economics admissions more data-driven than other social science programs
  2. Tradeoff between marginal cost of reading file vs. quality of marginal acceptance
     - Exploit tradeoff to calculate, in dollars, how much success is worth to committee
     - Alternatively, what if committee uses AI to read every applicant's file?
- **Future Research:** Two-sided matching model across multiple universities

Summary of Results Multiyear Models

Thank You!

A1: Synthetic Rows $\neq$ Real Subjects

Transition	Accept	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score
0	0	0	89	0	0	0	.	.
1	0	1	96	0	1	0	1	5
0	0	1	88	0	1	0	.	.
0	0	0	80	1	1	0	.	.
1	1	1	99	0	1	0	1	9
1	0	1	81	0	0	1	1	4
1	0	1	92	0	1	0	1	4
1	1	1	93	0	0	1	0	6
1	0	1	83	1	0	1	0	7
0	0	0	87	0	1	0	.	.

$$\Downarrow$$

Transition	Accept	Success	GRE Quantitative Percentile	Gender	Demographic 1	Demographic 2	Advanced Math	Committee Score
1	0	0	90	1	1	0	0	5
1	0	0	83	0	0	1	1	6
1	1	1	91	0	0	1	0	5
1	1	1	92	0	0	1	0	5
0	0	1	83	0	0	0	.	.
1	0	0	93	0	0	1	0	9
0	0	0	84	0	0	1	.	.
1	1	1	88	0	1	0	1	4
1	0	1	86	0	1	0	0	6
0	0	1	86	1	0	1	.	.

▸▸ Synthetic Data Protocol

A1: Dependent Variables

Table 1a: Dependent Variables

Admission	Success
$\mathbb{1}$(Transition)	**$\mathbb{1}$(Complete PhD Program)**
$\mathbb{1}$(Accept)	**$\mathbb{1}$(Academic Placement)**
$\mathbb{1}$(Waitlist)	**$\mathbb{1}$(Government Placement)**
$\mathbb{1}$(Open House)	**$\mathbb{1}$(Consulting Placement)**
$\mathbb{1}$(Matriculate)	**$\mathbb{1}$(Tech Placement)**
$\mathbb{1}$(Attend PhD Program)	**Placement U.S. Ranking**
PhD Program U.S. Ranking	**Placement Global Ranking**
PhD Program Global Ranking	**$\mathbb{1}$(Good Placement)**

- Economics-only variables bold, non-economics-only variables italic
- U.S. News PhD program rankings, IDEAS/RePEc placement rankings
  - For non-tenure track, use weighted average of university's and maximum rankings
  - For non-academia, use average of ChatGPT, Claude, and Gemini's rankings

Data Description

A1: Objective Variables

Table 1b: Objective Independent Variables

Demographic	Performance	Economics-Only Performance
$\mathbb{1}$(Covid Year)	GRE Verbal Percentile	**Masters GPA**
Winsorized Age	GRE Quantitative Percentile	**$\mathbb{1}$(Missing Masters GPA)**
$\mathbb{1}$(Female)	GRE Analytic Percentile	**$\mathbb{1}$(Elite U.S. Masters)**
$\mathbb{1}$(U.S. African/Hispanic/Native)	$\mathbb{1}$(Missing GRE Percentiles)	**$\mathbb{1}$(Elite International Masters)**
$\mathbb{1}$(U.S. Asian)	Undergraduate GPA	**$\mathbb{1}$(Other International Masters)**
$\mathbb{1}$(International Asian)	$\mathbb{1}$(Missing Undergraduate GPA)	**$\mathbb{1}$(Research Experience)**
$\mathbb{1}$(International Other)	TOEFL/IELTS Score	**Objective Score**
	$\mathbb{1}$(Missing TOEFL/IELTS Score)	**Relative Committee Score**
	$\mathbb{1}$(Elite U.S. University)	
	$\mathbb{1}$(Elite U.S. Liberal Arts College)	
	$\mathbb{1}$(Elite International University)	
	$\mathbb{1}$(Other International University)	
	$\mathbb{1}$(Graduate Degree)	
	$\mathbb{1}$(Work Experience)	
	Number of Professor Recommenders	
	Number of Prolific Recommenders	
	Concentration Dummy Variables	

Data Description

A1: Subjective Variables

Table 1c: Subjective Independent Variables

Recommendation Letter	Essay/CV/Supplemental Form
Relative Letter Length	**Essay Length (Essay)**
Relative Letter Features	**$\mathbb{1}$(Specific Topic) (Essay)**
Positive Words (YS 2012)	**$\mathbb{1}$(Specific Professor) (Essay)**
Negative Words (YS 2012)	**$\mathbb{1}$(Economics Work Experience) (CV)**
Standout Words (BEMS 2022)	**$\mathbb{1}$(Working Paper) (CV)**
Ability Words (BEMS 2022)	**$\mathbb{1}$(Publication) (CV)**
Research Words (BEMS 2022)	**$\mathbb{1}$(Coding) (CV)**
Grindstone Words (BEMS 2022)	**Number of Math Courses (Supplemental)**
Teaching Words (BEMS 2022)	**$\mathbb{1}$(Advanced Math) (Supplemental)**
Communal Words (BEMS 2022)	**$\mathbb{1}$(Theory Field) (Essay/CV)**
Agentic Words (MHM 2009)	**$\mathbb{1}$(Applied Field) (Essay/CV)**

- Dictionaries from Madera, Hebl & Martin (2009), Young & Soroka (2012), and Bai, Esche, MacLeod & Shi (2022)
- #MathCourses and $\mathbb{1}$(AdvMath) objective after 2020

Data Description

A1: Academia-Equivalent Rankings (Government)

- **European Institutions:** European Central Bank (40), HM Treasury (80)
- **Foreign Central Banks:** Banco Central do Brasil (93), Banco de México (93), Bank of Canada (80), Bank of England (67), Bank of Israel (93), Bank of Italy (80), Bank of Japan (80), Bank of Spain (80), Banque de France (80), Central Bank of Chile (93), Central Bank of Colombia (93), Central Bank of Korea (93), Central Bank of Turkey (93), Norges Bank (80), Reserve Bank of Australia (80), Reserve Bank of India (93), Reserve Bank of New Zealand (93), Sveriges Riksbank (80), Swiss National Bank (80)
- **Multilaterals/International:** African Development Bank (93), Asian Development Bank (93), Bank for International Settlements (53), European Bank for Reconstruction and Development (93), European Investment Bank (93), Inter-American Development Bank (93), International Monetary Fund (40), Statistics Canada (107), World Bank (40)
- **Policy Research Institutes:** American Enterprise Institute (93), American Institutes for Research (107), Becker Friedman Institute (80), Brookings Institution (80), Center for Global Development (93), Center for Migration Studies in New York (112), Center for Strategic and International Studies (107), Cowles Foundation (80), Harris School Policy Labs (93), Hoover Institution (80), IDinsight (112), Innovations for Poverty Action (93), J-PAL (80), MDRC (107), National Bureau of Economic Research (67), Peterson Institute for International Economics (80), Pew Research Center (107), RAND Corporation (80), Resources for the Future (93), Urban Institute (107)
- **U.S. Executive/Cabinet:** Department of Commerce (107), Department of Justice (80), Department of Labor (107), Department of the Treasury (80)
- **U.S. Executive Policy Offices:** Council of Economic Advisers (80), Office of Management and Budget (93)
- **U.S. Federal Reserve System:** Board of Governors (40), Atlanta (80), Boston (80), Chicago (80), Cleveland (80), Dallas (80), Kansas City (80), Minneapolis (80), New York (67), Philadelphia (80), Richmond (80), San Francisco (80), St. Louis (80)
- **U.S. GSE/Housing Finance:** Fannie Mae (120), Freddie Mac (120)
- **U.S. Independent Agencies:** Commodity Futures Trading Commission (93), Federal Deposit Insurance Corporation (93), Federal Energy Regulatory Commission (93), Federal Housing Finance Agency (93), Federal Trade Commission (80), Office of Financial Research (93), Securities and Exchange Commission (93)
- **U.S. Legislative Committees:** Congressional Budget Office (80), Congressional Research Service (107), Joint Committee on Taxation (93), Joint Economic Committee (107)
- **U.S. Science and Research:** National Science Foundation (107)
- **U.S. Statistical Agencies:** Bureau of Economic Analysis (93), Bureau of Labor Statistics (93), Census Bureau (93)

Data Description

A1: Academia-Equivalent Rankings (Consulting/Tech/Other)

- **Consulting:** Aecon Consulting (130), AlixPartners (125), Analysis Group (120), Bain & Compnay (120), Bates White (120), Berkeley Research Group (120), Boston Consulting Group (120), Brattle Group (120), Charles River Associates (120), Coleman Research (130), Compass Lexecon (120), Cornerstone Research (120), Deloitte Economic and Financial Advisory (125), Econic Partners (130), Economists Incorporated (120), Edgeworth Economics (120), Ernst & Young Economic Advisory (125), Frontier Economics (120), FTI Consulting (120), Guidehouse (130), KPMG Economics & Valuation (125), Matrix Economics (130), McKinsey & Company (120), Mercer (135), NERA Economic Consulting (120), Oliver Wyman (120), PwC Economics (125), Roland Berger (125), Willis Towers Watson (135)
- **Tech (Athey & Luca 2019):** Airbnb (107), Alibaba (120), Amazon (93), AppNexus (135), Apple (107), Block/Square (120), ByteDance/TikTok (120), CoreLogic (135), Coursera (120), DStillery (135), Didichuxing (125), Digonex (135), DoorDash (107), eBay (107), ECONorthwest (125), Expedia (120), Forkcast (135), Glassdoor (120), Google (93), Granular (135), Groupon (125), Houzz (130), Huawei (120), IBM (120), Indeed (120), ING (130), Instacart (120), Intel (120), Kensho (125), Lending Club (135), LinkedIn (107), Lyft (120), Meta/Facebook (93), Microsoft (93), Netflix (93), Nuna (135), Nvidia (120), Pandora (135), PayPal (120), Pinterest (120), Prattle (135), Quantco (130), Quora (130), Redfin (120), Ripple (135), Roblox (120), Rover (135), Salesforce (120), Shopify (120), Spotify (120), Stripe (107), Trulia (125), Uber (107), Upwork (135), Visa (125), Walmart (125), Wealthfront (135), Yahoo (125), Yelp (125), Zillow (120)
- **Tech Labs:** Amazon Science (40), Apple Machine Learning Research (80), DeepMind (80), Google Research (40), Meta Research (80), Microsoft Research (40), Netflix Research (93), OpenAI Research (80), Uber Research (93)
- **Other:** AIG (135), Bank of America (125), Bloomberg (125), Boeing (135) Capital One (125), Exelon (135), ExxonMobil (135), Ford (135), General Electric (135), General Motors (135), Goldman Sachs (125), JPMorgan Chase (125), Liberty Mutual (135), Moody's Analytics (125), Morningstar (125), Nielsen (135), PNC (135), Progressive (135), S&P Global (125), Shell (135), Swiss Re (130), Toyota (130), Other (135), No Placement (150)

Data Description

A1: Dictionaries

- **Recommendation Letters:** Porter stemmer reduces words to their roots
  - **Positive/Negative Words:** Young & Soroka (2012)'s Lexicoder Sentiment Dictionary
  - **Standout, Ability, Research, Grindstone, Communal Words:** Bai, Esche, MacLeod & Shi (2022)
  - **Agentic Words:** assertive, confident, aggressive, ambitious, dominant, forceful, independent, daring, outspoken, intellectual (Madera, Hebl & Martin 2009); **AND** audacious, bold, command, compete, decisive, determine, drive, enterprise, fearless, influential, leader, pioneer, powerful, proactive, resilient, resourceful, self-assured, strategic, tenacious, vision
  - **Teaching Words:** academic, assess, clarity, coach, collaborate, communicate, cultivate, curriculum, demonstrate, develop, educate, encourage, engage, enlighten, evaluate, explain, facilitate, foster, guide, impart, inspire, insight, instruct, knowledge, learn, lesson, mentor, nurture, participate, pedagogy, present, scholar, support, teach, train, tutor, workshop
- **Essays and CVs:**
  - **Specific Topic:** applied, asset, behavior, climate, computation, data, development, dynamic programming, economics, education, empirical, environment, experiment, finance, fiscal, game theory, health, immigration, industrial, inequality, international, io, labor, linear programming, macro, math, metrics, micro, monetary, policy, political, poverty, pricing, probability, public, sports, statistic, theory, trade, urban
  - **Economics Work Experience:** assistant, fed, imf, international monetary fund, predoc, ra, research, treasury, world bank
  - **Working Paper:** arxiv, center for economic policy research, cepr, econpapers, national bureau of economic research, nber, research papers in economics, repec, social science research network, ssrn, working paper
  - **Publication:** Journal names from IDEAS/RePEc Aggregate Rankings for Journals
  - **Coding:** c/c++, eviews, fortran, java, jax, julia, latex, matlab, python, r, sas, stata, webscrape

Data Description

A1: Inference vs. Prediction

- Challenging to obtain consistent estimates due to omitted variables
  - **Example:** Don't observe effort (e.g. hours worked). Can bias coefficients upward if effort correlated with variables and with success irrespective of those variables
  - Existing literature also suffers from this problem, though BEMS (2022) parse recommendation letters for "grindstone terms" (e.g. depend*, diligen*)
- Research questions emphasize prediction over inference
  - Mullainathan & Spiess (2017) distinguish $\hat{\beta}$ vs. $\hat{y}$ problems
  - If GRE predicts success, committee should use it regardless of why

Data Description

A1: Sample Selection

- Determinants of success conditional on applying vs. matriculating
- BEMS (2022) prove that if $P(\text{Success}|\text{Skill})$ is an S-curve, then $\hat{\beta}$ will be smaller for more selected samples. Consistent with empirical results
- If there aren't some types of outcomes data (e.g. grades, passing comps) for all applicants, or such data are too hard to obtain, use Heckman correction
  - Let $y_n^* = x_n\beta + u_n$, $d_n = \mathbb{1}(z_n\gamma + v_n > 0)$, and $y_n = d_n y_n^*$
  - **Step 1:** Run probit of d on z and compute inverse Mills $\hat{\lambda}_n = \frac{\phi(z_n\hat{\gamma})}{\Phi(z_n\hat{\gamma})}$
  - **Step 2:** Run OLS, probit, or ordered probit of y on $(x, \hat{\lambda})$ to get $\hat{\beta}$
  - **Instruments:** Enthusiasm score, average acceptance rate/quality

Data Description

A1: Insufficiency of Comparing AMEs

- Cannot test optimality by comparing AMEs across regressions. Suppose:
  - Math skills have large positive effect on success, coding skills smaller positive effect
  - 25% of applicants have both skills, 25% math only, 25% coding only, 25% neither
  - Committee can accept half of applicants, and other half attend different university
- Optimizing committee accepts half with math and ignores coding
  - **Acceptance Regression:** $\text{AME}_{\text{Math}} = 1$, $\text{AME}_{\text{Coding}} = 0$
  - **Success Regression:** $1 > \text{AME}_{\text{Math}} > \text{AME}_{\text{Coding}} > 0$
  - Despite optimality, acceptance and success AMEs not equal across regressions
- **Upshot:** Relationship between AMEs must be interpreted through a model

Acceptance vs. Success AMEs

A1: Wald Tests

- Test if admission and success AMEs are same across programs
  - **Example:** Does elite undergraduate university predict admission and/or success more or less for economics than for other programs?
  - **Data:** Simulated from reduced-form-calibrated data-generating parameters
- Can I test if acceptance AMEs equal success AMEs within programs?
  - Predictors of acceptance only (success only) may be overvalued (undervalued), provided committee wants to select based on a "success" model
  - **Problem:** Not a test if admissions decisions are optimal
  - Need structural model of admissions to conclude about optimality

Acceptance vs. Success AMEs

A1: Logit Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\big|X$. Let $P\left(Y_n^E = 1\big|x_n^E;\theta^E\right) = \frac{\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}{1+\exp(x_{n,0}^E\theta_0^E + x_{n,1}^E\theta_1^E)}$, and $P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right) = \frac{\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}{1+\exp(x_{n,0}^{NE}\theta_0^{NE} + x_{n,1}^{NE}\theta_1^{NE})}$. Finally, let $N = N_E + N_{NE}$. Then:

$$\mathcal{L}_N^U\left(y\big|x;\theta\right) = \prod_{n=1}^{N_E} P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)^{Y_n^E}\left[1 - P\left(Y_n^E = 1\big|x_n^E;\theta^E\right)\right]^{1-Y_n^E}$$
$$\prod_{n=1}^{N_{NE}} P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)^{Y_n^{NE}}\left[1 - P\left(Y_n^{NE} = 1\big|x_n^{NE};\theta^{NE}\right)\right]^{1-Y_n^{NE}}.$$

- $Y \in \{\mathbb{1}(\text{Accept}), \mathbb{1}(\text{Attend Program}), \mathbb{1}(\text{Complete Program}), \mathbb{1}(\text{Good Placement})\}$
- **Test:** Let $g(\hat{\theta}) = \frac{1}{N_E}\sum_{n=1}^{N_E}\hat{P}_n^E(1-\hat{P}_n^E)\hat{\theta}_1^E - \frac{1}{N_{NE}}\sum_{n=1}^{N_{NE}}\hat{P}_n^{NE}(1-\hat{P}_n^{NE})\hat{\theta}_1^{NE}$. Then: $W = g(\hat{\theta})'[\nabla_\theta g(\hat{\theta})\hat{V}(\hat{\theta})\nabla_\theta g(\hat{\theta})']^{-1}g(\hat{\theta}) \sim \chi_Q^2$. Compare AMEs per Mood (2010)

Acceptance vs. Success AMEs

A1: Linear Wald Test Across Programs

Assume $\left(Y^E \perp\!\!\!\perp Y^{NE}\right)\Big|X$, i.e. $\left(\text{Economics} \perp\!\!\!\perp \text{Non-Economics}\right)\Big|\text{Predictors}$.

Let $f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) = \frac{1}{\sqrt{2\pi\sigma_E^2}} \exp\left[\frac{-1}{2\sigma_E^2}\left(y_n^E - x_{n,0}^E\beta_0^E - x_{n,1}^E\beta_1^E\right)^2\right]$, and

$f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_S^2\right) = \frac{1}{\sqrt{2\pi\sigma_{NE}^2}} \exp\left[\frac{-1}{2\sigma_{NE}^2}\left(y_n^{NE} - x_{n,0}^{NE}\beta_0^{NE} - x_{n,1}^{NE}\beta_1^{NE}\right)^2\right]$. Then:

$$\mathcal{L}_N^U\left(y \middle| x; \beta, \sigma^2\right) = \prod_{n=1}^{N_E} f\left(y_n^E \middle| x_n^E; \beta^E, \sigma_E^2\right) \prod_{n=1}^{N_{NE}} f\left(y_n^{NE} \middle| x_n^{NE}; \beta^{NE}, \sigma_{NE}^2\right).$$

- $Y \in \{\mathbb{1}(\text{Standardized Committee Score}), \mathbb{1}(\text{Standardized Placement Rank})\}$
- **Test:** Let $g(\hat{\beta}_1) = \hat{\beta}_1^E - \hat{\beta}_1^{NE}$. (Can also do likelihood-ratio test.)
  Then $W = g(\hat{\beta}_1)'[\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)\hat{V}(\hat{\beta}, \hat{\sigma}^2)\nabla_{\beta,\sigma^2} g(\hat{\beta}_1)']^{-1} g(\hat{\beta}_1) \sim \chi_Q^2$
- **Assumption:** Homoskedasticity (i.e. $\forall n,\ \sigma_n^2 = \sigma^2$)

Acceptance vs. Success AMEs

A1: Additional Questions

- Are there different determinants of different types of success? (AKKLP 2007)
  - Need different skills at different stages in the program, such as grades vs. placement
  - For some programs, initial placement matters less than for economics
- Different determinants for international applicants? (KW 2000)
  - **Admission:** May have different preparation and/or interests
  - **Completion:** Depending on country, may have fewer outside options
  - **Placement:** May return to native country after graduation
- Different determinants over time? (2015 to 2023)
  - COVID-19 and other macro factors; changes in professions (e.g. more empirics)
  - For economics, increased importance of elite predocs and Fed RAships

Acceptance vs. Success AMEs

A2.1: Portfolio Choice to Cutoff Rule (BEMS 2022)

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_n = A) \leq N_A$$

$$\mathcal{L}(d_1, \ldots, d_N; \lambda) = \sum_{n=1}^{N} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] + \lambda(N_A - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\},\ d_n(x_n, z_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda^* \end{cases}$$

Upshot: λ^* is cutoff probability. Denote it hereafter as $P(y^* = S)$

- $P(y^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Model Framework

A2.1: Models of Admissions

1. **Static Model:** Committee ranks applicants based on objective variables, reads files of applicants above Cutoff 1, rejects the rest, re-ranks surviving applicants based on all variables, accepts surviving applicants above Cutoff 2, and rejects the rest

2. **Dynamic Model:** Committee forms conditional expectation in Round 1 about applicants who survive to Round 2. Favors applicants in Round 1 whose objective variables are positively correlated with subjective variables that predict success

3. **Waitlist-Hybrid/Dynamic Models:** After Round 1, committee accepts or waitlists applicants in Round 2. After accepted applicants matriculate or don't matriculate, committee accepts or rejects waitlisted applicants in Round 3. Favors likely matriculants in Round 2 to increase selectivity in Round 3

Cutoff Rule Problem

A2.1: Description of Models

- **Round 2:** If accept applicant, get success probability $P(y = S|x, z)$, where Success $= \mathbb{1}$(Complete Program) or $\mathbb{1}$(Good Placement). If reject applicant, get $P(y_2^* = S|t) = P(\text{Success})$ of year t's marginal accepted applicant

- **Round 1:** If reject applicant, get $P(y_1^* = S|t) = P(\text{Success})$ of year t's marginal transitioned applicant. Can be less or greater than $P(y_2^* = S|t)$ depending on value placed on committee's time vs. admitting a successful cohort

  - **Static:** If transition applicant, get $P(y = S|x)$, or $P(\text{Success})$ unconditional on z. It may differ from $P(y = S|x, z)$, though it still accounts for x's effect on z

  - **Dynamic:** If transition applicant, get $\mathbb{E}[v_2(x, z, t, \epsilon_2)|x, t] =$ expected value of Round 2, where z is integrated out conditional on x based on $F(z|x)$

  - **Myopic:** In $\mathbb{E}[v_2(x, z, t, \epsilon_2)|x, t]$, replace $F(z|x)$ with $F(z)$

Cutoff Rule Problem

A2.1: Portfolio Choice to Cutoff Rule, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A), \text{ such that: } \sum_{n=1}^{N_T} \mathbb{1}(d_{2,n} = A) \leq N_A$$

$$\mathcal{L}(d_{2,1}, \ldots, d_{2,N_T}; \lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|x_n, z_n)\mathbb{1}(d_{2,n} = A) + \lambda_2 \mathbb{1}(d_{2,n} = R)\right] + \lambda_2(N_A - N_T)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N_T\},\ d_{2,n}(x_n, z_n|\lambda_2^*) = \begin{cases} A & \text{if } P(y_n = S|x_n, z_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|x_n, z_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|x_n, z_n) < \lambda_2^* \end{cases}$$

Upshot: λ_2^* is cutoff probability. Denote it hereafter as $P(y_2^* = S)$

- $P(y_2^* = S) \in \left[P(y_{N_A+1} = S|x_{N_A+1}, z_{N_A+1}), P(y_{N_A} = S|x_{N_A}, z_{N_A})\right]$

Dynamic Model Static Model

A2.1: Portfolio Choice to Cutoff Rule, Round 1 (Static)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\dots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\dots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[P(y_n = S|x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\dots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } P(y_n = S|x_n) > \lambda_1^* \\ \{T,R\} & \text{if } P(y_n = S|x_n) = \lambda_1^* \\ R & \text{if } P(y_n = S|x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff probability. Denote it hereafter as $P(y_1^* = S)$

- $P(y_1^* = S) \in \left[P(y_{N_T+1} = S|x_{N_T+1}), P(y_{N_T} = S|x_{N_T})\right]$

Static Model

A2.1: Static Parameterization

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, t; \theta_A\right) = \frac{\exp(f_A(x,z,t)\theta_A)}{1 + \exp(f_A(x,z,t)\theta_A)}$$

Success Probabilities:

$$P_{y_1} = P\left(y = S \middle| x; \theta_{S_1}\right) = \frac{\exp(f_{S_1}(x)\theta_{S_1})}{1 + \exp(f_{S_1}(x)\theta_{S_1})}$$

$$P_{y_2} = P\left(y = S \middle| x, z; \theta_{S_2}\right) = \frac{\exp(f_{S_2}(x,z)\theta_{S_2})}{1 + \exp(f_{S_2}(x,z)\theta_{S_2})}$$

Cutoff Probabilities:

$$P_{y_1^*} = P\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \frac{\exp(f_{S_1^*}(t)\theta_{S_1^*})}{1 + \exp(f_{S_1^*}(t)\theta_{S_1^*})}$$

$$P_{y_2^*} = P\left(y_2^* = S \middle| t; \theta_{S_2^*}\right) = \frac{\exp(f_{S_2^*}(t)\theta_{S_2^*})}{1 + \exp(f_{S_2^*}(t)\theta_{S_2^*})}$$

▸▸ Static Model

A2.1: Portfolio Choice to Cutoff Rule, Round 1 (Dynamic)

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)1(d_{1,n}=T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n}=T) \leq N_T$$

$$\mathcal{L}(d_{1,1},\ldots,d_{1,N};\lambda_1) = \sum_{n=1}^{N} \left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n}=T) + \lambda_1 \mathbb{1}(d_{1,n}=R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T,R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Upshot: λ_1^* is cutoff level. Denote it hereafter as $L(y_1^* = S)$

- $L(y_1^* = S) \in \left[EV_{2,N_T+1}, EV_{2,N_T}\right]$. **Note:** Committee takes optimal λ_2^* as given

Dynamic Model

A2.1: Dynamic Parameterization

$$\mathbb{E}[v_2(x,z,t,\epsilon_2)|x,t] = \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A,R\}} e^{u_2(x,\tilde{z},t,\tilde{d}_2)/\sigma} \right) dF(\tilde{z}|x)$$

$$= \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \Big[\cdot \Big] dF_1(\tilde{z}_1|\tilde{z}_2,\ldots \tilde{z}_J, x) \cdots dF_J(\tilde{z}_J|x)$$

- **Continuous:** $f_{z|x}(z_j|x;\beta_{F_{z|x,j}},\sigma^2_{F_{z|x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,j}}}(z_j - g(x)\beta_{F_{z|x,j}})^2\right]$
- **Binary:** $P(z_j = 1|x;\theta_{F_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F_{z|x,j}}\right)}{1+\exp\left(g(x)\theta_{F_{z|x,j}}\right)}$
- **Multinomial:** $P(z_j = (k \neq K)|x;\theta_{F^k_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F^k_{z|x,j}}\right)}{1+\sum_{\ell=1}^{K-1}\exp\left(g(x)\theta_{F^\ell_{z|x,j}}\right)}$

Dynamic Model

A2.1: Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \underbrace{f(z_n|x_n;\theta_F)}_{\text{Density of } z|x} \underbrace{P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}}_{\text{Conditional choice probabilities}}$$

$$\underbrace{P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}}_{\text{Outcome probabilities}} \prod_{n=N_T+1}^{N} \underbrace{f(z_n|x_n;\theta_F)}_{\text{Density of } z|x} \quad \underbrace{(1 - P_{d_{1,n}}(\theta_T))}_{\text{Conditional choice probability}}$$

- **Problem:** I don't observe success for recent years
    - **Solution:** Separate block for recent years without success subblocks
    - **Assumption:** Older years representative of recent years
- If years are similar, can make $(\theta_{S_1^*}, \theta_{S_2^*})$ in $\mathcal{L}^R$ intercept only

Maximum Likelihood Estimation

A2.1: Heckman-Corrected Dynamic Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_T} \Phi(x_n, t_n; \theta_H) f(z_n | x_n, \frac{\phi(x_n, t_n; \theta_H)}{\Phi(x_n, t_n; \theta_H)}; \theta_F) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1 - P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)}$$

$$P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^{N} (1 - \Phi(x_n, t_n; \theta_H))(1 - P_{d_{1,n}}(\theta_T))$$

- z may be unobservable for applicants who didn't survive Round 1
- Estimate Heckman (1979)-corrected full-information likelihood
- **Heckman Instrument:** Year t affects d_1 but not z

Maximum Likelihood Estimation

A2.1: Conditional Independence Assumption

- Should I assume success is independent of admission conditional on predictors?
  - This university aside, if you don't get into anywhere good, you're unlikely to succeed
  - Can bias $\hat{\theta}_S$ upward since without loss of generality, a low GRE is correlated with not getting in anywhere good, which itself prevents you from succeeding
- So should I replace $P\left(y = S \middle| x; \theta_S\right)$ with $P\left(y = S \middle| d, x; \theta_S\right)$?
  - **Problem 1:** If I condition y on $d = \mathbb{1}(\text{Accept})$, I would be "logistically regressing" d on y on d in restricted likelihood, which will fail
  - **Problem 2:** Because d is a function of x, even conditioning y on x and $d = \mathbb{1}(\text{Attend Program})$ will "collapse" $\hat{\theta}_S$ (except for $\hat{\theta}_S^d$)
  - ★ **Example:** In AKKLP (2007), quantitative GRE predicts course grades but not when Committee Score is included; coefficient even goes negative for Metrics Grade
  - **Problem 3:** Committee doesn't know *ex-ante* whether d will be 1 or 0

⏩ Maximum Likelihood Estimation

A2.1: Empirical Considerations

- **Selection Effects:** Track success for all applicants, including non-matriculants
  - In selected samples, success parameters are biased down (e.g. quantitative GRE)
- **Potential Outcomes:** Track which graduate program all applicants attended, and add expected program ranking conditional on (x, z) as control function
  - If a variable only predicts success by helping applicant get into elite graduate program, then committee should not use that variable to rank applicants
- **Unobservables:** I may not observe all information that committee observes
  - **Example:** Recommendation letter has nuances that algorithms can't detect
  - Include subjective committee score in z as a proxy for this information

▸▸ Cutoff Probability Identification

A2.1: Cutoff Rule with Matriculation

Portfolio Choice Problem:

$$\max_{(d_1,\ldots,d_N)\in\{A,R\}^N} \sum_{n=1}^{N} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_n = A),$$

$$\text{such that: } \sum_{n=1}^{N} P(m_n = M|w_n)\mathbb{1}(d_n = A) \leq N_M$$

$$\mathcal{L}(d_1,\ldots,d_N;\lambda) = \sum_{n=1}^{N} \left[P(y_n = S|w_n)\mathbb{1}(d_n = A) + \lambda\mathbb{1}(d_n = R)\right] P(m_n = M|w_n) + \lambda(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N\},\ d_n(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda^* \\ \{A,R\} & \text{if } P(y_n = S|w_n) = \lambda^* \\ R & \text{if } P(y_n = S|w_n) < \lambda^* \end{cases}$$

Portfolio Choice with Matriculation

A2.1: Cutoff Rule with Matriculation (Large N)

Portfolio Choice Problem:

$$\max_{\delta \in \{\delta | \delta : \mathbb{R}^K \to [0,1]\}} \int_{w \in \mathbb{R}^K} P(y = S|w) P(m = M|w) \delta(w) f(w) dw,$$

$$\text{such that: } \int_{w \in \mathbb{R}^K} P(m = M|w) \delta(w) f(w) dw \leq N_M$$

$$\mathcal{L}(\delta; \lambda) = \int_{w \in \mathbb{R}^K} \left[P(y = S|w) \delta(w) + \lambda (1 - \delta(w)) \right] P(m = M|w) f(w) dw + \lambda (N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall w \in \mathbb{R}^K, \; \delta(w|\lambda^*) = \begin{cases} 1 & \text{if } P(y = S|w) > \lambda^* \\ [0,1] & \text{if } P(y = S|w) = \lambda^* \\ 0 & \text{if } P(y = S|w) < \lambda^* \end{cases}$$

Upshot: Point-identified cutoff probability λ^* is with respect to success probabilities

Portfolio Choice with Matriculation

A2.1: Cutoff Rule with Matriculation, Round 2

Portfolio Choice Problem:

$$\max_{(d_{2,1},\ldots,d_{2,N_T})\in\{A,R\}^{N_T}} \sum_{n=1}^{N_T} P(y_n = S|w_n)P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A),$$

$$\text{such that: } \sum_{n=1}^{N_T} P(m_n = M|w_n)\mathbb{1}(d_{2,n} = A) \leq N_M$$

$$\mathcal{L}(d_{2,1},\ldots,d_{2,N_T};\lambda_2) = \sum_{n=1}^{N_T} \left[P(y_n = S|w_n)\mathbb{1}(d_{2,n} = A) + \lambda_2\mathbb{1}(d_{2,n} = R)\right] P(m_n = M|w_n) + \lambda_2(N_M - \overline{N_M})$$

Cutoff Rule Solution:

$$\forall n \in \{1,\ldots,N_T\},\ d_{2,n}(w_n|\lambda^*) = \begin{cases} A & \text{if } P(y_n = S|w_n) > \lambda_2^* \\ \{A, R\} & \text{if } P(y_n = S|w_n) = \lambda_2^* \\ R & \text{if } P(y_n = S|w_n) < \lambda_2^* \end{cases}$$

▸▸ Portfolio Choice with Matriculation

A2.1: Cutoff Rule with Matriculation, Round 1 (Dynamic)

Assumption: Committee defers matriculation considerations to Round 2

$$EV_{2,n}(x_n) = \int_{\widetilde{z}_n} \max\{P(y = S|x_n, \widetilde{z}_n), \lambda_2^*\} dF_{z|x}(\widetilde{z}_n|x_n)$$

Portfolio Choice Problem:

$$\max_{(d_{1,1},\ldots,d_{1,N})\in\{T,R\}^N} \sum_{n=1}^{N} EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T), \text{ such that: } \sum_{n=1}^{N} \mathbb{1}(d_{1,n} = T) \leq N_T$$

$$\mathcal{L}(d_{1,1}, \ldots, d_{1,N}; \lambda_1) = \sum_{n=1}^{N} \left[EV_{2,n}(x_n)\mathbb{1}(d_{1,n} = T) + \lambda_1 \mathbb{1}(d_{1,n} = R)\right] + \lambda_1(N_T - N)$$

Cutoff Rule Solution:

$$\forall n \in \{1, \ldots, N\}, \ d_{1,n}(x_n|\lambda_1^*) = \begin{cases} T & \text{if } EV_{2,n}(x_n) > \lambda_1^* \\ \{T, R\} & \text{if } EV_{2,n}(x_n) = \lambda_1^* \\ R & \text{if } EV_{2,n}(x_n) < \lambda_1^* \end{cases}$$

Portfolio Choice with Matriculation

A2.1: Waitlist Model, Round 1

Round 1:

$$v_1(x,t,\epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x,t,d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x,t,d_1) = \underbrace{\mathbb{E}[v_2(x,z,\ddot{m}_2,\epsilon_2)|x]\mathbb{1}(d_1 = T)}_{\text{If transition, get } \mathbb{E}[\text{Social Surplus}]} + \underbrace{L(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } L(\text{Cutoff})}, \text{ such that:}$$

$$\mathbb{E}[v_2(x,z,\ddot{m}_2,\epsilon_2)|x] = \int_{\widetilde{z}} \int_{\widetilde{\ddot{m}_2}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,W\}} e^{u_2(x,\widetilde{z},\widetilde{\ddot{m}_2},\widetilde{d}_2)/\sigma} \right) dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}_2}|x,\widetilde{z}) dF_{z|x}(\widetilde{z}|x)$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Waitlist Model, Round 2

A2.1: Waitlist Model, Round 1

Round 1 (Hybrid):

$$v_1(x, t, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, t, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, t, d_1) = \underbrace{P(y = S|x)\mathbb{1}(d_1 = T)}_{\text{If transition, get } P(\text{Success})} + \underbrace{P(y_1^* = S|t)\mathbb{1}(d_1 = R)}_{\text{If reject, get } P(\text{Cutoff})}$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Waitlist Model, Round 2

A2.1: Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, \ddot{m}_2; \theta_{A_2}\right) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}{1 + \exp(f_{A_2}(x,z,\ddot{m}_2)\theta_{A_2})}$$

$$P_{d_3} = P\left(d_3 = A \middle| x, z, \overline{m_{2A}}; \theta_{A_3}\right) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}{1 + \exp(f_{A_3}(x,z,\overline{m_{2A}})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P\left(y = S \middle| x, z; \theta_S\right) = \frac{\exp(f_S(x,z)\theta_S)}{1 + \exp(f_S(x,z)\theta_S)}, P_{m_2} = P\left(m_2 = M \middle| x, z, t; \theta_{M_2}\right) = \frac{\exp(f_{M_2}(x,z,t)\theta_{M_2})}{1 + \exp(f_{M_2}(x,z,t)\theta_{M_2})}$$

Cutoff Level/Probability:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\left(y_3^* = S \middle| \overline{m_{2A}}; \theta_{S_3^*}\right) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}})\theta_{S_3^*})}$$

Waitlist Model, Round 2

A2.1: Waitlist-Dynamic Parameterization 2

★ Assume z and $\ddot{m}_2$ are conditionally independent:

$$\mathbb{E}[v_2(x, z, \ddot{m}_2, \epsilon_2)|x] = \int_{\tilde{z}} \int_{\widetilde{\ddot{m}_2}} \sigma \log \left(\sum_{\tilde{d}_2 \in \{A,R\}} e^{u_2(x,\tilde{z},\widetilde{\ddot{m}_2},\tilde{d}_2)/\sigma} \right) dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}_2}|x, \tilde{z}) dF_{z|x}(\tilde{z}|x)$$

$$= \int_{\tilde{z}_1} \cdots \int_{\widetilde{z_{J_z}}} \int_{\widetilde{\ddot{m}_2}} \left[\cdot \right] dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}_2}|x, \tilde{z}) dF_{z|x,1}(\tilde{z}_1|\tilde{z}_2, \ldots, \widetilde{z_{J_z}}, x) \cdots dF_{z|x,J_z}(\widetilde{z_{J_z}}|x)$$

$$\mathbb{E}[v_3(x, z, \overline{m_{2A}}, \epsilon_3)|x, z, \ddot{m}_2] = \int_{\widetilde{\overline{m_{2A}}}} \sigma \log \left(\sum_{\tilde{d}_3 \in \{A,R\}} e^{u_3(x,z,\widetilde{\overline{m_{2A}}},\tilde{d}_3)/\sigma} \right) dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x}(z_j|x; \beta_{F_{z|x,j}}, \sigma^2_{F_{z|x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,j}}}(z_j - g(x)\beta_{F_{z|x,j}})^2\right]$
- **Binary:** $P(z_j = 1|x; \theta_{F_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F_{z|x,j}}\right)}{1+\exp\left(g(x)\theta_{F_{z|x,j}}\right)}$
- **Multinomial:** $P(z_j = (k \neq K)|x; \theta_{F^k_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F^k_{z|x,j}}\right)}{1+\sum_{\ell=1}^{K-1}\exp\left(g(x)\theta_{F^\ell_{z|x,j}}\right)}$

Waitlist Model, Round 2

A2.1: Waitlist-Dynamic Likelihood

$$\mathcal{L}^{U}_{N_1}(\theta) = \prod_{n=1}^{N_W} f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{\ddot{m}_2|x,z}(\ddot{m_{2,n}}|x_n,z_n;\theta_{F_{\ddot{m}_2|x,z}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m_{2,n}};\theta_{F_{\overline{m_{2A}}|\ddot{m}_2}})$$

$$P_{d_{1,n}}(\theta_T)(1-P_{d_{2,n}}(\theta_{A_2}))P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)}P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}$$

$$\prod_{n=N_W+1}^{N_T} f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{\ddot{m}_2|x,z}(\ddot{m_{2,n}}|x_n,z_n;\theta_{F_{\ddot{m}_2|x,z}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m_{2,n}};\theta_{F_{\overline{m_{2A}}|\ddot{m}_2}})$$

$$P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^{N} f_{z|x}(z_n|x_n;\theta_{F_{z|x}})(1-P_{d_{1,n}}(\theta_T))$$

- Estimate θ_{M_2} with 1st-stage likelihood to get $m_{2,n}=$ matriculation probability $P_{m_{2,n}}$: $\mathcal{L}^{M_2}_{N_T-N_W}(\theta_{M_2}) = \prod_{n=N_W+1}^{N_T} P_{m_{2,n}}(\theta_{M_2})^{\mathbb{1}(m_{2,n}=M)}(1-P_{m_{2,n}}(\theta_{M_2}))^{\mathbb{1}(m_{2,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ^U_F,θ^U_S) as initial guess of (θ^R_F,θ^R_S)

Maximum Likelihood Estimation

A2.1: Waitlist Notes

- **Estimation:** Estimate $f_{\overline{m_{2A}}|\ddot{m}_2}$ on all transitioned applicants; slope parameter's expected sign is positive from variation in $\overline{m_2}$ across years (e.g. due to Covid)
    - **Assumption:** Effect of $\ddot{m}_2$ on $\overline{m_{2A}}$ is stationary across years, and therefore can be used to forecast changes in $\overline{m_{2A}}$ based on changes in $\ddot{m}_2$ from changes in d_2
- **Dynamics:** What if committee doesn't think dynamically?
    - If $\ddot{m}_2$ increasing increases $\overline{F}_{\overline{m_{2A}}|\ddot{m}_2}$, which increases $\mathbb{E}[P(\text{Cutoff})_3]$, then it should
    - Don't need success data for all years to identify these two effects
- **Over/Undershooting:** Committee may wish to avoid both
    - To Round 3's $P(\text{Cutoff})$, add target cohort size minus number of matriculants
    - (Negative) parameter may be unidentifiable, so may have to calibrate it

Waitlist Dynamics

A2.1: Symmetric Waitlist Model 1

Round 3:

$$v_3(x,z,\overline{m_{2A}},\overline{m_3},\epsilon_3) = \max_{d_3\in\{A,R\}} u_3(x,z,\overline{m_{2A}},\overline{m_3},d_3) + \epsilon_3(d_3), \text{ such that:}$$

$$u_3(x,z,\overline{m_{2A}},\overline{m_3},d_3) = P(y=S|x,z)\mathbb{1}(d_3=A) + P(y_3^*=S|\overline{m_{2A}},\overline{m_3})\mathbb{1}(d_3=R)$$

Round 2:

$$v_2(x,z,\ddot{m}_2,m_3,\epsilon_2) = \max_{d_2\in\{A,W\}} u_2(x,z,\ddot{m}_2,m_3,d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$u_2(x,z,\ddot{m}_2,m_3,d_2) = P(y=S|x,z)\mathbb{1}(d_2=A)+\beta\mathbb{E}[v_3(x,z,\overline{m_{2A}},\overline{m_3},\epsilon_3)|x,z,\ddot{m}_2,m_3]\mathbb{1}(d_2=W), \text{such that:}$$

$$\mathbb{E}[v_3(x,z,\overline{m_{2A}},\widetilde{\overline{m_3}},\epsilon_3)|x,z,\ddot{m}_2,m_3] =$$
$$\int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma \log\left(\sum_{\widetilde{d_3}\in\{A,R\}} e^{u_3(x,z,\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\widetilde{d_3})/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)\, dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

Waitlist Dynamics

A2.1: Symmetric Waitlist Model 2

Round 1:

$$v_1(x,t) = \max_{d_1 \in \{T,R\}} u_1(x,t,d_1) + \epsilon_1(d_1), \text{ such that:}$$

- **Hybrid:** $u_1(x,t,d_1) = P(y = S|x)\mathbb{1}(d_1 = T) + P(y_1^* = S|t)\mathbb{1}(d_1 = R)$
- **Dynamic:** $u_1(x,t,d_1) = \mathbb{E}[v_2(x,z,\ddot{m}_2,m_3,\epsilon_2)|x]\mathbb{1}(d_1 = T) + L(y_1^* = S|t)\mathbb{1}(d_1 = R)$, such that:

$$\mathbb{E}[v_2(x,z,\ddot{m}_2,m_3,\epsilon_2)|x] =$$

$$\int_{\widetilde{z}} \int_{\widetilde{\ddot{m}_2}} \int_{\widetilde{m_3}} \sigma \log \left(\sum_{\widetilde{d}_2 \in \{A,R\}} e^{u_2(x,\widetilde{z},\widetilde{\ddot{m}_2},\widetilde{m_3},\widetilde{d}_2)/\sigma} \right) dF_{m_3|x,z}(\widetilde{m}_3|x,\widetilde{z}) dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}}_2|x,\widetilde{z}) dF_{z|x}(\widetilde{z}|x)$$

Conditional Choice Probabilities:

$$P(d_r|\text{states}) = \frac{\exp(u_r(\text{states}, d_r)/\sigma)}{\sum_{\widetilde{d}_r \in D_r} \exp(u_r(\text{states}, \widetilde{d}_r)/\sigma)}$$

Waitlist Dynamics

A2.1: Symmetric Waitlist-Dynamic Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t; \theta_T\right) = \frac{\exp(f_T(x,t)\theta_T)}{1 + \exp(f_T(x,t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, \ddot{m}_2, m_3; \theta_{A_2}\right) = \frac{\exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}{1 + \exp(f_{A_2}(x,z,\ddot{m}_2,m_3)\theta_{A_2})}$$

$$P_{d_3} = P\left(d_3 = A \middle| x, z, \overline{m_{2A}}, \overline{m_3}; \theta_{A_3}\right) = \frac{\exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}{1 + \exp(f_{A_3}(x,z,\overline{m_{2A}},\overline{m_3})\theta_{A_3})}$$

Success/Matriculation Probabilities:

$$P_y = P\left(y = S \middle| x, z; \theta_S\right) = \frac{\exp(f_S(x,z)\theta_S)}{1 + \exp(f_S(x,z)\theta_S)}, P_{m_3} = P\left(m_3 = M \middle| x, z, t; \theta_{M_3}\right) = \frac{\exp(f_{M_3}(x,z,t)\theta_{M_3})}{1 + \exp(f_{M_3}(x,z,t)\theta_{M_3})}$$

Cutoff Levels/Probabilities:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = \exp(f_{S_1^*}(t)\theta_{S_1^*})$$

$$P_{y_3^*} = P\left(y_3^* = S \middle| \overline{m_{2A}}, \overline{m_3}; \theta_{S_3^*}\right) = \frac{\exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}{1 + \exp(f_{S_3^*}(\overline{m_{2A}},\overline{m_3})\theta_{S_3^*})}$$

Waitlist Dynamics

A2.1: Symmetric Waitlist-Dynamic Parameterization 2

★ Assume $(z, \ddot{m}_2, m_3)$ and $(\overline{m_{2A}}, \overline{m_3})$ are mutually conditionally independent:

$$\mathbb{E}[v_2(x,z,\ddot{m}_2,m_3,\epsilon_2)|x] = \int_{\tilde{z}}\int_{\widetilde{\ddot{m}_2}}\int_{\widetilde{m_3}} \sigma \log\left(\sum_{\tilde{d}_2\in\{A,R\}} e^{u_2(x,\tilde{z},\widetilde{\ddot{m}_2},\widetilde{m_3},\tilde{d}_2)/\sigma}\right) dF_{m_3|x,z}(\widetilde{m_3}|x,\tilde{z})dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}_2}|x,\tilde{z})dF_{z|x}(\tilde{z}|x)$$

$$= \int_{\tilde{z}_1}\cdots\int_{\widetilde{z_{J_z}}}\int_{\widetilde{\ddot{m}_2}}\int_{\widetilde{m_3}}\left[\cdot\right] dF_{m|x,z}(\widetilde{m_3}|x,\tilde{z})dF_{\ddot{m}_2|x,z}(\widetilde{\ddot{m}_2}|x,\tilde{z})dF_{z|x,1}(\tilde{z}_1|\tilde{z}_2,\ldots,\widetilde{z_{J_z}},x)\cdots dF_{z|x,J_z}(\widetilde{z_{J_z}}|x)$$

$$\mathbb{E}[v_3(x,z,\overline{m_{2A}},\overline{m_3},\epsilon_3)|x,z,\ddot{m}_2,m_3] = \int_{\widetilde{\overline{m_{2A}}}}\int_{\widetilde{\overline{m_3}}} \sigma \log\left(\sum_{\tilde{d}_3\in\{A,R\}} e^{u_3(x,z,\widetilde{\overline{m_{2A}}},\widetilde{\overline{m_3}},\tilde{d}_3)/\sigma}\right) dF_{\overline{m_3}|m_3}(\widetilde{\overline{m_3}}|m_3)dF_{\overline{m_{2A}}|\ddot{m}_2}(\widetilde{\overline{m_{2A}}}|\ddot{m}_2)$$

- **Continuous:** $f_{z|x}(z_j|x;\beta_{F_{z|x,j}},\sigma^2_{F_{z|x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,j}}}}\exp\left[\frac{-1}{2\sigma^2_{F_{z|x,j}}}(z_j - g(x)\beta_{F_{z|x,j}})^2\right]$
- **Binary:** $P(z_j = 1|x;\theta_{F_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F_{z|x,j}}\right)}{1+\exp\left(g(x)\theta_{F_{z|x,j}}\right)}$
- **Multinomial:** $P(z_j = (k \neq K)|x;\theta_{F^k_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F^k_{z|x,j}}\right)}{1+\sum_{\ell=1}^{K-1}\exp\left(g(x)\theta_{F^\ell_{z|x,j}}\right)}$

Waitlist Dynamics

A2.1: Symmetric Waitlist-Dynamic Likelihood

$$\mathcal{L}^{U}_{N_1}(\theta) = \prod_{n=1}^{N_W} f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{\ddot{m}_2|x,z}(\ddot{m_2}_{,n}|x_n,z_n;\theta_{F_{\ddot{m}_2|x,z}}) f_{m_3|x,z}(m_{3,n}|x_n,z_n;\theta_{F_{m_3|x,z}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m_2}_{,n};\theta_{F_{\overline{m_{2A}}|\ddot{m}_2}})$$

$$f_{\overline{m_3}|m_3}(\overline{m_{3,n}}|m_{3,n};\theta_{F_{\overline{m_3}|m_3}}) P_{d_{1,n}}(\theta_T)(1-P_{d_{2,n}}(\theta_{A_2}))P_{d_{3,n}}(\theta_{A_3})^{\mathbb{1}(d_{3,n}=A)}(1-P_{d_{3,n}}(\theta_{A_3}))^{\mathbb{1}(d_{3,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}$$

$$\prod_{n=N_W+1}^{N_T} f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{\ddot{m}_2|x,z}(\ddot{m_2}_{,n}|x_n,z_n;\theta_{F_{\ddot{m}_2|x,z}}) f_{m_3|x,z}(m_{3,n}|x_n,z_n;\theta_{F_{m|x,z}}) f_{\overline{m_{2A}}|\ddot{m}_2}(\overline{m_{2A,n}}|\ddot{m_2}_{,n};\theta_{F_{\overline{m_{2A}}|\ddot{m}_2}})$$

$$f_{\overline{m_3}|m_3}(\overline{m_{3,n}}|m_{3,n};\theta_{F_{\overline{m_3}|m_3}}) P_{d_{1,n}}(\theta_T)P_{d_{2,n}}(\theta_{A_2})P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)} \prod_{n=N_T+1}^{N} f_{z|x}(z_n|x_n;\theta_{F_{z|x}})(1-P_{d_{1,n}}(\theta_T))$$

- Estimate θ_{M_3} with 1st-stage likelihood to get $m_{3,n}=$ matriculation probability $P_{m_{3,n}}$: $\mathcal{L}^{M_3}_{N_W}(\theta_{M_3}) = \prod_{n=1}^{N_W} P_{m_{3,n}}(\theta_{M_3})^{\mathbb{1}(d_{3,n}=A,m_{3,n}=M)}(1-P_{m_{3,n}}(\theta_{M_3}))^{\mathbb{1}(d_{3,n}=A,m_{3,n}=D)}$
- Use Murphy-Topel SEs, and for $\mathcal{L}_R$, use (θ^U_F,θ^U_S) as initial guess of (θ^R_F,θ^R_S)

Maximum Likelihood Estimation

A2.1: Optimality Tests Notes

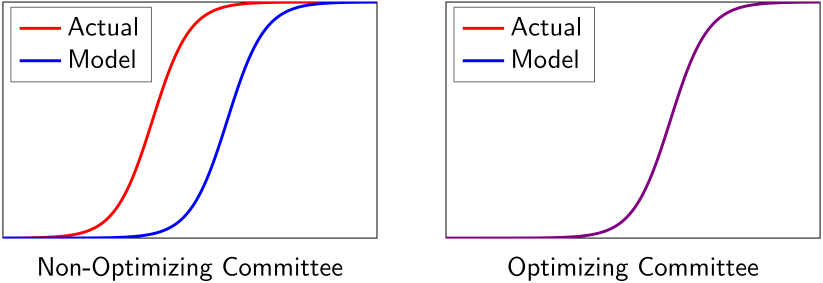
1. **Likelihood-Ratio:** Restricted model not nested in unrestricted model (Vuong 1989)
   - **Program Fit:** Does model fit program's behavior?
     - $d_1 = \mathbb{1}(\text{Transition})$, $d_2 = \mathbb{1}(\text{Accept})$
     - If not, could program be behaving suboptimally (Simon 1956)?
   - **Profession Fit:** Does model fit profession's behavior?
     - $d_2 = \mathbb{1}(\text{AttendProgram})$ but d_1 unobservable, so must estimate one-stage model

2. **Deviation Gains:** Adjust θ_{S^*} so transition and acceptance rates are unchanged
   - **Static:** Rank all applicants by $P(y = S|x)$, set $\theta_{S_1^*}$ to match transition rate, rank survivors by $P(S|x, z)$, and set $\theta_{S_2^*}$ to match acceptance rate
   - **Dynamic:** Use MSM to estimate new $(\theta_{S_1^*}, \theta_{S_2^*})$; moments are each year's transition and acceptance rates. But counterfactual does not require $(\hat{\theta}_{S_1^*}, \hat{\theta}_{S_2^*})$

Optimality Tests

A2.1: How Deviation Gains Test Works

1. Write Bellman equations for committee's optimization problem
2. Estimate parameters. Rationality assumption distorts estimates
3. Re-estimate parameters without rationality, and use to solve model
4. Compare selections from model's decision rule to actual decision rule
5. To prevent overfitting, use perturbations of estimates for comparison



Deviation Gains Interpretation

A2.1: Deviation Gains Algorithm 1

Static/Dynamic/Myopic:

1. Rank applicants in each year by $P(y = S|x)$ for static, or $\mathbb{E}[v_2(x, z, \epsilon_2)|x]$ for dynamic and myopic. Transition top N_T applicants, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z)$, and accept top N_A survivors. Compute $\overline{P(y = S|x, z)}$ of acceptance set, and compare to committee's set

Deviation Gains Interpretation

A2.1: Deviation Gains Algorithm 2

Waitlist-Hybrid/Dynamic:

1. Rank applicants in each year by $P(y = S|x)$ for hybrid, or $\mathbb{E}[v_2(x, z, \ddot{m}_2, \epsilon_2)|x]$ for dynamic. Transition top N_T applicants, and reject the rest
2. Rank surviving applicants by $P(y = S|x, z) - \beta\mathbb{E}[v_3(x, z, \overline{m_{2A}}, \epsilon_3)|x, z, \ddot{m}_2]$. Accept top N_{2A} survivors, and waitlist the rest
3. Rank early acceptances by matriculation probability m_2. All early acceptances with m_2 at least as high as committee's marginal early matriculant matriculate
4. Rank waitlisted applicants by $P(y = S|x, z)$. Increase/decrease N_{3A} to N'_{3A} depending on how many spots remain relative to committee
5. Rank late acceptances by m_3. Top ones matriculate until cohort is filled. Compute $\overline{P(y = S|x, z)}$ of matriculation set, and compare to committee's set

Deviation Gains Interpretation

A2.1: Randomization Tests 1

Theorem 4 (Proof)

Let d^R be the model's decision rule, and d^U the actual decision rule. By optimality:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$\geq \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n; \hat{\theta}_S^U\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}^U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- But for perturbation $\widetilde{\theta_{S,t}^U}$ of $\hat{\theta}_S^U$'s asymptotic distribution, can have:

$$\frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^R\left[P_{y_n}\left(\hat{\theta}_S^U\right)\right] = A\right\}$$
$$< \frac{1}{N_A}\sum_{n=1}^{N} P\left(y_n = S \middle| x_n, z_n; \widetilde{\theta_{S,t}^U}\right) \mathbb{1}\left\{d_{1,n}\left[P_{y_n}\left(\hat{\theta}_U\right)\right] = T, d_{2,n}^U\left[P_{y_n}\left(\hat{\theta}_A^U\right)\right] = A\right\}$$

- For waitlist models, compare matriculants, not acceptances

Deviation Gains Interpretation

A2.1: Randomization Tests 2

- **Question:** Are deviation gains robust to estimation error?
- **Test 1:** Across T perturbations, plot CDFs of $\overline{P(\text{Success})}$ of both acceptance sets. Does the model's CDF stochastically dominate the actual decision rule's CDF?
- **Test 2:** Two-sample Kolmogorov-Smirnov test of equality of CDFs:
  - Let $X = \overline{P_y\left[d\left(\hat{\theta}^U\right), \widetilde{\theta_S^U}\right]}$, and $F_T(x)$ be X's CDF over T perturbations
  - Then $D_T = \sup_x \left|F_T^R(x) - F_T^U(x)\right|$. Reject equality if $D_T > \sqrt{\frac{-\log(\alpha/2)}{T}}$
- **Open Question:** Would deviation gains be borne out in practice?

Deviation Gains Interpretation

A2.1: Summary of Results

- **Results:** Estimation results are well behaved
  - **Static:** Likelihood-ratio test rejects and deviation gains statistically significant only for simulated dataset where committee misvalues applicant characteristics
  - **Dynamic:** Likelihood-ratio test always rejects, but gains exist only for non-optimizing dataset. Gains are slightly larger than in myopic model
  - **Waitlist-Hybrid/Dynamic:** Matriculation estimates are as expected
- Example of cutoff probability reversal if committee transitions few files in Round 1:
  - Three applicants: A, B, and C. Committee transitions two, accepts one
  - A is best and C is worst overall, but A is worst in objective variables
  - Committee rejects A in Round 1 and accepts B, who is worse than A
  - By assuming h hours to read application file and w hourly wage, and calculating model's $\overline{P(\text{Success})}$ induced by all possible N_T, can put dollar value on success

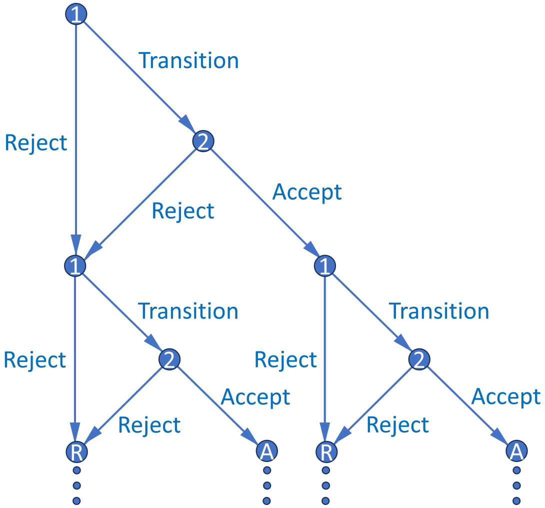
Conclusion

A2.1: Extensions

- **Extension 1:** Create matriculation probability uncorrelated with $P(\text{Success})$
  - For accepted applicants, use matriculation predictors in essays/rec letters
  - Can help committee find "people that we can afford" (*Moneyball*)
- **Extension 2:** Relax assumption that committee maximizes $P(\text{Success})$
  - Instead maximize convex combination of $P(\text{Success})$ and other objectives
  - **Example:** Committee may want diverse cohort (e.g. across fields): $\alpha P(\text{Success}) + (1-\alpha)\text{Diversity}$, where $\alpha \in [0,1]$ can be estimated
- **Extension 3:** Multiyear models (see Appendix 2.2)
  - Two-round models repeat over infinite horizon with discounting
  - States contain info about past years/rounds, and committee values future years/rounds (memory includes matriculation probability and diversity)

Conclusion

A2.2: Multiyear Decision Graph



Conclusion

A2.2: Multiyear Memory (Matriculation)

- Suppose committee maximizes discounted "lifetime" cohort success
- Respecify $P(y_2^* = S|t)$, where $t =$ year, as $L(y_2^* = S|\overline{q}, M, c_{-1})$
  - $\overline{q} =$ current year's average surviving applicant quality (i.e. average $P(\text{Success})$)
  - $M =$ current year's number of surviving applicants
  - $c_{-1} =$ previous year's cohort size
  - $L(\text{Cutoff}_2)$ increases in all three. In particular, in increases in c_{-1} because committee must keep number of total students in program relatively constant
- What information available at start of Round 2 determines c?
  - **Always:** $\overline{m} =$ average surviving applicant matriculation probability
  - **If and only if $\boldsymbol{d_2 = A}$:** $m =$ applicant's matriculation probability

Conclusion

A2.2: Multiyear Dynamics (Matriculation)

Theorem 5 (Proof)

Success probability equal, an applicant's Round 2 acceptance probability is strictly increasing in their matriculation probability.

- With memory, optimal dynamic decision may not equal optimal myopic one
  - High m means high $\overline{F}_{c|m}$, which means high $\mathbb{E}[L(\text{Cutoff}_2)']$ and future EV
  - Accepting likely matriculants above this year's margin means that next year in expectation, committee won't have to accept as many marginal applicants
- Harder to estimate since multiyear problem has no closed-form solution
  - Reduced-form results and factor analysis can help reduce dimensionality

Conclusion

A2.2: Multiyear Model (Matriculation), Round 2

$$v_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, d_2) = &\{P(y = S|x, z) + \beta \mathbb{E}[v_1(x', t', c, {\epsilon_1}')|m, \overline{m}]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|\overline{q}, M, c_{-1}) + \beta \mathbb{E}[v_1(x', t', c, {\epsilon_1}')|\overline{m}]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x', t', c, {\epsilon_1}')|(m), \overline{m}] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{{d_1}' \in \{T,R\}} e^{u_1(x', t', c, {d_1}')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m),\overline{m}}(c|(m), \overline{m})$$

Conditional Choice Probabilities:

$$P(d_2|x, z, m, \overline{m}, \overline{q}, M, c_{-1}) = \frac{\exp(u_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, d_2)/\sigma)}{\sum_{\widetilde{d_2} \in \{A,R\}} \exp(u_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, \widetilde{d_2})/\sigma)}$$

Conclusion

A2.2: Multiyear Model (Matriculation), Round 1

$$v_1(x,t,c_{-1},\epsilon_1) = \max_{d_1\in\{T,R\}} u_1(x,t,c_{-1},d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$\begin{aligned} u_1(x,t,c_{-1},d_1) = \mathbb{E}[v_2(x,z,m,\overline{m},\overline{q},M,c_{-1},\epsilon_2)|x,c_{-1}]\mathbb{1}(d_1 = T) \\ + \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x',t',c,{\epsilon_1}')|\overline{m}]\}\mathbb{1}(d_1 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_2(x,z,m,\overline{m},\overline{q},M,c_{-1},\epsilon_2)|x,c_{-1}] =$$

$$\int_{\widetilde{M}}\int_{\widetilde{\overline{q}}}\int_{\widetilde{\overline{m}}}\int_{\widetilde{m}}\int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},\widetilde{m},\widetilde{\overline{m}},\widetilde{\overline{q}},\widetilde{M},c_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x}(\widetilde{z}|x)dF_{m|x}(\widetilde{m}|x)dF_{\overline{m}}(\widetilde{\overline{m}})dF_{\overline{q}}(\widetilde{\overline{q}})dF_M(\widetilde{M})$$

Conditional Choice Probabilities:

$$P(d_1|x,t,c_{-1}) = \frac{\exp(u_1(x,t,c_{-1},d_1)/\sigma)}{\sum_{\widetilde{d}_1\in\{T,R\}}\exp(u_1(x,t,c_{-1},\widetilde{d}_1)/\sigma)}$$

Conclusion

A2.2: Multiyear (Matriculation) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, t, c_{-1}; \theta_T\right) = \frac{\exp(f_T(x, t, c_{-1})\theta_T)}{1 + \exp(f_T(x, t, c_{-1})\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, m, \overline{m}, \overline{q}, M, c_{-1}; \theta_A\right) = \frac{\exp(f_A(x, z, m, \overline{m}, \overline{q}, M, c_{-1})\theta_A)}{1 + \exp(f_A(x, z, m, \overline{m}, \overline{q}, M, c_{-1})\theta_A)}$$

Success Probability:

$$P_y = P\left(y = S \middle| x, z; \theta_S\right) = \frac{\exp(f_S(x, z)\theta_S)}{1 + \exp(f_S(x, z)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = f_{S_1^*}(t)\theta_{S_1^*}$$

$$L_{y_2^*} = L\left(y_2^* = S \middle| \overline{q}, M, c_{-1}; \theta_{S_2^*}\right) = f_{S_2^*}(\overline{q}, M, c_{-1})\theta_{S_2^*}$$

Conclusion

A2.2: Multiyear (Matriculation) Parameterization 2.1

★ Assume (x, t) mutually independent, and next year independent of current year:

$$\mathbb{E}[v_1(x', t', c, \epsilon_1')|(m), \overline{m}] = \int_c \int_{t'} \int_{x'} \sigma \log \left(\sum_{d_1' \in \{T,R\}} e^{u_1(x',t',c,d_1')/\sigma} \right) dF_x(x') dF_t(t') dF_{c|(m),\overline{m}}(c|(m), \overline{m})$$

$$= \int_c \int_{t_1'} \cdots \int_{t_{J_t}'} \int_{x_1'} \cdots \int_{x_{J_x}'} \left[\cdot \right] dF_{x,1}(x_1'|x_2', \ldots, x_{J_x}') \cdots dF_{x,J_x}(x_{J_x}') dF_{t,1}(t_1'|t_2', \ldots, t_{J_t}') \cdots dF_{t,J_t}(t_{J_t}') dF_{c|(m),\overline{m}}(c|(m), \overline{m})$$

- **Continuous:** $f_x(x_j; \beta_{F_{x,j}}, \sigma^2_{F_{x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{x,j}}} (x_j - \beta_{F_{x,j}})^2 \right]$
- **Binary:** $P(x_j = 1; \theta_{F_{x,j}}) = \frac{\exp\left(\theta_{F_{x,j}}\right)}{1 + \exp\left(\theta_{F_{x,j}}\right)}$
- **Multinomial:** $P(x_j = (k \neq K); \theta_{F^k_{x,j}}) = \frac{\exp\left(\theta_{F^k_{x,j}}\right)}{1 + \sum_{\ell=1}^{K-1} \exp\left(\theta_{F^\ell_{x,j}}\right)}$

▸▸ Conclusion

A2.2: Multiyear (Matriculation) Parameterization 2.2

★ Assume $(z, m, \overline{m}, \overline{q}, M)$ mutually independent such that $(z, m)|x$:

$$\mathbb{E}[v_2(x, z, m, \overline{m}, \overline{q}, M, c_{-1}, \epsilon_2)|x, c_{-1}] =$$

$$\int_{\widetilde{M}}\int_{\widetilde{\overline{q}}}\int_{\widetilde{\overline{m}}}\int_{\widetilde{m}}\int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2\in\{A,R\}} e^{u_2(x,\widetilde{z},\widetilde{m},\widetilde{\overline{m}},\widetilde{\overline{q}},\widetilde{M},c_{-1},\widetilde{d}_2)/\sigma}\right) dF_{z|x}(\widetilde{z}|x)dF_{m|x}(\widetilde{m}|x)dF_{\overline{m}}(\widetilde{\overline{m}})dF_{\overline{q}}(\widetilde{\overline{q}})dF_M(\widetilde{M})$$

$$= \int_{\widetilde{M}}\int_{\widetilde{\overline{q}}}\int_{\widetilde{\overline{m}}}\int_{\widetilde{m}}\int_{\widetilde{z}_1}\cdots\int_{\widetilde{z_{J_z}}}\Big[\cdot\Big] dF_{z|x,1}(\widetilde{z}_1|\widetilde{z}_2,\ldots,\widetilde{z_{J_z}},x)\cdots dF_{z|x,J_z}(\widetilde{z_{J_z}}|x)dF_{m|x}(\widetilde{m}|x)dF_{\overline{m}}(\widetilde{\overline{m}})dF_{\overline{q}}(\widetilde{\overline{q}})dF_M(\widetilde{M})$$

- **Continuous:** $f_{z|x}(z_j|x; \beta_{F_{z|x,j}}, \sigma^2_{F_{z|x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,j}}}(z_j - g(x)\beta_{F_{z|x,j}})^2\right]$
- **Binary:** $P(z_j = 1|x; \theta_{F_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F_{z|x,j}}\right)}{1+\exp\left(g(x)\theta_{F_{z|x,j}}\right)}$
- **Multinomial:** $P(z_j = (k \neq K)|x; \theta_{F^k_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F^k_{z|x,j}}\right)}{1+\sum_{\ell=1}^{K-1}\exp\left(g(x)\theta_{F^\ell_{z|x,j}}\right)}$

⏩ Conclusion

A2.2: Multiyear (Matriculation) Likelihood

$$\mathcal{L}^U_N(\theta) = \prod_{n=1}^{N_T} f_x(x_n;\theta_{F_x}) f_t(t_n;\theta_{F_t}) f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{m|x}(m_n|x_n;\theta_{F_{m|x}}) f_{\overline{m}}(\overline{m}_n;\theta_{F_{\overline{m}}}) f_{\overline{q}}(\overline{q}_n;\theta_{F_{\overline{q}}}) f_M(M_n;\theta_{F_M})$$

$$f_{c|m,\overline{m}}(c_n|m_n,\overline{m}_n;\theta_{F_{c|m,\overline{m}}}) P_{d_{1,n}}(\theta_T) P_{d_{2,n}}(\theta_A)^{\mathbb{1}(d_{2,n}=A)} (1-P_{d_{2,n}}(\theta_A))^{\mathbb{1}(d_{2,n}=R)} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)} (1-P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}$$

$$\prod_{n=N_T+1}^{N} f_x(x_n;\theta_{F_x}) f_t(t_n;\theta_{F_t}) f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_{m|x}(m_n|x_n;\theta_{F_{m|x}}) (1 - P_{d_{1,n}}(\theta_T))$$

- For $\mathcal{L}^R$ and $\mathcal{L}^U$, estimate θ^U_S with 1st-stage likelihood to get $\overline{q}$ = current year's $\overline{P_{y_n}}$: $\mathcal{L}^S_{N_T}(\theta_S) = \prod_{n=1}^{N_T} P_{y_n}(\theta_S)^{\mathbb{1}(y_n=S)}(1 - P_{y_n}(\theta_S))^{\mathbb{1}(y_n=F)}$
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ^U_F, θ^U_S) as initial guess of (θ^R_F, θ^R_S)

Conclusion

A2.2: Multiyear Memory and Dynamics (Diversity)

- Let additional state $a =$ any diversity variable affecting success
  - For example, $a = \mathbb{1}(\text{TheoryField})$; theory is less common than applied
  - Let $\bar{a}_{-1} =$ percentage of last year's acceptance set that's theory
  - If $|a - \bar{a}_{-1}|$ is small, $P(\text{Success})$ may decrease because future advisor may be too busy (effect may be unidentifiable if program never gets too imbalanced)

Theorem 6 (Proof)

Consider any two applicants, one with $a = 1$ and one with $a = 0$, but with equal success probabilities and therefore different values of x or z. Suppose $P(a' = 1) = p > 0$. Then for all sufficiently small p, the $a = 1$ applicant has a strictly greater Round 2 acceptance probability.

- Optimal dynamic decision may not equal optimal myopic one. May accept diverse applicant with marginally lower $P(\text{Success})$ to increase $P(\text{Success})$ of next year's pool
  - Relative to $a = 0$, $a = 1 \Rightarrow$ higher $\overline{F}_{\bar{a}|a} \Rightarrow$ higher $\mathbb{E}[P(\text{Success})']$ and future EV

Conclusion

A2.2: Multiyear Model (Diversity), Round 2

$$v_2(x, z, a, \bar{a}_{-1}, t, \epsilon_2) = \max_{d_2 \in \{A,R\}} u_2(x, z, a, \bar{a}_{-1}, t, d_2) + \epsilon_2(d_2), \text{ such that:}$$

$$\begin{aligned} u_2(x, z, a, \bar{a}_{-1}, t, d_2) = &\{P(y = S|x, z, |a - \bar{a}_{-1}|) + \beta\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')|a]\}\mathbb{1}(d_2 = A) \\ &+ \{L(y_2^* = S|t) + \beta\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')]\}\mathbb{1}(d_2 = R), \text{ such that:} \end{aligned}$$

$$\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')|a] = \int_{\widetilde{t'}} \int_{\widetilde{\bar{a}}} \int_{\widetilde{x'}} \sigma \log \left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{\bar{a}}, \widetilde{t'}, d_1')/\sigma} \right) dF_{x'}(\widetilde{x'}) dF_{\bar{a}|a}(\widetilde{\bar{a}}|a) dF_{t'}(\widetilde{t'})$$

Conditional Choice Probabilities:

$$P(d_2|x, z, a, \bar{a}_{-1}, t) = \frac{\exp(u_2(x, z, a, \bar{a}_{-1}, t, d_2)/\sigma)}{\sum_{\widetilde{d}_2 \in \{A,R\}} \exp(u_2(x, z, a, \bar{a}_{-1}, t, \widetilde{d}_2)/\sigma)}$$

Conclusion

A2.2: Multiyear Model (Diversity), Round 1

$$v_1(x, \bar{a}_{-1}, t, \epsilon_1) = \max_{d_1 \in \{T,R\}} u_1(x, \bar{a}_{-1}, t, d_1) + \epsilon_1(d_1), \text{ such that:}$$

$$u_1(x, \bar{a}_{-1}, t, d_1) = \mathbb{E}[v_2(x, z, a, \bar{a}_{-1}, t, \epsilon_2)|x, \bar{a}_{-1}, t]\mathbb{1}(d_1 = T)$$
$$+ \{L(y_1^* = S|t) + \beta\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')]\}\mathbb{1}(d_1 = R), \text{ such that:}$$

$$\mathbb{E}[v_2(x, z, a, \bar{a}_{-1}, t, \epsilon_2)|x, \bar{a}_{-1}, t] = \int_{\widetilde{a}}\int_{\widetilde{z}} \sigma \log\left(\sum_{\widetilde{d}_2 \in A,R} e^{u_2(x,\widetilde{z},\widetilde{a},\bar{a}_{-1},t,\widetilde{d}_2)/\sigma}\right) dF_{z|x}(\widetilde{z}|x)dF_a(\widetilde{a})$$

$$\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')] = \int_{\widetilde{t'}}\int_{\widetilde{\bar{a}}}\int_{\widetilde{x'}} \sigma \log\left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'},\widetilde{\bar{a}},\widetilde{t'},d_1')/\sigma}\right) dF_{x'}(\widetilde{x'})dF_{\bar{a}}(\widetilde{\bar{a}})dF_{t'}(\widetilde{t'})$$

Conditional Choice Probabilities:

$$P(d_1|x, \bar{a}_{-1}, t) = \frac{\exp(u_1(x, \bar{a}_{-1}, t, d_1)/\sigma)}{\sum_{\widetilde{d}_1 \in \{T,R\}} \exp(u_1(x, \bar{a}_{-1}, t, \widetilde{d}_1)/\sigma)}$$

Conclusion

A2.2: Multiyears (Diversity) Parameterization 1

Unrestricted CCPs:

$$P_{d_1} = P\left(d_1 = T \middle| x, \bar{a}_{-1}, t; \theta_T\right) = \frac{\exp(f_T(x, \bar{a}_{-1}, t)\theta_T)}{1 + \exp(f_T(x, \bar{a}_{-1}, t)\theta_T)}$$

$$P_{d_2} = P\left(d_2 = A \middle| x, z, a, \bar{a}_{-1}, t; \theta_A\right) = \frac{\exp(f_A(x, z, a, \bar{a}_{-1}, t)\theta_A)}{1 + \exp(f_A(x, z, a, \bar{a}_{-1}, t)\theta_A)}$$

Success Probability:

$$P_y = P\left(y = S \middle| x, z, |a - \bar{a}_{-1}|; \theta_S\right) = \frac{\exp(f_S(x, z, |a - \bar{a}_{-1}|)\theta_S)}{1 + \exp(f_S(x, z, |a - \bar{a}_{-1}|)\theta_S)}$$

Cutoff Levels:

$$L_{y_1^*} = L\left(y_1^* = S \middle| t; \theta_{S_1^*}\right) = f_{S_1^*}(t)\theta_{S_1^*}$$
$$L_{y_2^*} = L\left(y_2^* = S \middle| t; \theta_{S_2^*}\right) = f_{S_2^*}(t)\theta_{S_2^*}$$

Conclusion

A2.2: Multiyear (Diversity) Parameterization 2.1

★ Assume $(x, \bar{a}, t)$ mutually independent such that $\bar{a}|a$, and t' independent of t:

$$\mathbb{E}[v_1(x', \bar{a}, t', \epsilon_1')|a] = \int_{\widetilde{t'}} \int_{\widetilde{\bar{a}}} \int_{\widetilde{x'}} \sigma \log \left(\sum_{d_1' \in \{T,R\}} e^{u_1(\widetilde{x'}, \widetilde{\bar{a}}, \widetilde{t'}, d_1')/\sigma} \right) dF_{x'}(\widetilde{x'}) dF_{\bar{a}|a}(\widetilde{\bar{a}}|a) dF_{t'}(\widetilde{t'}) =$$

$$\int_{\widetilde{t_1'}} \cdots \int_{\widetilde{t_{J_t}'}} \int_{\widetilde{\bar{a}}} \int_{\widetilde{x_1'}} \cdots \int_{\widetilde{x_{J_x}'}} \left[\cdot \right] F_{x,1}({x_1}'|{x_2}', \ldots, {x_{J_x}}') \cdots dF_{x,J_x}({x_{J_x}}') dF_{\bar{a}|a}(\widetilde{\bar{a}}|a) dF_{t,1}({t_1}'|{t_2}', \ldots, {t_{J_t}}') \cdots dF_{t,J_t}({t_{J_t}}')$$

- **Continuous:** $f_x(x_j; \beta_{F_{x,j}}, \sigma^2_{F_{x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{x,j}}}(x_j - \beta_{F_{x,j}})^2\right]$
- **Binary:** $P(x_j = 1; \theta_{F_{x,j}}) = \frac{\exp\left(\theta_{F_{x,j}}\right)}{1+\exp\left(\theta_{F_{x,j}}\right)}$
- **Multinomial:** $P(x_j = (k \neq K); \theta_{F^k_{x,j}}) = \frac{\exp\left(\theta_{F^k_{x,j}}\right)}{1+\sum_{\ell=1}^{K-1} \exp\left(\theta_{F^\ell_{x,j}}\right)}$

Conclusion

A2.2: Multiyear (Diversity) Parameterization 2.2

★ Assume (z, a) independent, such that $z|x$:

$$\mathbb{E}[v_2(x, z, a, \bar{a}_{-1}, t, \epsilon_2)|x, \bar{a}_{-1}, t] = \int_{\tilde{a}} \int_{\tilde{z}} \sigma \log \left(\sum_{\tilde{d}_2 \in A,R} e^{u_2(x,\tilde{z},\tilde{a},\bar{a}_{-1},t,\tilde{d}_2)/\sigma} \right) dF_{z|x}(\tilde{z}|x) dF_a(\tilde{a})$$

$$= \int_{\tilde{a}} \int_{\tilde{z}_1} \cdots \int_{\tilde{z}_J} \left[\cdot \right] dF_1(\tilde{z}_1|\tilde{z}_2, \ldots \tilde{z}_J, x) \cdots dF_J(\tilde{z}_J|x) dF_a(\tilde{a})$$

- **Continuous:** $f_{z|x}(z_j|x; \beta_{F_{z|x,j}}, \sigma^2_{F_{z|x,j}}) = \frac{1}{\sqrt{2\pi\sigma^2_{F_{z|x,j}}}} \exp\left[\frac{-1}{2\sigma^2_{F_{z|x,j}}}(z_j - g(x)\beta_{F_{z|x,j}})^2\right]$
- **Binary:** $P(z_j = 1|x; \theta_{F_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F_{z|x,j}}\right)}{1+\exp\left(g(x)\theta_{F_{z|x,j}}\right)}$
- **Multinomial:** $P(z_j = (k \neq K)|x; \theta_{F^k_{z|x,j}}) = \frac{\exp\left(g(x)\theta_{F^k_{z|x,j}}\right)}{1+\sum_{\ell=1}^{K-1} \exp\left(g(x)\theta_{F^\ell_{z|x,j}}\right)}$

⏩ Conclusion

A2.2: Multiyear (Diversity) Likelihood

$$\mathcal{L}_N^U(\theta) = \prod_{n=1}^{N_A} f_x(x_n;\theta_{F_x}) f_t(t_n;\theta_{F_t}) f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_a(a) f_{\bar{a}}(\bar{a}) f_{\bar{a}|a}(\bar{a}|a) P_{d_{2,n}} P_{y_n}^{\mathbb{1}(y_n=S)} (1-P_{y_n})^{\mathbb{1}(y_n=F)}$$

$$\prod_{n=N_A+1}^{N_{T,R}} f_x(x_n;\theta_{F_x}) f_t(t_n;\theta_{F_t}) f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_a(a) f_{\bar{a}}(\bar{a}) (1-P_{d_{2,n}}) P_{y_n}^{\mathbb{1}(y_n=S)} (1-P_{y_n})^{\mathbb{1}(y_n=F)}$$

$$\prod_{n=N_{T,R}+1}^{N} f_x(x_n;\theta_{F_x}) f_t(t_n;\theta_{F_t}) f_{z|x}(z_n|x_n;\theta_{F_{z|x}}) f_a(a) f_{\bar{a}}(\bar{a}) (1-P_{d_{1,n}})$$

- Estimate $f_{\bar{a}|a}(\bar{a}|a)$ on N_A accepted applicants only
- As with waitlist-dynamic model, for $\mathcal{L}_R$, use (θ_F^U, θ_S^U) as initial guess of (θ_F^R, θ_S^R)

Conclusion

A3: Full Post-Lasso Logistic Regression Results

Table 2a: Post-Lasso Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3960**	-0.4191	-0.5330	-0.5737	
	(0.1996)	(0.2609)	(0.3435)	(0.3824)	
Winsorized Age	-0.0977***	0.0473*	0.0440		
	(0.0333)	(0.0248)	(0.0325)		
Female	0.4106**	0.6570***	0.6797**		-0.3204
	(0.1670)	(0.2281)	(0.2932)		(0.4908)
U.S. African/Hispanic/Native	0.4355	0.9687***	1.1594***		
	(0.5232)	(0.3521)	(0.4187)		
U.S. Asian	-0.5139	-0.5533		0.6970	0.7934
	(0.4149)	(0.6435)		(0.6906)	(0.6164)
International Asian	-0.7960***	-0.0979		0.3796	
	(0.2665)	(0.3808)		(0.4458)	
International Other	0.5490*	0.3749	1.1102**	-0.2036	
	(0.2838)	(0.3238)	(0.4378)	(0.4436)	
GRE Verbal Percentile	0.0155***	0.0335*	0.0423**		0.0498***
	(0.0059)	(0.0180)	(0.0188)		(0.0185)
GRE Quantitative Percentile	0.1549***	0.0156*	0.0255**	0.0157	
	(0.0184)	(0.0091)	(0.0106)	(0.0120)	
GRE Analytic Percentile	0.0177***	0.0232**		0.0159	
	(0.0041)	(0.0101)		(0.0121)	
Undergraduate GPA	2.2336***	0.7028	1.3004**	0.9096	
	(0.6626)	(0.5465)	(0.5090)	(0.9856)	
TOEFL/IELTS	0.3900	0.6701	1.0010*	0.8625*	
	(0.2878)	(0.4545)	(0.5503)	(0.5186)	
Elite U.S. University	0.8292***	0.3801	0.5964*		0.7688
	(0.2312)	(0.2728)	(0.3491)		(0.4721)
Elite U.S. Liberal Arts College	1.0523***	0.6501*	1.0178**		1.3415
	(0.3373)	(0.3553)	(0.5124)		(0.8540)
Elite Foreign University	0.4772*		0.5710	0.9748	
	(0.2702)		(0.5770)	(0.5963)	
Graduate Degree			0.2774		
			(0.3193)		
Work Experience	0.3826**	0.4139	0.5055		1.6031**
	(0.1881)	(0.2987)	(0.3290)		(0.6992)
#Professor Recommenders			0.5752**	0.4712*	
			(0.2597)	(0.2697)	
#Prolific Recommenders	0.2727***	0.3028**	0.2113	0.3359**	0.4341
	(0.0840)	(0.1229)	(0.1577)	(0.1456)	(0.2885)
#Math Courses	0.1715***				
	(0.0593)				
Advanced Math	0.2350				
	(0.1777)				
Missing GRE	0.1827			-0.0909	-0.8985
	(0.7660)			(0.3941)	(0.5569)
Missing TOEFL/IELTS	-0.3983*	0.5313	0.5689		
	(0.2266)	(0.4700)	(0.5289)		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2671	0.1500	0.1847	0.1722	0.1844
CV Pseudo-R^2	0.2168	0.0865	0.0715	0.0104	0.0445

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .05$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses

- **Economics Negative ($p < .05$):** Covid Year, Winsorized Age, International Asian

Post-Lasso Logistic Regression AMEs

A3: Full Post-Lasso Logistic Regression AMEs

Table 2b: Post-Lasso Logistic Regression AMEs

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.0303**	-0.0159	-0.0404	-0.0315	
	(0.0152)	(0.0100)	(0.0261)	(0.0212)	
Winsorized Age	-0.0075***	0.0018*	0.0033		
	(0.0025)	(0.0010)	(0.0025)		
Female	0.0314**	0.0249***	0.0515**		-0.0144
	(0.0128)	(0.0087)	(0.0220)		(0.0222)
U.S. African/Hispanic/Native	0.0333	0.0367***	0.0878***		
	(0.0400)	(0.0136)	(0.0318)		
U.S. Asian	-0.0394	-0.0210		0.0383	0.0357
	(0.0318)	(0.0245)		(0.0382)	(0.0270)
International Asian	-0.0610***	-0.0037		0.0208	
	(0.0203)	(0.0144)		(0.0245)	
International Other	0.0420*	0.0142	0.0841**	-0.0112	
	(0.0217)	(0.0123)	(0.0333)	(0.0245)	
GRE Verbal Percentile	0.0012***	0.0013*	0.0032**		0.0022**
	(0.0005)	(0.0007)	(0.0015)		(0.0009)
GRE Quantitative Percentile	0.0119***	0.0006*	0.0019**	0.0009	
	(0.0014)	(0.0004)	(0.0008)	(0.0007)	
GRE Analytic Percentile	0.0014***	0.0009**		0.0009	
	(0.0003)	(0.0004)		(0.0007)	
Undergraduate GPA	0.1710***	0.0267	0.0985**	0.0499	
	(0.0506)	(0.0208)	(0.0397)	(0.0542)	
TOEFL/IELTS	0.0299	0.0254	0.0758*	0.0474	
	(0.0219)	(0.0173)	(0.0420)	(0.0290)	
Elite U.S. University	0.0635***	0.0144	0.0452*		0.0346
	(0.0176)	(0.0104)	(0.0262)		(0.0214)
Elite U.S. Liberal Arts College	0.0806***	0.0247*	0.0771**		0.0604
	(0.0255)	(0.0136)	(0.0387)		(0.0398)
Elite Foreign University	0.0365*		0.0432	0.0535	
	(0.0207)		(0.0434)	(0.0331)	
Graduate Degree			0.0210		
			(0.0243)		
Work Experience	0.0293**	0.0157	0.0383		0.0722**
	(0.0143)	(0.0113)	(0.0250)		(0.0335)
#Professor Recommenders			0.0436**	0.0259*	
			(0.0195)	(0.0147)	
#Prolific Recommenders	0.0209***	0.0115**	0.0160	0.0184**	0.0195
	(0.0064)	(0.0047)	(0.0119)	(0.0080)	(0.0127)
#Math Courses	0.0131***				
	(0.0045)				
Advanced Math	0.0180				
	(0.0136)				
Missing GRE	0.0140			-0.0050	-0.0404
	(0.0587)			(0.0216)	(0.0253)
Missing TOEFL/IELTS	-0.0305*	0.0202	0.0431		
	(0.0175)	(0.0179)	(0.0399)		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .05$):**
  Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergradute GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .05$):**
  Covid Year, Winsorized Age, International Asian

Post-Lasso Logistic Regression AMEs

A3: Means by Program

Table 3a: Means by Program

	Economics	Government	History	Linguistics	Psychology
Accept	0.1121	0.0430	0.0986	0.0678	0.0535
Covid Year	0.2478	0.2792	0.3029	0.3460	0.3004
Winsorized Age	26.2782	27.9633	26.8478	28.0692	25.8230
Female	0.4201	0.4182	0.4232	0.6133	0.7490
U.S. African/Hispanic/Native	0.0289	0.0766	0.1159	0.0665	0.1728
U.S. Asian	0.0313	0.0421	0.0304	0.0353	0.0576
International Asian	0.5322	0.2510	0.1058	0.2714	0.1523
International Other	0.2517	0.2967	0.2203	0.3256	0.1584
GRE Verbal Percentile	71.8677	81.3486	82.6054	73.5451	75.6192
GRE Quantitative Percentile	86.2163	63.3428	49.3295	57.8455	58.0000
GRE Analytic Percentile	52.6351	72.1905	75.0460	63.6824	69.2077
Undergraduate GPA	3.6534	3.5675	3.6531	3.6567	3.5695
TOEFL/IELTS	7.5232	7.5422	7.4252	7.5444	7.3158
Elite U.S. University	0.1516	0.1847	0.2565	0.1954	0.2840
Elite U.S. Liberal Arts College	0.0423	0.0672	0.0696	0.0231	0.0535
Elite Foreign University	0.0717	0.0578	0.0478	0.0488	0.0206
Other Foreign University	0.5736	0.3953	0.2333	0.5102	0.1852
Graduate Degree	0.7478	0.7553	0.6652	0.7544	0.4259
Work Experience	0.5938	0.7463	0.6638	0.7544	0.7593
#Professor Recommenders	2.5173	2.3469	2.6261	2.4355	2.1831
#Prolific Recommenders	1.0760	0.6912	0.5043	0.7069	0.3128
#Math Courses	6.6396				
Advanced Math	0.4827				
Missing GRE	0.0255	0.4612	0.6217	0.6839	0.4650
Missing Undergraduate GPA	0.6607	0.4827	0.3072	0.5726	0.2490
Missing TOEFL/IELTS	0.5958	0.7714	0.8449	0.7707	0.8827
Observations	2078	2231	690	737	486

- **Economics Highest:** Accept, International Asian, GRE Quantitative Percentile, Elite Foreign University, Other Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, U.S. African/Hispanic/Native, GRE Verbal Percentile, GRE Analytic Percentile, Elite U.S. University, Work Experience, Missing GRE, Missing TOEFL/IELTS

Reduced-Form Summary

A3: Means by Program, Accept = 1

Table 3b: Means by Program, Accept = 1

	Economics	Government	History	Linguistics	Psychology
Covid Year	0.1888	0.2188	0.2059	0.2200	0.2308
Winsorized Age	25.0815	27.7292	27.2794	27.1000	25.7692
Female	0.4850	0.5000	0.5588	0.5800	0.6154
U.S. African/Hispanic/Native	0.0386	0.1458	0.1765	0.0400	0.0769
U.S. Asian	0.0429	0.0312	0.0147	0.0600	0.1154
International Asian	0.4163	0.1667	0.0882	0.3600	0.0769
International Other	0.2575	0.2292	0.3235	0.2000	0.1538
GRE Verbal Percentile	84.3980	88.8725	86.6328	77.9161	85.7729
GRE Quantitative Percentile	91.9289	68.3708	54.8113	64.5504	62.3462
GRE Analytic Percentile	69.0355	82.7222	78.0879	68.8958	74.6169
Undergraduate GPA	3.7336	3.6404	3.7204	3.7004	3.6341
TOEFL/IELTS	7.6419	7.6050	7.4730	7.6237	7.3178
Elite U.S. University	0.2833	0.3125	0.3382	0.2200	0.5385
Elite U.S. Liberal Arts College	0.1030	0.1458	0.1176	0.0200	0.1154
Elite Foreign University	0.1030	0.0521	0.0588	0.1000	0.0385
Other Foreign University	0.3863	0.2396	0.2647	0.4000	0.1154
Graduate Degree	0.5837	0.6771	0.7206	0.7400	0.4615
Work Experience	0.6309	0.8125	0.7647	0.7000	0.9231
#Professor Recommenders	2.5494	2.3750	2.8235	2.7200	2.2308
#Prolific Recommenders	1.2833	0.9167	0.6765	1.0800	0.4615
#Math Courses	7.2575				
Advanced Math	0.6781				
Missing GRE	0.0086	0.2917	0.6324	0.5800	0.1923
Missing Undergraduate GPA	0.4893	0.3125	0.3088	0.4800	0.1923
Missing TOEFL/IELTS	0.6567	0.8854	0.8529	0.7600	0.8846
Observations	233	96	68	50	26

- **Economics Highest:** International Asian, GRE Quantitative Percentile, Undergraduate GPA, TOEFL/IELTS, Elite Foreign University, #Prolific Recommenders, Missing Undergraduate GPA
- **Economics Lowest:** Covid Year, Winsorized Age, Female, U.S. African/Hispanic/Native, Work Experience, Missing GRE, Missing TOEFL/IELTS

Reduced-Form Summary

A3: Correlation Matrix

Table 4: Correlation Matrix

	Covid	Age	Fem	USAHN	USAsn	IntAsn	IntOth	GREV	GREQ	GREA	GPA	T/I	EUSUni	EUSLA	EFor	OthFor	Grad	Work	Prof	Pro	MGRE	MGPA	MT/I
Covid Year	1.00																						
Winsorized Age	0.01	1.00																					
Female	0.02	-0.13	1.00																				
U.S. African/Hispanic/Native	0.00	-0.04	0.04	1.00																			
U.S. Asian	-0.00	-0.06	0.05	-0.05	1.00																		
International Asian	-0.05	-0.07	0.02	-0.19	-0.14	1.00																	
International Other	0.01	0.18	-0.06	-0.17	-0.12	-0.42	1.00																
GRE Verbal Percentile	0.03	-0.06	-0.04	0.00	0.04	-0.12	-0.15	1.00															
GRE Quantitative Percentile	-0.02	-0.16	-0.10	-0.18	0.00	0.44	-0.10	0.16	1.00														
GRE Analytic Percentile	0.05	-0.06	0.03	0.08	0.08	-0.37	-0.08	0.60	-0.20	1.00													
Undergraduate GPA	-0.02	-0.29	0.07	-0.13	-0.02	0.02	0.00	0.06	0.12	0.05	1.00												
TOEFL/IELTS	-0.01	-0.07	0.01	-0.02	0.01	0.04	0.00	0.24	0.12	0.19	0.00	1.00											
Elite U.S. University	0.01	-0.12	0.02	0.12	0.14	-0.12	-0.22	0.16	-0.00	0.19	-0.01	-0.02	1.00										
Elite U.S. Liberal Arts College	-0.02	-0.02	0.04	0.04	0.04	-0.08	-0.09	0.12	-0.01	0.13	-0.02	-0.00	-0.11	1.00									
Elite Foreign University	-0.00	-0.04	-0.00	-0.06	-0.04	0.09	0.09	0.02	0.08	-0.02	0.01	0.04	-0.12	-0.06	1.00								
Other Foreign University	-0.02	0.20	-0.04	-0.23	-0.14	0.34	0.39	-0.26	0.20	-0.39	0.02	0.02	-0.43	-0.21	-0.22	1.00							
Graduate Degree	0.00	0.41	-0.08	-0.13	-0.09	0.21	0.15	-0.14	0.05	-0.20	-0.22	0.03	-0.21	-0.10	0.05	0.37	1.00						
Work Experience	0.01	0.33	0.05	0.05	0.01	-0.18	0.11	0.01	-0.15	0.11	-0.08	-0.03	-0.02	0.03	-0.03	-0.01	0.10	1.00					
#Professor Recommenders	-0.02	-0.19	-0.04	-0.04	0.01	0.11	-0.09	0.01	0.09	-0.04	0.14	0.03	-0.02	0.02	-0.00	-0.02	-0.03	-0.18	1.00				
#Prolific Recommenders	0.01	-0.06	-0.02	-0.07	-0.01	0.24	-0.10	-0.01	0.25	-0.11	0.04	0.10	-0.01	-0.01	0.06	0.08	0.17	-0.09	0.19	1.00			
Missing GRE	0.09	0.15	0.05	0.11	-0.01	-0.22	0.17	0.09	-0.33	0.17	-0.09	-0.10	-0.03	-0.04	-0.03	-0.01	0.02	0.10	-0.08	-0.20	1.00		
Missing Undergraduate GPA	-0.01	0.18	-0.04	-0.24	-0.15	0.36	0.42	-0.24	0.21	-0.38	0.02	0.04	-0.46	-0.23	0.23	0.83	0.37	-0.02	-0.03	0.10	-0.01	1.00	
Missing TOEFL/IELTS	0.05	-0.07	0.03	0.16	0.11	-0.32	-0.22	0.16	-0.24	0.32	-0.02	-0.02	0.29	0.14	-0.04	-0.58	-0.20	0.05	-0.07	-0.05	0.12	-0.58	1.00
Observations	6222																						

Reduced-Form Summary

A3: Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3966**	-0.4332*	-0.5752*	-0.5501	-0.1766
	(0.2010)	(0.2612)	(0.3479)	(0.4129)	(0.5545)
Winsorized Age	-0.0995***	0.0454*	0.0380	0.0222	-0.0520
	(0.0358)	(0.0271)	(0.0327)	(0.0532)	(0.0899)
Female	0.4079**	0.6723***	0.6539**	0.1311	-0.5060
	(0.1681)	(0.2299)	(0.2971)	(0.3452)	(0.5606)
U.S. African/Hispanic/Native	0.4221	1.0081***	1.1601***	0.0120	-0.8263
	(0.5258)	(0.3545)	(0.4426)	(0.8411)	(1.1611)
U.S. Asian	-0.5394	-0.5669	-0.3260	0.8638	1.1711
	(0.4199)	(0.6504)	(1.2512)	(0.7201)	(0.7276)
International Asian	-0.8445***	-0.1853	0.9405	0.4890	-0.7827
	(0.2945)	(0.4146)	(0.7233)	(0.6729)	(0.9323)
International Other	0.5034	0.3781	1.3523***	-0.1061	0.6239
	(0.3172)	(0.4131)	(0.5086)	(0.7483)	(1.1070)
GRE Verbal Percentile	0.0155***	0.0281*	0.0753**	-0.0092	0.0607***
	(0.0060)	(0.0160)	(0.0344)	(0.0154)	(0.0218)
GRE Quantitative Percentile	0.1547***	0.0161*	0.0210*	0.0211	-0.0138
	(0.0186)	(0.0086)	(0.0116)	(0.0150)	(0.0159)
GRE Analytic Percentile	0.0179***	0.0199**	0.0030	0.0175	-0.0202
	(0.0042)	(0.0094)	(0.0134)	(0.0127)	(0.0218)
Undergraduate GPA	2.2212***	0.6968	1.3648**	1.2645	1.7772*
	(0.7219)	(0.5476)	(0.5513)	(0.9703)	(0.9487)
TOEFL/IELTS	0.3878	0.6456	1.0963**	0.9655*	-0.4656
	(0.2883)	(0.4708)	(0.5526)	(0.5456)	(0.6112)
Elite U.S. University	0.8850***	0.3943	0.6541*	-0.6057	1.4303**
	(0.2960)	(0.3135)	(0.3802)	(0.4985)	(0.6192)
Elite U.S. Liberal Arts	1.1173***	0.6276*	1.0698**	-0.6088	2.3151**
	(0.3813)	(0.3754)	(0.5399)	(1.4186)	(1.0642)
Elite Foreign University	1.0061*	0.4759	1.7715**	1.6195*	0.3889
	(0.6060)	(0.8224)	(0.7938)	(0.8587)	(1.9889)
Other Foreign University	0.5442	0.5777	1.3733**	0.4188	-1.0737
	(0.5765)	(0.7686)	(0.6992)	(0.7157)	(1.3016)
Graduate Degree	0.0467	0.0733	0.3128	0.4049	0.4892
	(0.2238)	(0.2583)	(0.3280)	(0.4896)	(0.6399)
Work Experience	0.3932**	0.4027	0.4972	0.1495	1.8201**
	(0.1889)	(0.2993)	(0.3295)	(0.3778)	(0.8564)
#Professor Recommenders	0.0340	-0.0179	0.5511**	0.4781*	-0.0981
	(0.1317)	(0.1339)	(0.2554)	(0.2662)	(0.2660)
#Prolific Recommenders	0.2635***	0.3025**	0.1880	0.3284**	0.5901*
	(0.0854)	(0.1262)	(0.1576)	(0.1544)	(0.3145)
#Math Courses	0.1689***				
	(0.0596)				
Advanced Math	0.2319				
	(0.1790)				
Missing GRE	0.1899	-0.3259	0.6282	-0.0831	-1.1514**
	(0.7646)	(0.3100)	(0.4275)	(0.3761)	(0.5702)
Missing Undergraduate GPA	-0.4745	-0.4917	-1.5492**	-1.1097*	0.4690
	(0.4891)	(0.5440)	(0.6893)	(0.6481)	(1.1831)
MissingTOEFL/IELTS	-0.3786	0.6188	0.6745	-0.0440	-1.7224*
	(0.2432)	(0.5657)	(0.5190)	(0.5099)	(0.9271)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2677	0.1525	0.1982	0.1871	0.2475
CV Pseudo-R^2	0.2102	0.0720	0.0604	-0.0265	-0.2787

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Table 5a: Logistic Regression Results

- **Economics Positive ($p < .05$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses
- **Economics Negative ($p < .05$):** Covid Year, Winsorized Age, International Asian

Post-Lasso Logistic Regression AMEs

A3: Logistic Regression AMEs

Table 5b: Logistic Regression AMEs

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0303**	-0.0164	-0.0431*	-0.0298	-0.0075	-0.0218***
	(0.0153)	(0.0100)	(0.0261)	(0.0224)	(0.0238)	(0.0084)
Winsorized Age	-0.0076***	0.0017*	0.0028	0.0012	-0.0022	0.0016*
	(0.0027)	(0.0010)	(0.0025)	(0.0029)	(0.0038)	(0.0009)
Female	0.0312**	0.0254***	0.0490**	0.0071	-0.0216	0.0207***
	(0.0129)	(0.0088)	(0.0220)	(0.0187)	(0.0239)	(0.0075)
U.S. African/Hispanic/Native	0.0323	0.0381***	0.0869***	0.0007	-0.0352	0.0384***
	(0.0402)	(0.0137)	(0.0332)	(0.0455)	(0.0487)	(0.0119)
U.S. Asian	-0.0413	-0.0215	-0.0244	0.0468	0.0499	0.0062
	(0.0322)	(0.0247)	(0.0936)	(0.0391)	(0.0313)	(0.0185)
International Asian	-0.0646***	-0.0070	0.0704	0.0265	-0.0334	0.0069
	(0.0224)	(0.0157)	(0.0543)	(0.0363)	(0.0393)	(0.0132)
International Other	0.0385	0.0143	0.1013***	-0.0057	0.0266	0.0238*
	(0.0243)	(0.0157)	(0.0385)	(0.0406)	(0.0471)	(0.0132)
GRE Verbal Percentile	0.0012***	0.0011*	0.0056**	-0.0005	0.0026***	0.0015***
	(0.0005)	(0.0006)	(0.0026)	(0.0008)	(0.0010)	(0.0005)
GRE Quantitative Percentile	0.0118***	0.0006*	0.0016*	0.0011	-0.0006	0.0006**
	(0.0014)	(0.0003)	(0.0009)	(0.0008)	(0.0007)	(0.0003)
GRE Analytic Percentile	0.0014***	0.0008**	0.0002	0.0009	-0.0009	0.0007**
	(0.0003)	(0.0004)	(0.0010)	(0.0007)	(0.0009)	(0.0003)
Undergraduate GPA	0.1699***	0.0264	0.1022**	0.0684	0.0757*	0.0488***
	(0.0551)	(0.0208)	(0.0423)	(0.0522)	(0.0418)	(0.0176)
TOEFL/IELTS	0.0297	0.0244	0.0821**	0.0523*	-0.0198	0.0326***
	(0.0219)	(0.0179)	(0.0417)	(0.0301)	(0.0259)	(0.0123)
Elite U.S. University	0.0677***	0.0149	0.0490*	-0.0328	0.0610**	0.0233**
	(0.0225)	(0.0119)	(0.0281)	(0.0264)	(0.0271)	(0.0098)
Elite U.S. Liberal Arts College	0.0855***	0.0238*	0.0801**	-0.0330	0.0987**	0.0420***
	(0.0288)	(0.0143)	(0.0402)	(0.0761)	(0.0480)	(0.0136)
Elite Foreign University	0.0770*	0.0180	0.1327**	0.0877*	0.0166	0.0479**
	(0.0463)	(0.0311)	(0.0591)	(0.0481)	(0.0850)	(0.0241)
Other Foreign University	0.0416	0.0219	0.1028*	0.0227	-0.0458	0.0273
	(0.0441)	(0.0291)	(0.0527)	(0.0391)	(0.0556)	(0.0221)
Graduate Degree	0.0036	0.0028	0.0234	0.0219	0.0208	0.0106
	(0.0171)	(0.0098)	(0.0247)	(0.0262)	(0.0273)	(0.0089)
Work Experience	0.0301**	0.0152	0.0372	0.0081	0.0776**	0.0221**
	(0.0144)	(0.0113)	(0.0248)	(0.0205)	(0.0368)	(0.0089)
#Professor Recommenders	0.0026	-0.0007	0.0413**	0.0259*	-0.0042	0.0063
	(0.0101)	(0.0051)	(0.0190)	(0.0142)	(0.0113)	(0.0047)
#Prolific Recommenders	0.0202***	0.0114**	0.0141	0.0178**	0.0251*	0.0133***
	(0.0065)	(0.0048)	(0.0118)	(0.0082)	(0.0130)	(0.0038)
#Math Courses	0.0129***					
	(0.0045)					
Advanced Math	0.0177					
	(0.0137)					
Missing GRE	0.0145	-0.0123	0.0470	-0.0045	-0.0491*	-0.0062
	(0.0585)	(0.0118)	(0.0319)	(0.0204)	(0.0252)	(0.0088)
Missing Undergraduate GPA	-0.0363	-0.0186	-0.1160**	-0.0601*	0.0200	-0.0291
	(0.0375)	(0.0206)	(0.0524)	(0.0355)	(0.0505)	(0.0179)
Missing TOEFL/IELTS	-0.0290	0.0234	0.0505	-0.0024	-0.0734*	0.0108
	(0.0188)	(0.0215)	(0.0388)	(0.0276)	(0.0417)	(0.0135)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	115.8837	23	0.0000			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .05$):** Female, GRE Verbal Percentile, GRE Quantitative Percentile, GRE Analytic Percentile, Undergraduate GPA, Elite U.S. University, Elite U.S. Liberal Arts College, Work Experience, #Prolific Recommenders, #Math Courses

- **Economics Negative ($p < .05$):** Covid Year, Winsorized Age, International Asian

Post-Lasso Logistic Regression AMEs

A3: Logistic Regression Results (Synthetic)

Table 6a: Logistic Regression Results (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.1997	-0.0323	0.2028	-0.4791	-0.5946
	(0.1633)	(0.2215)	(0.3078)	(0.3477)	(0.4520)
Winsorized Age	-0.0509**	-0.0198	0.0576*	-0.0164	0.1203***
	(0.0201)	(0.0174)	(0.0307)	(0.0357)	(0.0400)
Female	0.1941	0.0923	0.1389	0.2595	0.2119
	(0.1402)	(0.2004)	(0.2526)	(0.3255)	(0.4310)
U.S. African/Hispanic/Native	0.3053	-0.1457	-0.2379	-1.6401	0.5413
	(0.3724)	(0.4303)	(0.4670)	(1.0720)	(0.5306)
U.S. Asian	0.5663	0.6016	-0.7934	-0.4305	0.7122
	(0.3642)	(0.4452)	(1.1675)	(0.7936)	(0.8863)
International Asian	-0.1428	0.1630	0.5780	-0.4965	0.1868
	(0.2169)	(0.2803)	(0.3740)	(0.4275)	(0.6866)
International Other	-0.1231	0.1395	0.1097	-0.1602	-1.0807
	(0.2311)	(0.3046)	(0.3237)	(0.3832)	(0.7583)
GRE Verbal Percentile	0.0167***	0.0247**	0.0164	-0.0029	0.0132
	(0.0048)	(0.0117)	(0.0127)	(0.0100)	(0.0184)
GRE Quantitative Percentile	0.0233***	-0.0027	0.0026	0.0304**	0.0097
	(0.0090)	(0.0051)	(0.0088)	(0.0137)	(0.0099)
GRE Analytic Percentile	0.0042	0.0004	-0.0073	-0.0042	0.0069
	(0.0031)	(0.0059)	(0.0092)	(0.0101)	(0.0103)
Undergraduate GPA	0.2292	0.6327	1.3552**	-1.3171**	-0.1002
	(0.4745)	(0.4985)	(0.5796)	(0.5584)	(0.6152)
TOEFL/IELTS	-0.4673**	0.5141	-0.7648	0.2446	-0.7354
	(0.2157)	(0.4185)	(0.6814)	(0.7498)	(0.8025)
Elite U.S. University	-0.4850*	-0.2243	0.0572	0.4764	0.3941
	(0.2560)	(0.3258)	(0.3421)	(0.4788)	(0.4782)
Elite U.S. Liberal Arts College	0.2429	0.1759	0.2358	0.4860	0.8652
	(0.3049)	(0.4042)	(0.4964)	(0.9631)	(0.7261)
Elite Foreign University	-0.1622	-1.1929	0.7404	0.4801	3.0530***
	(0.3015)	(0.7484)	(0.5668)	(0.7841)	(0.8673)
Other Foreign University	-0.1087	0.2043	0.2721	0.1731	0.3023
	(0.1979)	(0.2732)	(0.3391)	(0.4292)	(0.5926)
Graduate Degree	-0.0198	0.1770	-0.1601	-0.8698***	-0.2769
	(0.1624)	(0.2439)	(0.2765)	(0.3320)	(0.3814)
Work Experience	0.2525*	-0.0391	-0.0736	-0.2650	-1.1936**
	(0.1441)	(0.2396)	(0.2629)	(0.3417)	(0.4805)
#Professor Recommenders	-0.1059	0.0911	-0.1828	0.2181	-0.4923**
	(0.0909)	(0.1138)	(0.1473)	(0.1817)	(0.1956)
#Prolific Recommenders	0.0039	0.2189**	-0.1743	0.0673	0.5134**
	(0.0656)	(0.1108)	(0.1723)	(0.1642)	(0.2213)
#Math Courses	0.0366				
	(0.0421)				
Advanced Math	0.1961				
	(0.1416)				
Missing GRE	-0.1993	-0.2029	-0.0011	-0.0433	-0.6040
	(0.4328)	(0.2195)	(0.2689)	(0.3503)	(0.4017)
Missing Undergraduate GPA	0.1531	0.0895	-0.2455	0.6298**	-0.3574
	(0.1548)	(0.2291)	(0.3015)	(0.3125)	(0.4342)
Missing TOEFL/IELTS	-0.2950*	0.4898	0.5752	0.1346	-2.0781***
	(0.1640)	(0.3480)	(0.5466)	(0.4345)	(0.7633)
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.0524	0.0366	0.0753	0.1046	0.2230
CV Pseudo-R^2	0.0139	-0.0189	-0.0887	0.0055	-0.0014

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Economics Positive ($p < .05$):** GRE Verbal Percentile, GRE Quantitative Percentile
- **Economics Negative ($p < .05$):** Winsorized Age, TOEFL/IELTS

▶▶ Post-Lasso Logistic Regression AMEs

A3: Logistic Regression AMEs (Synthetic)

Table 6b: Logistic Regression AMEs (Synthetic)

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology	Non-Economics
Covid Year	-0.0204	-0.0014	0.0192	-0.0298	-0.0345	-0.0089
	(0.0167)	(0.0099)	(0.0291)	(0.0216)	(0.0262)	(0.0086)
Winsorized Age	-0.0052**	-0.0009	0.0054*	-0.0010	0.0070***	0.0004
	(0.0021)	(0.0008)	(0.0029)	(0.0022)	(0.0023)	(0.0008)
Female	0.0198	0.0041	0.0131	0.0162	0.0123	0.0099
	(0.0143)	(0.0089)	(0.0239)	(0.0204)	(0.0251)	(0.0078)
U.S. African/Hispanic/Native	0.0312	-0.0065	-0.0225	-0.1021	0.0314	-0.0137
	(0.0381)	(0.0192)	(0.0441)	(0.0673)	(0.0306)	(0.0147)
U.S. Asian	0.0579	0.0268	-0.0750	-0.0268	0.0413	0.0241
	(0.0372)	(0.0199)	(0.1101)	(0.0495)	(0.0516)	(0.0179)
International Asian	-0.0146	0.0073	0.0546	-0.0309	0.0108	0.0075
	(0.0222)	(0.0125)	(0.0353)	(0.0263)	(0.0399)	(0.0112)
International Other	-0.0126	0.0062	0.0104	-0.0100	-0.0627	-0.0014
	(0.0236)	(0.0136)	(0.0306)	(0.0237)	(0.0443)	(0.0106)
GRE Verbal Percentile	0.0017***	0.0011**	0.0015	-0.0002	0.0008	0.0008**
	(0.0005)	(0.0005)	(0.0012)	(0.0006)	(0.0010)	(0.0004)
GRE Quantitative Percentile	0.0024***	-0.0001	0.0003	0.0019**	0.0006	0.0003
	(0.0009)	(0.0002)	(0.0008)	(0.0009)	(0.0006)	(0.0002)
GRE Analytic Percentile	0.0004	0.0000	-0.0007	-0.0003	0.0004	0.0000
	(0.0003)	(0.0003)	(0.0009)	(0.0006)	(0.0006)	(0.0002)
Undergraduate GPA	0.0234	0.0282	0.1281**	-0.0820**	-0.0058	0.0130
	(0.0485)	(0.0223)	(0.0553)	(0.0358)	(0.0357)	(0.0158)
TOEFL/IELTS	-0.0478**	0.0229	-0.0723	0.0152	-0.0427	0.0050
	(0.0221)	(0.0187)	(0.0645)	(0.0466)	(0.0469)	(0.0187)
Elite U.S. University	-0.0496*	-0.0100	0.0054	0.0297	0.0229	0.0036
	(0.0261)	(0.0145)	(0.0324)	(0.0297)	(0.0274)	(0.0108)
Elite U.S. Liberal Arts College	0.0248	0.0078	0.0223	0.0303	0.0502	0.0217
	(0.0312)	(0.0180)	(0.0469)	(0.0602)	(0.0418)	(0.0160)
Elite Foreign University	-0.0166	-0.0532	0.0700	0.0299	0.1771***	0.0158
	(0.0308)	(0.0337)	(0.0537)	(0.0490)	(0.0484)	(0.0179)
Other Foreign University	-0.0111	0.0091	0.0257	0.0108	0.0175	0.0116
	(0.0202)	(0.0122)	(0.0321)	(0.0267)	(0.0344)	(0.0103)
Graduate Degree	-0.0020	0.0079	-0.0151	-0.0542***	-0.0161	-0.0093
	(0.0166)	(0.0109)	(0.0261)	(0.0206)	(0.0220)	(0.0085)
Work Experience	0.0258*	-0.0017	-0.0070	-0.0165	-0.0692**	-0.0093
	(0.0148)	(0.0107)	(0.0248)	(0.0213)	(0.0275)	(0.0086)
#Professor Recommenders	-0.0108	0.0041	-0.0173	0.0136	-0.0286***	-0.0027
	(0.0093)	(0.0051)	(0.0140)	(0.0114)	(0.0110)	(0.0042)
#Prolific Recommenders	0.0004	0.0098*	-0.0165	0.0042	0.0298**	0.0083*
	(0.0067)	(0.0050)	(0.0164)	(0.0102)	(0.0128)	(0.0045)
#Math Courses	0.0037					
	(0.0043)					
Advanced Math	0.0201					
	(0.0145)					
Missing GRE	-0.0204	-0.0090	-0.0001	-0.0027	-0.0350	-0.0098
	(0.0443)	(0.0098)	(0.0254)	(0.0218)	(0.0239)	(0.0079)
Missing Undergraduate GPA	0.0157	0.0040	-0.0232	0.0392*	-0.0207	0.0019
	(0.0158)	(0.0102)	(0.0285)	(0.0201)	(0.0254)	(0.0083)
Missing TOEFL/IELTS	-0.0302*	0.0218	0.0544	0.0084	-0.1205***	0.0063
	(0.0168)	(0.0156)	(0.0517)	(0.0272)	(0.0434)	(0.0130)
Concentration FE	No	Yes	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486	4144
Wald Statistic | DF | P-Value	37.4760	23	0.0290			

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Post-Lasso Logistic Regression AMEs

- **Economics Positive ($p < .05$):** GRE Verbal Percentile, GRE Quantitative Percentile
- **Economics Negative ($p < .05$):** Winsorized Age, TOEFL/IELTS

Table 7: Lasso Logistic Regression Results

Depdendent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3330	-0.3123	-0.3498	-0.2378	
Winsorized Age	-0.0788	0.0266	0.0200		
Female	0.3352	0.5096	0.5068		-0.0654
U.S. African/Hispanic/Native	0.2638	0.8041	0.7157		
U.S. Asian	-0.2802	-0.2918		0.0413	0.1155
International Asian	-0.6744	-0.0726		0.0465	
International Other	0.4729	0.1946	0.7240	-0.1087	
GRE Verbal Percentile	0.0139	0.0252	0.0272		0.0343
GRE Quantitative Percentile	0.1349	0.0119	0.0187	0.0154	
GRE Analytic Percentile	0.0166	0.0208		0.0080	
Undergraduate GPA	1.9977	0.4621	0.7414	0.1714	
TOEFL/IELTS	0.4009	0.3295	0.5199	0.5151	
Elite U.S. University	0.6886	0.2709	0.3267		0.4606
Elite U.S. Liberal Arts College	0.8956	0.5199	0.5985		0.4470
Elite Foreign University	0.3724		0.0414	0.4343	
Other Foreign University					
Graduate Degree			0.0978		
Work Experience	0.2835	0.3447	0.3273		0.5559
#Professor Recommenders			0.3739	0.2322	
#Prolific Recommenders	0.2243	0.2362	0.1651	0.2213	0.0632
#Math Courses	0.1493				
Advanced Math	0.2272				
Missing GRE		-0.2304		-0.0527	-0.5416
Missing Undergraduate GPA					
Missing TOEFL/IELTS	-0.2364	0.2852	0.0796		
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2642	0.1456	0.1589	0.1486	0.1362
CV Pseudo-R^2	0.2297	0.0912	0.0592	0.0850	0.0537

Post-Lasso Logistic Regression AMEs

A3: Elastic-Net Logistic Regression Results

Table 8: Elastic-Net Logistic Regression Results

Dependent Variable = Accept	Economics	Government	History	Linguistics	Psychology
Covid Year	-0.3323	-0.3013	-0.3652	-0.2642	-0.0428
Winsorized Age	-0.0786	0.0231	0.0235		
Female	0.3316	0.4214	0.4373		-0.2716
U.S. African/Hispanic/Native	0.2622	0.7009	0.6038		-0.2580
U.S. Asian	-0.2741	-0.4137	-0.3622	0.1731	0.4670
International Asian	-0.6634	-0.1850	0.1309	0.1086	-0.2639
International Other	0.4732	0.1832	0.5739	-0.1280	0.0824
GRE Verbal Percentile	0.0140	0.0187	0.0188		0.0236
GRE Quantitative Percentile	0.1324	0.0117	0.0166	0.0136	
GRE Analytic Percentile	0.0165	0.0157	0.0055	0.0089	
Undergraduate GPA	1.9971	0.5294	0.7165	0.3104	0.3833
TOEFL/IELTS	0.4029	0.4179	0.6717	0.5298	
Elite U.S. University	0.6894	0.3109	0.3740		0.5669
Elite U.S. Liberal Arts College	0.8906	0.5376	0.6277		0.7887
Elite Foreign University	0.3738	0.0332	0.3675	0.5020	0.3558
Other Foreign University		0.0233	0.1382		
Graduate Degree		0.0315	0.1873		0.1018
Work Experience	0.2822	0.3230	0.3192		0.5851
#Professor Recommenders		-0.0179	0.3250	0.2333	
#Prolific Recommenders	0.2238	0.2164	0.1665	0.2212	0.1970
#Math Courses	0.1489				
Advanced Math	0.2309				
Missing GRE		-0.3089	0.1123	-0.0864	-0.5616
Missing Undergraduate GPA		-0.1040	-0.1046		
Missing TOEFL/IELTS	-0.2369	0.2950	0.2603		-0.1782
Concentration FE	No	Yes	Yes	Yes	Yes
Observations	2078	2231	690	737	486
Pseudo-R^2	0.2640	0.1422	0.1626	0.1519	0.1653
CV Pseudo-R^2	0.2297	0.0999	0.0786	0.0859	0.0582
α^*	0.9	0.0	0.0	0.4	0.1

▸▸ Post-Lasso Logistic Regression AMEs

A3: Economics Means (Matriculation)

Table 9: Economics Means (Matriculation)

Mean	All Observations	Matriculate = 1
Matriculate	0.2814	
Covid Year	0.1905	0.2000
Winsorized Age	25.0303	25.1692
Female	0.4848	0.4154
International	0.6710	0.7385
GRE Verbal Percentile	84.5065	79.7385
GRE Quantitative Percentile	91.9784	91.2308
GRE Analytic Percentile	69.1775	65.4615
Undergraduate GPA	3.7160	3.6868
TOEFL/IELTS	7.6394	7.5953
Elite Undergraduate	0.4892	0.4000
Graduate Degree	0.5801	0.6308
Work Experience	0.6320	0.5231
#Professor Recommenders	2.5498	2.5846
#Prolific Recommenders	1.2727	1.1692
Advanced Math	0.6753	0.5077
Missing Undergraduate GPA	0.4892	0.5077
Missing TOEFL/IELTS	0.6537	0.6000
Observations	231	65

Reduced-Form Summary

A3: Correlation Matrix (Matriculation)

Table 10: Correlation Matrix (Matriculation)

	Covid	Age	Fem	Int	GREV	GREQ	GREA	GPA	T/I	Elite	Grad	Work	Prof	Pro	Math	MGPA	MT/I
Covid Year	1.00																
Winsorized Age	0.03	1.00															
Female	0.04	-0.09	1.00														
International	-0.04	0.11	0.02	1.00													
GRE Verbal Percentile	-0.04	-0.14	-0.04	-0.34	1.00												
GRE Quantitative Percentile	0.03	-0.03	-0.15	0.26	0.04	1.00											
GRE Analytic Percentile	0.08	-0.09	-0.01	-0.50	0.44	-0.19	1.00										
Undergraduate GPA	-0.00	-0.37	0.10	-0.40	0.21	-0.13	0.23	1.00									
TOEFL/IELTS	-0.06	-0.05	-0.03	0.24	0.23	0.15	0.02	-0.21	1.00								
Elite Undergraduate	-0.03	-0.17	0.11	-0.37	0.33	-0.00	0.27	0.19	-0.18	1.00							
Graduate Degree	-0.06	0.42	-0.12	0.43	-0.25	0.12	-0.25	-0.48	0.17	-0.38	1.00						
Work Experience	-0.04	0.34	0.00	-0.19	0.08	-0.17	0.30	-0.02	-0.03	0.05	-0.07	1.00					
#Professor Recommenders	-0.06	-0.24	-0.01	0.23	-0.05	0.09	-0.11	0.11	0.02	-0.07	0.05	-0.30	1.00				
#Prolific Recommenders	0.04	-0.02	0.14	0.14	-0.02	0.11	-0.08	-0.11	0.10	-0.11	0.21	-0.02	0.22	1.00			
Advanced Math	0.01	-0.17	0.01	-0.21	0.24	0.06	0.17	0.15	-0.12	0.22	-0.25	0.01	0.04	0.14	1.00		
Missing Undergraduate GPA	0.01	0.28	-0.12	0.65	-0.34	0.18	-0.39	-0.64	0.32	-0.58	0.57	-0.04	0.03	0.19	-0.26	1.00	
Missing TOEFL/IELTS	0.05	-0.20	0.12	-0.51	0.27	-0.08	0.38	0.45	-0.46	0.49	-0.36	0.12	0.01	-0.00	0.23	-0.71	1.00
Observations	231																

Reduced-Form Summary

A3: Logistic Regression Results (Matriculation)

Table 11a: Logistic Regression Results (Matriculation)

Dependent Variable = Matriculate	Coefficient	AME	Lasso	Post-Lasso Coefficient	Post-Lasso AME
Covid Year	0.1384	0.0232			
	(0.4306)	(0.0724)			
Winsorized Age	0.0261	0.0044			
	(0.0900)	(0.0152)			
Female	-0.6244*	-0.1049*	-0.3914	-0.6158*	-0.1037*
	(0.3570)	(0.0589)		(0.3471)	(0.0573)
International	0.7372	0.1239	0.1687	0.6690	0.1127
	(0.5175)	(0.0856)		(0.4693)	(0.0779)
GRE Verbal Percentile	-0.0224*	-0.0038*	-0.0194	-0.0215*	-0.0036*
	(0.0130)	(0.0021)		(0.0123)	(0.0020)
GRE Quantitative Percentile	-0.0609*	-0.0102*	-0.0425	-0.0631*	-0.0106*
	(0.0348)	(0.0058)		(0.0341)	(0.0056)
GRE Analytic Percentile	0.0034	0.0006			
	(0.0091)	(0.0015)			
Undergraduate GPA	-4.1670***	-0.7001***	-2.2674	-4.1144***	-0.6932***
	(1.6083)	(0.2570)		(1.4501)	(0.2319)
TOEFL/IELTS	-0.3597	-0.0604	-0.1628	-0.2931	-0.0494
	(0.5858)	(0.0984)		(0.5263)	(0.0887)
Elite Undergraduate	-0.9465*	-0.1590*	-0.3943	-0.9245*	-0.1558*
	(0.4954)	(0.0827)		(0.4844)	(0.0816)
Graduate Degree	-0.2158	-0.0363			
	(0.4760)	(0.0799)			
Work Experience	-0.8483**	-0.1425**	-0.6164	-0.7994**	-0.1347**
	(0.4006)	(0.0666)		(0.3458)	(0.0572)
#Professor Recommenders	0.0296	0.0050			
	(0.2316)	(0.0389)			
#Prolific Recommenders	0.0268	0.0045			
	(0.1948)	(0.0328)			
Advanced Math	-1.0856***	-0.1824***	-0.8427	-1.0371***	-0.1747***
	(0.3627)	(0.0574)		(0.3463)	(0.0548)
Missing Undergraduate GPA	-2.1263***	-0.3573***	-0.9377	-2.0496***	-0.3453***
	(0.8113)	(0.1304)		(0.7114)	(0.1135)
Missing TOEFL/IELTS	-0.1722	-0.0289			
	(0.6110)	(0.1026)			
Observations	231	231	231	231	231
Pseudo-R^2	0.1469		0.1288	0.1445	
CV Pseudo-R^2	-0.0928		0.0112	0.0485	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

- **Positive ($p < .05$):** None
- **Negative ($p < .05$):** Undergraduate GPA, Work Experience, Advanced Math, Missing Undergraduate GPA

Post-Lasso Logistic Regression AMEs

A3: Simulated Wald Test Summary Statistics (Different)

Table 12a: Simulated Wald Test Summary Statistics (Different)

	Economics		Non-Economics 1		Non-Economics 2	
	Mean	SD	Mean	SD	Mean	SD
Accept	0.1335	0.3401	0.0452	0.2078	0.0543	0.2266
Success	0.3000	0.4583	0.3126	0.4636	0.1314	0.3379
GRE Quantitative Percentile	84.8390	7.5405	65.1343	7.4495	65.6114	7.6706
Gender	0.3935	0.4885	0.3952	0.4889	0.5914	0.4916
Demographic 1	0.5005	0.5000	0.3091	0.4621	0.2600	0.4386
Demographic 2	0.2505	0.4333	0.2957	0.4563	0.3414	0.4742
Advanced Math	0.5095	0.4999				
Committee Score	4.9325	1.9877	6.0191	2.0205	6.0243	2.0038
Observations	2000		2300		700	

Acceptance vs. Success AMEs

A3: Simulated Wald Test Summary Statistics (Same)

Table 12b: Simulated Wald Test Summary Statistics (Same)

	Economics		Non-Economics 1		Non-Economics 2	
	Mean	SD	Mean	SD	Mean	SD
Accept	0.1335	0.3401	0.0900	0.2862	0.0786	0.2691
Success	0.3000	0.4583	0.4057	0.4910	0.1157	0.3199
GRE Quantitative Percentile	84.8390	7.5405	65.1343	7.4495	65.6114	7.6706
Gender	0.3935	0.4885	0.3952	0.4889	0.5914	0.4916
Demographic 1	0.5005	0.5000	0.3091	0.4621	0.2600	0.4386
Demographic 2	0.2505	0.4333	0.2957	0.4563	0.3414	0.4742
Advanced Math	0.5095	0.4999				
Committee Score	4.9325	1.9877	6.0191	2.0205	6.0243	2.0038
Observations	2000		2300		700	

Acceptance vs. Success AMEs

A3: Simulated Wald Test Correlation Matrix (Different)

Table 13a: Simulated Wald Test Correlation Matrix (Different)

	Acc	Suc	GRE	Gen	Dem1	Dem2	Com	NEcon1	NEcon2
Accept	1								
Success	0.0885	1							
GRE Quantitative Percentile	0.2376	0.1167	1						
Gender	0.0582	-0.0220	-0.0795	1					
Demographic 1	0.0152	-0.0004	0.3377	-0.0136	1				
Demographic 2	0.0078	0.0003	-0.0876	-0.0209	-0.4918	1			
Committee Score	0.1437	0.1500	-0.0770	0.0980	-0.1567	-0.0359	1		
Non-Economics 1	-0.1232	0.0624	-0.6017	-0.0501	-0.1326	0.0238	0.1930	1	
Non-Economics 2	-0.0405	-0.1352	-0.2472	0.1384	-0.0988	0.0514	0.0854	-0.3724	1
Observations	5000								

Acceptance vs. Success AMEs

A3: Simulated Wald Test Correlation Matrix (Same)

Table 13b: Simulated Wald Test Correlation Matrix (Same)

	Acc	Suc	GRE	Gen	Dem1	Dem2	Com	NEcon1	NEcon2
Accept	1								
Success	0.1603	1							
GRE Quantitative Percentile	0.2772	0.1665	1						
Gender	0.0484	-0.0581	-0.0795	1					
Demographic 1	-0.0233	0.0199	0.3377	-0.0136	1				
Demographic 2	0.0487	0.0148	-0.0876	-0.0209	-0.4918	1			
Committee Score	0.2045	0.1486	-0.0770	0.0980	-0.1567	-0.0359	1		
Non-Economics 1	-0.0474	0.1636	-0.6017	-0.0501	-0.1326	0.0238	0.1930	1	
Non-Economics 2	-0.0357	-0.1787	-0.2472	0.1384	-0.0988	0.0514	0.0854	-0.3724	1
Observations	5000								

Acceptance vs. Success AMEs

A3: Simulated Wald Test Results (Different)

Table 14a: Simulated Wald Test Results (Different)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3291***	-1.5784***	-8.4753***	-0.3642***	-10.4581***	-1.9104***	-3.2236***	-0.5997***
	(1.1274)	(0.0955)	(0.8673)	(0.0412)	(0.7237)	(0.1070)	(0.4018)	(0.0728)
GRE Quantitative Percentile	0.1502***	0.0145***	0.0402***	0.0017***	0.1058***	0.0193***	0.0034	0.0006
	(0.0131)	(0.0012)	(0.0128)	(0.0006)	(0.0088)	(0.0014)	(0.0062)	(0.0012)
Gender	0.4702***	0.0454***	0.5942***	0.0255***	-0.1372	-0.0251	0.0375	0.0070
	(0.1473)	(0.0140)	(0.1820)	(0.0079)	(0.1095)	(0.0200)	(0.0868)	(0.0161)
Demographic 1	-0.6246***	-0.0604***	-0.2541	-0.0109	-0.3544**	-0.0647**	-0.1152	-0.0214
	(0.2085)	(0.0200)	(0.2393)	(0.0103)	(0.1467)	(0.0267)	(0.1103)	(0.0205)
Demographic 2	0.1555	0.0150	0.0081	0.0003	0.1713	0.0313	-0.0562	-0.0105
	(0.2070)	(0.0200)	(0.2109)	(0.0091)	(0.1557)	(0.0284)	(0.1030)	(0.0192)
Advanced Math	0.2886*	0.0279*			0.3284***	0.0600***		
	(0.1611)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0217***	0.3817***	0.0164***	0.1026***	0.0187***	0.1808***	0.0336***
	(0.0413)	(0.0039)	(0.0528)	(0.0025)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			-0.0452	-0.0019			1.1393***	0.2120***
			(0.1995)	(0.0086)			(0.1219)	(0.0219)
Observations | Pseudo-R^2	5000	0.1488			5000	0.0758		
LL | Parameters | AIC	-1155.7	14	2339.5		-2747.2	14	5522.4	
Wald Statistic | DF | P-Value	112.5897	5	0.0000		124.7379	5	0.0000	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance vs. Success AMEs

A3: Simulated Wald Test Results (Same)

Table 14b: Simulated Wald Test Results (Same)

	Dependent Variable = Accept				Dependent Variable = Success			
	Economics		Non-Economics		Economics		Non-Economics	
	Coef	AME	Coef	AME	Coef	AME	Coef	AME
Intercept	-16.3848***	-1.5822***	-28.5524***	-1.3677***	-10.4586***	-1.9105***	-10.2610***	-1.8514***
	(1.1315)	(0.0956)	(1.4963)	(0.0500)	(0.7236)	(0.1070)	(0.4815)	(0.0628)
GRE Quantitative Percentile	0.1509***	0.0146***	0.3245***	0.0155***	0.1058***	0.0193***	0.1154***	0.0208***
	(0.0131)	(0.0012)	(0.0179)	(0.0007)	(0.0088)	(0.0014)	(0.0070)	(0.0011)
Gender	0.4714***	0.0455***	0.7880***	0.0377***	-0.1384	-0.0253	-0.0497	-0.0090
	(0.1475)	(0.0140)	(0.1696)	(0.0081)	(0.1095)	(0.0200)	(0.0895)	(0.0161)
Demographic 1	-0.6287***	-0.0607***	-1.8488***	-0.0886***	-0.3486**	-0.0637**	-0.3079***	-0.0556***
	(0.2088)	(0.0200)	(0.2445)	(0.0115)	(0.1467)	(0.0267)	(0.1123)	(0.0202)
Demographic 2	0.1531	0.0148	0.3703*	0.0177*	0.1756	0.0321	0.0282	0.0051
	(0.2072)	(0.0200)	(0.1929)	(0.0091)	(0.1558)	(0.0284)	(0.1069)	(0.0193)
Advanced Math	0.2889*	0.0279*			0.3267***	0.0597***		
	(0.1612)	(0.0156)			(0.1112)	(0.0202)		
Committee Score	0.2241***	0.0216***	0.3994***	0.0191***	0.1026***	0.0187***	0.0757***	0.0137***
	(0.0413)	(0.0039)	(0.0506)	(0.0023)	(0.0287)	(0.0052)	(0.0228)	(0.0041)
Non-Economics 1			0.8376***	0.0401***			1.9483***	0.3515***
			(0.2297)	(0.0107)			(0.1236)	(0.0205)
Observations | Pseudo-R^2	5000	0.3251			5000	0.1431		
LL | Parameters | AIC	-1130.4	14	2288.8		-2691.4	14	5410.8	
Wald Statistic | DF | P-Value	3.1018	5	0.6843		3.0042	5	0.6993	

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Acceptance vs. Success AMEs

A4: Simulated Summary Statistics (Suboptimal)

Table 15a: Simulated Summary Statistics (Suboptimal)

	Mean	SD
Transition	0.3910	0.4880
Accept	0.1350	0.3417
Success	0.3040	0.4600
GRE Quantitative Percentile	84.9310	7.5286
Gender	0.3970	0.4893
Demographic 1	0.5410	0.4983
Demographic 2	0.2600	0.4386
Advanced Math	0.5300	0.4991
Committee Score	5.1470	2.0277
Year Transition More	0.5000	0.5000
Observations	1000	

▸▸ Static Unrestricted AMEs (Suboptimal)

A4: Simulated Summary Statistics (Optimal)

Table 15b: Simulated Summary Statistics (Optimal)

	Mean	SD
Transition	0.3940	0.4886
Accept	0.1400	0.3470
Success	0.3040	0.4600
GRE Quantitative Percentile	84.9310	7.5286
Gender	0.3970	0.4893
Demographic 1	0.5410	0.4983
Demographic 2	0.2600	0.4386
Advanced Math	0.5300	0.4991
Committee Score	5.1470	2.0277
Year Transition More	0.5000	0.5000
Observations	1000	

Static Unrestricted AMEs (Optimal)

A4: Simulated Correlation Matrix (Suboptimal)

Table 16a: Simulated Correlation Matrix (Suboptimal)

	Trans	Acc	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year
Transition	1									
Accept	0.4930	1								
Success	0.1164	0.0379	1							
GRE Quantitative Percentile	0.2550	0.2081	0.3225	1						
Gender	0.0200	0.0503	-0.0164	-0.0398	1					
Demographic 1	-0.0351	-0.0472	0.0023	0.2688	-0.0032	1				
Demographic 2	0.0670	0.0394	0.0642	-0.0579	-0.0476	-0.6435	1			
Advanced Math	0.0894	0.1140	0.1868	0.2799	-0.0181	0.1941	-0.0996	1		
Committee Score	0.0915	0.1662	0.1204	0.2029	0.0783	-0.0827	-0.0396	0.2372	1	
Year Transition More	0.1865	-0.0497	0.0217	-0.0410	-0.0225	-0.0341	0.0593	0.0521	0.0192	1
Observations	1000									

Static Unrestricted AMEs (Suboptimal)

A4: Simulated Correlation Matrix (Optimal)

Table 16b: Simulated Correlation Matrix (Optimal)

	Trans	Acc	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year
Transition	1									
Accept	0.5004	1								
Success	0.1300	0.0905	1							
GRE Quantitative Percentile	0.2876	0.3038	0.3225	1						
Gender	-0.0185	0.0084	-0.0164	-0.0398	1					
Demographic 1	-0.0006	0.0189	0.0023	0.2688	-0.0032	1				
Demographic 2	0.0493	0.0105	0.0642	-0.0579	-0.0476	-0.6435	1			
Advanced Math	0.0991	0.1605	0.1868	0.2799	-0.0181	0.1941	-0.0996	1		
Committee Score	0.0980	0.1271	0.1204	0.2029	0.0783	-0.0827	-0.0396	0.2372	1	
Year Transition More	0.1678	-0.0692	0.0217	-0.0410	-0.0225	-0.0341	0.0593	0.0521	0.0192	1
Observations	1000									

Dynamic Unrestricted AMEs (Optimal)

A4: Full Unrestricted Static Estimates (Suboptimal)

Table 17a: Unrestricted Static Estimates (Suboptimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-8.5777***	-1.8014***	-4.7849***	-0.9157***	-10.5425***	-1.9666***	-8.3580***	-1.7285***
	(0.9369)	(0.1610)	(1.6496)	(0.2990)	(1.0566)	(0.1529)	(1.7683)	(0.3127)
GRE Quantitative Percentile	0.0919***	0.0193***	0.0401**	0.0077**	0.1140***	0.0213***	0.0839***	0.0173***
	(0.0110)	(0.0020)	(0.0193)	(0.0036)	(0.0125)	(0.0019)	(0.0210)	(0.0039)
Gender	0.1789	0.0376	0.3374	0.0646	0.0044	0.0008	-0.0594	-0.0123
	(0.1417)	(0.0297)	(0.2315)	(0.0442)	(0.1509)	(0.0281)	(0.2300)	(0.0476)
Demographic 1	-0.4843**	-0.1017**	-0.5416	-0.1036	-0.2944	-0.0549	-0.2355	-0.0487
	(0.1988)	(0.0412)	(0.3380)	(0.0639)	(0.2115)	(0.0392)	(0.3420)	(0.0701)
Demographic 2	0.0338	0.0071	-0.0696	-0.0133	0.2505	0.0467	0.5612*	0.1161*
	(0.2101)	(0.0441)	(0.3531)	(0.0676)	(0.2214)	(0.0413)	(0.3209)	(0.0661)
Advanced Math			0.4165	0.0797*			0.6465***	0.1337***
			(0.2555)	(0.0482)			(0.2502)	(0.0504)
Committee Score			0.2219***	0.0425***			0.0059	0.0012
			(0.0668)	(0.0120)			(0.0583)	(0.0121)
Year Transition More	0.8980***	0.1886***	-1.2106***	-0.2317***				
	(0.1416)	(0.0273)	(0.2388)	(0.0397)				
Observations	1000							
LL | Parameters | AIC	-1616.1	26	3284.3					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Suboptimal)

A4: Full Unrestricted Static Estimates (Optimal)

Table 17b: Unrestricted Static Estimates (Optimal)

	Transition		Accept		Success 1		Success 2	
	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-9.1577***	-1.9164***	-9.2245***	-1.6855***	-10.5425***	-1.9666***	-8.4374***	-1.7657***
	(0.9469)	(0.1576)	(1.8167)	(0.2808)	(1.0566)	(0.1529)	(1.7772)	(0.3171)
GRE Quantitative Percentile	0.0992***	0.0208***	0.0971***	0.0177***	0.1140***	0.0213***	0.0847***	0.0177***
	(0.0111)	(0.0019)	(0.0207)	(0.0033)	(0.0125)	(0.0019)	(0.0210)	(0.0039)
Gender	0.0002	0.0000	0.1897	0.0347	0.0044	0.0008	-0.0412	-0.0086
	(0.1422)	(0.0298)	(0.2448)	(0.0448)	(0.1509)	(0.0281)	(0.2299)	(0.0481)
Demographic 1	-0.3518*	-0.0736*	-0.4248	-0.0776	-0.2944	-0.0549	-0.2785	-0.0583
	(0.2004)	(0.0416)	(0.3637)	(0.0662)	(0.2115)	(0.0392)	(0.3423)	(0.0709)
Demographic 2	0.0435	0.0091	-0.1174	-0.0215	0.2505	0.0467	0.5504*	0.1152*
	(0.2137)	(0.0447)	(0.3788)	(0.0693)	(0.2214)	(0.0413)	(0.3256)	(0.0678)
Advanced Math			0.7137***	0.1304***			0.5665**	0.1185**
			(0.2626)	(0.0463)			(0.2501)	(0.0513)
Committee Score			0.0952	0.0174			0.0237	0.0050
			(0.0628)	(0.0113)			(0.0572)	(0.0119)
Year Transition More	0.8248***	0.1726***	-1.2833***	-0.2345***				
	(0.1419)	(0.0276)	(0.2467)	(0.0378)				
Observations	1000							
LL | Parameters | AIC	-1611.7	26	3275.4					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Optimal)

A4: Full Restricted Static Estimates (Suboptimal)

Table 18a: Restricted Static Estimates (Suboptimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	0.1539	53.84%	-0.5306**	37.04%	-10.3327***	-7.2584***	
	(0.2749)	(40.49%, 66.66%)	(0.2154)	(27.83%, 47.29%)	(1.9333)	(2.2514)	
GRE Quantitative Percentile					0.1123***	0.0663**	
					(0.0227)	(0.0260)	
Gender					0.1010	0.0839	
					(0.1144)	(0.1650)	
Demographic 1					-0.4233**	-0.3725	
					(0.1783)	(0.2547)	
Demographic 2					0.1588	0.2935	
					(0.1773)	(0.2525)	
Advanced Math						0.5525***	
						(0.1957)	
Committee Score						0.1161***	
						(0.0447)	
Year Transition More	-0.9253***	31.62%	1.2103***	66.37%			
	(0.2622)	(26.62%, 37.07%)	(0.4178)	(47.08%, 81.40%)			
Sigma							0.2473***
							(0.0680)
Observations	1000						
LL | Parameters | AIC	-1625.8	17	3285.6				
LR Statistic | DF | P-Value	19.2965	9	0.0228				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Suboptimal)

A4: Full Restricted Static Estimates (Optimal)

Table 18b: Restricted Static Estimates (Optimal)

	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success 1	Success 2	Sigma
Intercept	-0.0094	49.77%	-0.5492***	36.60%	-10.0587***	-8.9338***	
	(0.2357)	(38.43%, 61.13%)	(0.1882)	(28.53%, 45.50%)	(2.2082)	(2.4363)	
GRE Quantitative Percentile					0.1090***	0.0882***	
					(0.0259)	(0.0275)	
Gender					0.0070	0.0220	
					(0.1080)	(0.1620)	
Demographic 1					-0.3346*	-0.2945	
					(0.1803)	(0.2626)	
Demographic 2					0.1444	0.2881	
					(0.1702)	(0.2608)	
Advanced Math						0.6055***	
						(0.1854)	
Committee Score						0.0657	
						(0.0419)	
Year Transition More	-0.7509***	31.86%	1.1488***	64.56%			
	(0.2200)	(27.27%, 36.83%)	(0.3847)	(47.28%, 78.72%)			
Sigma							0.2149***
							(0.0597)
Observations	1000						
LL | Parameters | AIC	-1615.9	17	3265.9				
LR Statistic | DF | P-Value	8.4729	9	0.4873				

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Optimal)

A4: Unrestricted Dynamic Estimates (Suboptimal)

Table 19a: Unrestricted Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.1586***	-1.3913***	0.5480	-8.6113***	-1.8072***	-4.6409***	-0.8906***	-8.4142***	-1.7374***
	(0.8297)	(0.1654)	(0.7283)	(0.9390)	(0.1610)	(1.6387)	(0.2989)	(1.7756)	(0.3129)
GRE Quantitative Percentile	0.0710***	0.0160***	0.0581***	0.0923***	0.0194***	0.0386**	0.0074**	0.0846***	0.0175***
	(0.0099)	(0.0020)	(0.0088)	(0.0110)	(0.0020)	(0.0191)	(0.0036)	(0.0211)	(0.0039)
Gender	-0.0381	-0.0086	0.3310***	0.1796	0.0377	0.3556	0.0682	-0.0468	-0.0097
	(0.1363)	(0.0308)	(0.1204)	(0.1418)	(0.0297)	(0.2307)	(0.0441)	(0.2303)	(0.0475)
Demographic 1	0.5198***	0.1174***	-1.2830***	-0.4878**	-0.1024**	-0.5489	-0.1053*	-0.2506	-0.0517
	(0.1800)	(0.0400)	(0.1529)	(0.1990)	(0.0412)	(0.3363)	(0.0638)	(0.3422)	(0.0700)
Demographic 2	-0.0415	-0.0094	-0.9475***	0.0313	0.0066	-0.0869	-0.0167	0.5529*	0.1142*
	(0.2000)	(0.0452)	(0.1759)	(0.2102)	(0.0441)	(0.3517)	(0.0675)	(0.3207)	(0.0659)
Advanced Math			0.8895***			0.4169	0.0800*	0.6588***	0.1360***
			(0.1239)			(0.2549)	(0.0482)	(0.2507)	(0.0504)
Committee Score						0.2201***	0.0422***	0.0048	0.0010
						(0.0666)	(0.0120)	(0.0584)	(0.0121)
Year Transition More				0.8992***	0.1887***	-1.1992***	-0.2301***		
				(0.1418)	(0.0273)	(0.2380)	(0.0398)		
$\sqrt{\text{MSE}}$			1.8844***						
			(0.0398)						
Observations	1000								
LL | Parameters | AIC	-3758.1	33	7582.1						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Suboptimal)

A4: Unrestricted Dynamic Estimates (Optimal)

Table 19b: Unrestricted Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.1391***	-1.3875***	0.5474	-9.2064***	-1.9245***	-8.8857***	-1.6359***	-8.7036***	-1.8098***
	(0.8288)	(0.1654)	(0.7284)	(0.9500)	(0.1576)	(1.7825)	(0.2813)	(1.8073)	(0.3173)
GRE Quantitative Percentile	0.0708***	0.0160***	0.0581***	0.0998***	0.0209***	0.0937***	0.0173***	0.0879***	0.0183***
	(0.0099)	(0.0020)	(0.0088)	(0.0112)	(0.0019)	(0.0204)	(0.0033)	(0.0213)	(0.0039)
Gender	-0.0384	-0.0087	0.3312***	0.0009	0.0002	0.1963	0.0361	-0.0426	-0.0089
	(0.1363)	(0.0308)	(0.1204)	(0.1424)	(0.0298)	(0.2429)	(0.0448)	(0.2313)	(0.0481)
Demographic 1	0.5189***	0.1173***	-1.2807***	-0.3536*	-0.0739*	-0.4632	-0.0853	-0.3130	-0.0651
	(0.1800)	(0.0400)	(0.1529)	(0.2007)	(0.0416)	(0.3599)	(0.0659)	(0.3443)	(0.0707)
Demographic 2	-0.0429	-0.0097	-0.9447***	0.0434	0.0091	-0.1716	-0.0316	0.5339	0.1110
	(0.1999)	(0.0452)	(0.1759)	(0.2140)	(0.0447)	(0.3749)	(0.0691)	(0.3263)	(0.0676)
Advanced Math			0.8894***			0.7280***	0.1340***	0.5854**	0.1217**
			(0.1239)			(0.2611)	(0.0463)	(0.2519)	(0.0512)
Committee Score						0.0908	0.0167	0.0227	0.0047
						(0.0623)	(0.0113)	(0.0574)	(0.0119)
Year Transition More				0.8263***	0.1727***	-1.2645***	-0.2328***		
				(0.1421)	(0.0276)	(0.2443)	(0.0379)		
$\sqrt{\text{MSE}}$			1.8844***						
			(0.0398)						
Observations	1000								
LL | Parameters | AIC	-3753.7	33	7573.4						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Optimal)

A4: Restricted Dynamic Estimates (Suboptimal)

Table 20a: Restricted Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-6.1931***	0.5166	-0.4477	0.6391	-0.6118***	35.17%	-8.4870	
	(0.8371)	(0.7436)	(0.6919)	(0.1647, 2.4802)	(0.1802)	(27.59%, 43.57%)	(12.5846)	
GRE Quantitative Percentile	0.0715***	0.0586***					0.0799	
	(0.0100)	(0.0090)					(0.1379)	
Gender	-0.0410	0.3267***					0.1457	
	(0.1361)	(0.1202)					(0.1426)	
Demographic 1	0.5215***	-1.2825***					-0.3775	
	(0.1802)	(0.1530)					(0.5271)	
Demographic 2	-0.0349	-0.9447***					0.2483	
	(0.1999)	(0.1760)					(0.5059)	
Advanced Math		0.8948***					0.4141	
		(0.1239)					(0.4431)	
Committee Score							0.1340**	
							(0.0628)	
Year Transition More			-0.0096	0.6330	0.8913	56.94%		
			(0.0626)	(0.1709, 2.3444)	(1.3618)	(9.04%, 94.62%)		
$\sqrt{\text{MSE}}$		1.8840***						
		(0.0399)						
Sigma								0.1819
								(0.2909)
Observations	1000							
LL | Parameters | AIC	-3785.0	24	7618.0					
LR Statistic | DF | P-Value	53.8131	9	0.0000					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Suboptimal)

A4: Restricted Dynamic Estimates (Optimal)

Table 20b: Restricted Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	L(Cutoff)$_1$	Cutoff 2	P(Cutoff)$_2$	Success	Sigma
Intercept	-6.1768***	0.5240	-0.5344***	0.5860	-0.5960***	35.52%	-8.7931***	
	(0.8229)	(0.7313)	(0.1440)	(0.4419, 0.7771)	(0.1541)	(28.94%, 42.70%)	(3.0648)	
GRE Quantitative Percentile	0.0713***	0.0584***					0.0876***	
	(0.0099)	(0.0089)					(0.0334)	
Gender	-0.0391	0.3305***					0.0238	
	(0.1359)	(0.1204)					(0.1132)	
Demographic 1	0.5230***	-1.2810***					-0.2855	
	(0.1803)	(0.1531)					(0.2036)	
Demographic 2	-0.0371	-0.9441***					0.1727	
	(0.1999)	(0.1760)					(0.1999)	
Advanced Math		0.8913***					0.5028**	
		(0.1239)					(0.1953)	
Committee Score							0.0659*	
							(0.0384)	
Year Transition More			0.0245	0.6006	0.7803**	54.59%		
			(0.0515)	(0.4481, 0.8049)	(0.3189)	(40.35%, 68.12%)		
$\sqrt{\text{MSE}}$		1.8849***						
		(0.0398)						
Sigma								0.1473***
								(0.0541)
Observations	1000							
LL | Parameters | AIC	-3773.5	24	7595.0					
LR Statistic | DF | P-Value	39.6494	9	0.0000					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Optimal)

A4: Unrestricted Dynamic Estimates, z Missing (Suboptimal)

Table 21a: Unrestricted Dynamic Estimates, z Missing (Suboptimal)

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-5.1788***	-1.7932***	-2.9927*	-1.0548*	2.7507	-8.5772***	-1.8014***	-4.8978***	-0.9362***	-8.2901***	-1.7168***
	(0.5553)	(0.1624)	(1.7053)	(0.5911)	(2.3336)	(0.9369)	(0.1610)	(1.6572)	(0.2992)	(1.7618)	(0.3127)
GRE Quantitative Percentile	0.0554***	0.0192***	0.0409**	0.0144**	0.0355	0.0918***	0.0193***	0.0415**	0.0079**	0.0832***	0.0172***
	(0.0066)	(0.0020)	(0.0167)	(0.0057)	(0.0226)	(0.0110)	(0.0020)	(0.0193)	(0.0036)	(0.0209)	(0.0039)
Gender	0.1094	0.0379	-0.1976	-0.0696	0.1937	0.1789	0.0376	0.3418	0.0653	-0.0589	-0.0122
	(0.0858)	(0.0296)	(0.1380)	(0.0482)	(0.1947)	(0.1417)	(0.0297)	(0.2318)	(0.0441)	(0.2296)	(0.0475)
Demographic 1	-0.2859**	-0.0990**	0.2982	0.1051	-1.1999***	-0.4842**	-0.1017**	-0.5561	-0.1063*	-0.2415	-0.0500
	(0.1186)	(0.0406)	(0.1976)	(0.0691)	(0.2734)	(0.1988)	(0.0412)	(0.3383)	(0.0638)	(0.3411)	(0.0700)
Demographic 2	0.0241	0.0083	-0.1464	-0.0516	-0.9202***	0.0339	0.0071	-0.0826	-0.0158	0.5526*	0.1144*
	(0.1266)	(0.0438)	(0.2000)	(0.0703)	(0.2915)	(0.2101)	(0.0441)	(0.3531)	(0.0675)	(0.3201)	(0.0660)
Advanced Math					0.8786***			0.4132	0.0790	0.6481***	0.1342***
					(0.2057)			(0.2558)	(0.0482)	(0.2499)	(0.0504)
Committee Score								0.2220***	0.0424***	0.0053	0.0011
								(0.0668)	(0.0120)	(0.0582)	(0.0121)
Year Transition More	0.5446***	0.1886***				0.8980***	0.1886***	-1.2072***	-0.2307***		
	(0.0850)	(0.0274)				(0.1416)	(0.0273)	(0.2391)	(0.0397)		
Inverse Mills			-0.4189	-0.1476	-0.1894						
			(0.3872)	(0.1361)	(0.5612)						
$\sqrt{\text{MSE}}$					1.8925***						
					(0.0662)						
Observations	1000										
LL | Parameters | AIC	-2715.9	41	5513.9								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Suboptimal)

A4: Unrestricted Dynamic Estimates, z Missing (Optimal)

Table 21b: Unrestricted Dynamic Estimates, z Missing (Optimal)

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-5.5333***	-1.9103***	-2.6740	-0.9412	2.9426	-9.1587***	-1.9166***	-9.4836***	-1.7254***	-8.2448***	-1.7310***
	(0.5668)	(0.1614)	(1.9361)	(0.6738)	(2.6765)	(0.9470)	(0.1576)	(1.8420)	(0.2807)	(1.7584)	(0.3170)
GRE Quantitative Percentile	0.0599***	0.0207***	0.0373**	0.0131**	0.0342	0.0992***	0.0208***	0.0998***	0.0182***	0.0824***	0.0173***
	(0.0067)	(0.0020)	(0.0188)	(0.0065)	(0.0258)	(0.0111)	(0.0019)	(0.0210)	(0.0033)	(0.0208)	(0.0039)
Gender	0.0025	0.0009	-0.1822	-0.0641	0.1863	0.0002	0.0000	0.1969	0.0358	-0.0434	-0.0091
	(0.0860)	(0.0297)	(0.1380)	(0.0482)	(0.1918)	(0.1422)	(0.0298)	(0.2459)	(0.0448)	(0.2292)	(0.0481)
Demographic 1	-0.2056*	-0.0710*	0.2973	0.1047	-1.2445***	-0.3519*	-0.0736*	-0.4210	-0.0766	-0.2654	-0.0557
	(0.1193)	(0.0409)	(0.1954)	(0.0683)	(0.2697)	(0.2004)	(0.0416)	(0.3655)	(0.0662)	(0.3411)	(0.0709)
Demographic 2	0.0312	0.0108	-0.1242	-0.0437	-1.0245***	0.0433	0.0091	-0.1066	-0.0194	0.5579*	0.1171*
	(0.1280)	(0.0442)	(0.2051)	(0.0721)	(0.2997)	(0.2137)	(0.0447)	(0.3807)	(0.0693)	(0.3250)	(0.0679)
Advanced Math					0.8964***			0.7077***	0.1288***	0.5718**	0.1200**
					(0.2088)			(0.2634)	(0.0463)	(0.2495)	(0.0513)
Committee Score								0.0968	0.0176	0.0233	0.0049
								(0.0631)	(0.0113)	(0.0570)	(0.0120)
Year Transition More	0.4989***	0.1722***				0.8249***	0.1726***	-1.2813***	-0.2331***		
	(0.0852)	(0.0277)				(0.1419)	(0.0276)	(0.2479)	(0.0378)		
Inverse Mills			-0.4587	-0.1615	-0.2035						
			(0.4255)	(0.1495)	(0.6179)						
$\sqrt{\text{MSE}}$					1.9059***						
					(0.0667)						
Observations	1000										
LL | Parameters | AIC	-2720.3	41	5522.6								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

⏩ Unrestricted Static AMEs (Optimal)

A4: Restricted Dynamic Estimates, z Missing (Suboptimal)

Table 22a: Restricted Dynamic Estimates, z Missing (Suboptimal)

	Selection	Advanced Math	Committee Score	Cutoff 1	L(Cutoff)$_1$	Cutoff 2	P(Cutoff)$_2$	Success	Sigma
Intercept	-5.1849***	-3.0077*	2.7390	-0.4529	0.6358	-0.6156***	35.08%	-8.3629	
	(0.5600)	(1.7125)	(2.3203)	(0.6380)	(0.1821, 2.2201)	(0.1808)	(27.49%, 43.51%)	(11.5276)	
GRE Quantitative Percentile	0.0554***	0.0418**	0.0363					0.0786	
	(0.0066)	(0.0168)	(0.0226)					(0.1266)	
Gender	0.1106	-0.1963	0.1960					0.1511	
	(0.0858)	(0.1363)	(0.1938)					(0.1374)	
Demographic 1	-0.2846**	0.2917	-1.2040***					-0.3901	
	(0.1188)	(0.1960)	(0.2717)					(0.5285)	
Demographic 2	0.0225	-0.1595	-0.9260***					0.2394	
	(0.1268)	(0.1973)	(0.2899)					(0.4445)	
Advanced Math			0.8890***					0.4501	
			(0.2057)					(0.4458)	
Committee Score								0.1300**	
								(0.0579)	
Year Transition More	0.5474***			-0.0042	0.6331	0.8901	56.82%		
	(0.0861)			(0.0616)	(0.1890, 2.1209)	(1.2635)	(10.79%, 93.47%)		
Inverse Mills		-0.4754	-0.2427						
		(0.3913)	(0.5594)						
$\sqrt{\text{MSE}}$			1.8935***						
			(0.0666)						
Sigma									0.1805
									(0.2674)
Observations	1000								
LL | Parameters | AIC	-2742.5	32	5549.0						
LR Statistic | DF | P-Value	53.1755	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Suboptimal)

A4: Restricted Dynamic Estimates, z Missing (Optimal)

Table 22b: Restricted Dynamic Estimates, z Missing (Optimal)

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-5.5371***	-2.6870	2.9441	-0.5459***	0.5793	-0.5863***	35.75%	-8.3634***	
	(0.5732)	(1.9557)	(2.6682)	(0.1161)	(0.4615, 0.7273)	(0.1500)	(29.31%, 42.74%)	(2.3193)	
GRE Quantitative Percentile	0.0599***	0.0382**	0.0345					0.0826***	
	(0.0068)	(0.0190)	(0.0257)					(0.0250)	
Gender	0.0032	-0.1802	0.1850					0.0447	
	(0.0860)	(0.1357)	(0.1912)					(0.1096)	
Demographic 1	-0.2061*	0.2902	-1.2455***					-0.2408	
	(0.1194)	(0.1936)	(0.2693)					(0.1838)	
Demographic 2	0.0295	-0.1360	-1.0281***					0.1880	
	(0.1281)	(0.2022)	(0.2992)					(0.1935)	
Advanced Math			0.9026***					0.4942***	
			(0.2088)					(0.1764)	
Committee Score								0.0643*	
								(0.0372)	
Year Transition More	0.4995***			0.0281	0.5958	0.7546***	54.20%		
	(0.0857)			(0.0503)	(0.4697, 0.7558)	(0.2627)	(42.41%, 65.54%)		
Inverse Mills		-0.5095	-0.2301						
		(0.4343)	(0.6188)						
$\sqrt{\text{MSE}}$			1.9062***						
			(0.0667)						
Sigma									0.1418***
									(0.0414)
Observations	1000								
LL | Parameters | AIC	-2739.7	32	5543.4						
LR Statistic | DF | P-Value	38.8065	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Optimal)

A4: Unrestricted Myopic Estimates (Suboptimal)

Table 23a: Unrestricted Myopic Estimates (Suboptimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1201*	0.5300***	4.6362***	-8.5778***	-1.8014***	-4.7827***	-0.9153***	-8.3592***	-1.7287***
	(0.0634)	(0.0158)	(0.0879)	(0.9369)	(0.1610)	(1.6494)	(0.2990)	(1.7684)	(0.3127)
GRE Quantitative Percentile				0.0919***	0.0193***	0.0401**	0.0077**	0.0839***	0.0174***
				(0.0110)	(0.0020)	(0.0193)	(0.0036)	(0.0210)	(0.0039)
Gender				0.1788	0.0376	0.3374	0.0646	-0.0594	-0.0123
				(0.1417)	(0.0297)	(0.2315)	(0.0442)	(0.2300)	(0.0476)
Demographic 1				-0.4846**	-0.1018**	-0.5415	-0.1036	-0.2355	-0.0487
				(0.1988)	(0.0412)	(0.3380)	(0.0639)	(0.3420)	(0.0701)
Demographic 2				0.0335	0.0070	-0.0695	-0.0133	0.5612*	0.1161*
				(0.2101)	(0.0441)	(0.3531)	(0.0676)	(0.3209)	(0.0661)
Advanced Math			0.9638***			0.4166	0.0797*	0.6464***	0.1337***
			(0.1243)			(0.2555)	(0.0482)	(0.2502)	(0.0504)
Committee Score						0.2219***	0.0425***	0.0059	0.0012
						(0.0668)	(0.0120)	(0.0583)	(0.0121)
Year Transition More				0.8980***	0.1886***	-1.2106***	-0.2317***		
				(0.1416)	(0.0273)	(0.2388)	(0.0397)		
$\sqrt{\text{MSE}}$			1.9698***						
			(0.0413)						
Observations	1000								
LL | Parameters | AIC	-3851.0	25	7752.1						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Suboptimal)

A4: Unrestricted Myopic Estimates (Optimal)

Table 23b: Unrestricted Myopic Estimates (Optimal)

	Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.1201*	0.5300***	4.6362***	-9.1582***	-1.9165***	-9.2354***	-1.6874***	-8.4285***	-1.7639***
	(0.0634)	(0.0158)	(0.0879)	(0.9469)	(0.1576)	(1.8175)	(0.2808)	(1.7765)	(0.3171)
GRE Quantitative Percentile				0.0992***	0.0208***	0.0972***	0.0178***	0.0846***	0.0177***
				(0.0111)	(0.0019)	(0.0207)	(0.0033)	(0.0210)	(0.0039)
Gender				0.0002	0.0000	0.1906	0.0348	-0.0419	-0.0088
				(0.1422)	(0.0298)	(0.2448)	(0.0448)	(0.2299)	(0.0481)
Demographic 1				-0.3522*	-0.0737*	-0.4229	-0.0773	-0.2773	-0.0580
				(0.2004)	(0.0416)	(0.3637)	(0.0662)	(0.3423)	(0.0709)
Demographic 2				0.0431	0.0090	-0.1152	-0.0211	0.5521*	0.1155*
				(0.2137)	(0.0447)	(0.3789)	(0.0693)	(0.3257)	(0.0678)
Advanced Math			0.9638***			0.7138***	0.1304***	0.5680**	0.1189**
			(0.1243)			(0.2625)	(0.0463)	(0.2501)	(0.0513)
Committee Score						0.0953	0.0174	0.0237	0.0050
						(0.0628)	(0.0113)	(0.0572)	(0.0119)
Year Transition More				0.8248***	0.1726***	-1.2820***	-0.2342***		
				(0.1419)	(0.0276)	(0.2467)	(0.0378)		
$\sqrt{\text{MSE}}$			1.9698***						
			(0.0413)						
Observations	1000								
LL | Parameters | AIC	-3846.6	25	7743.2						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Optimal)

A4: Restricted Myopic Estimates (Suboptimal)

Table 24a: Restricted Myopic Estimates (Suboptimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1291**	4.6418***	-0.4609	0.6307	-0.6100***	35.21%	-8.4650	
	(0.0636)	(0.0884)	(0.6357)	(0.1814, 2.1926)	(0.1827)	(32.88%, 37.60%)	(10.9416)	
GRE Quantitative Percentile							0.0819	
							(0.1199)	
Gender							0.1398	
							(0.1284)	
Demographic 1							-0.4103	
							(0.4518)	
Demographic 2							0.2056	
							(0.4868)	
Advanced Math		0.9652***					0.3557	
		(0.1243)					(0.4351)	
Committee Score							0.1123**	
							(0.0529)	
Year Transition More			-0.0046	0.6279	0.8797	56.70%		
			(0.0615)	(0.1876, 2.1019)	(1.2631)	(10.93%, 93.32%)		
$\sqrt{\text{MSE}}$		1.9675***						
		(0.0412)						
Sigma								0.1803
								(0.2674)
Observations	1000							
LL | Parameters | AIC	-3881.2	16	7794.4					
LR Statistic | DF | P-Value	60.3013	9	0.0000					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Suboptimal)

A4: Restricted Myopic Estimates (Optimal)

Table 24b: Restricted Myopic Estimates (Optimal)

	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	0.1317**	4.6389***	-0.5593***	0.5716	-0.5909***	35.64%	-8.5595***	
	(0.0635)	(0.0879)	(0.1114)	(0.4595, 0.7111)	(0.1494)	(34.08%, 37.24%)	(2.1793)	
GRE Quantitative Percentile							0.0869***	
							(0.0239)	
Gender							0.0343	
							(0.1085)	
Demographic 1							-0.2998*	
							(0.1787)	
Demographic 2							0.1207	
							(0.1762)	
Advanced Math		0.9645***					0.4241**	
		(0.1244)					(0.1693)	
Committee Score							0.0498	
							(0.0351)	
Year Transition More			0.0304	0.5892	0.7487***	53.94%		
			(0.0510)	(0.4695, 0.7394)	(0.2510)	(42.65%, 64.83%)		
$\sqrt{\text{MSE}}$		1.9718***						
		(0.0414)						
Sigma								0.1403***
								(0.0391)
Observations	1000							
LL | Parameters | AIC	-3869.1	16	7770.3					
LR Statistic | DF | P-Value	45.0826	9	0.0000					

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Optimal)

Table 25a: Unrestricted Myopic Estimates, z Missing (Suboptimal)

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.5244***	-0.1957***	0.3641	0.1419	4.8485***	-8.5775***	-1.8014***	-4.7843***	-0.9156***	-8.3574***	-1.7284***
	(0.0589)	(0.0192)	(0.3839)	(0.1490)	(0.5941)	(0.9369)	(0.1610)	(1.6495)	(0.2990)	(1.7682)	(0.3127)
GRE Quantitative Percentile						0.0919***	0.0193***	0.0401**	0.0077**	0.0839***	0.0173***
						(0.0110)	(0.0020)	(0.0193)	(0.0036)	(0.0210)	(0.0039)
Gender						0.1788	0.0376	0.3374	0.0646	-0.0592	-0.0122
						(0.1417)	(0.0297)	(0.2315)	(0.0442)	(0.2300)	(0.0476)
Demographic 1						-0.4842**	-0.1017**	-0.5415	-0.1036	-0.2351	-0.0486
						(0.1988)	(0.0412)	(0.3380)	(0.0639)	(0.3420)	(0.0701)
Demographic 2						0.0339	0.0071	-0.0695	-0.0133	0.5614*	0.1161*
						(0.2101)	(0.0441)	(0.3531)	(0.0676)	(0.3210)	(0.0661)
Advanced Math					0.9097***			0.4165	0.0797*	0.6464***	0.1337***
					(0.1967)			(0.2555)	(0.0482)	(0.2502)	(0.0504)
Committee Score								0.2219***	0.0425***	0.0059	0.0012
								(0.0668)	(0.0120)	(0.0583)	(0.0121)
Year Transition More	0.4793***	0.1788***				0.8980***	0.1886***	-1.2106***	-0.2317***		
	(0.0814)	(0.0287)				(0.1416)	(0.0273)	(0.2388)	(0.0397)		
Inverse Mills			-0.1547	-0.0603	-0.0030						
			(0.3964)	(0.1543)	(0.6083)						
$\sqrt{\text{MSE}}$					1.9491***						
					(0.0676)						
Observations	1000										
LL | Parameters | AIC	-2795.4	29	5648.9								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Suboptimal)

A4: Unrestricted Myopic Estimates, z Missing (Optimal)

Table 25b: Unrestricted Myopic Estimates, z Missing (Optimal)

	Selection		Advanced Math		Committee Score	Transition		Accept		Success	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.4902***	-0.1843***	0.4285	0.1664	4.9569***	-9.1577***	-1.9164***	-9.2245***	-1.6855***	-8.4374***	-1.7657***
	(0.0586)	(0.0195)	(0.4250)	(0.1643)	(0.6578)	(0.9469)	(0.1576)	(1.8167)	(0.2808)	(1.7772)	(0.3171)
GRE Quantitative Percentile						0.0992***	0.0208***	0.0971***	0.0177***	0.0847***	0.0177***
						(0.0111)	(0.0019)	(0.0207)	(0.0033)	(0.0210)	(0.0039)
Gender						0.0002	0.0000	0.1897	0.0347	-0.0412	-0.0086
						(0.1422)	(0.0298)	(0.2448)	(0.0448)	(0.2299)	(0.0481)
Demographic 1						-0.3519*	-0.0736*	-0.4248	-0.0776	-0.2785	-0.0583
						(0.2004)	(0.0416)	(0.3637)	(0.0662)	(0.3423)	(0.0709)
Demographic 2						0.0434	0.0091	-0.1174	-0.0214	0.5504*	0.1152*
						(0.2137)	(0.0447)	(0.3788)	(0.0693)	(0.3256)	(0.0678)
Advanced Math					0.9270***			0.7136***	0.1304***	0.5665**	0.1185**
					(0.1981)			(0.2626)	(0.0463)	(0.2501)	(0.0513)
Committee Score								0.0952	0.0174	0.0237	0.0050
								(0.0628)	(0.0113)	(0.0572)	(0.0119)
Year Transition More	0.4300***	0.1617***				0.8248***	0.1726***	-1.2833***	-0.2345***		
	(0.0811)	(0.0292)				(0.1419)	(0.0276)	(0.2467)	(0.0378)		
Inverse Mills			-0.2068	-0.0803	-0.1171						
			(0.4399)	(0.1706)	(0.6787)						
$\sqrt{\text{MSE}}$					1.9628***						
					(0.0678)						
Observations	1000										
LL | Parameters | AIC	-2805.9	29	5669.8								

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Static AMEs (Optimal)

A4: Restricted Myopic Estimates, z Missing (Suboptimal)

Table 26a: Restricted Myopic Estimates, z Missing (Suboptimal)

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-0.5219***	0.3660	4.8574***	-0.4584	0.6323	-0.6106***	35.19%	-8.4597	
	(0.0589)	(0.3889)	(0.6005)	(0.4992)	(0.2377, 1.6820)	(0.1783)	(27.68%, 43.51%)	(8.6873)	
GRE Quantitative Percentile								0.0814	
								(0.0950)	
Gender								0.1396	
								(0.1268)	
Demographic 1								-0.4047	
								(0.3758)	
Demographic 2								0.2066	
								(0.3980)	
Advanced Math			0.9145***					0.3654	
			(0.1964)					(0.3542)	
Committee Score								0.1172**	
								(0.0505)	
Year Transition More	0.4753***			-0.0079	0.6274	0.8704	56.46%		
	(0.0813)			(0.0590)	(0.2422, 1.6247)	(0.9919)	(16.77%, 89.30%)		
Inverse Mills		-0.1441	0.0077						
		(0.4020)	(0.6141)						
$\sqrt{\text{MSE}}$			1.9501***						
			(0.0677)						
Sigma									0.1785
									(0.2089)
Observations	1000								
LL | Parameters | AIC	-2824.9	20	5689.7						
LR Statistic | DF | P-Value	58.8213	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Suboptimal)

Table 26b: Restricted Myopic Estimates, z Missing (Optimal)

	Selection	Advanced Math	Committee Score	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 2	$P(\text{Cutoff})_2$	Success	Sigma
Intercept	-0.4903***	0.4347	4.9623***	-0.5519***	0.5758	-0.5852***	35.77%	-8.5581***	
	(0.0585)	(0.4277)	(0.6606)	(0.1068)	(0.4671, 0.7099)	(0.1489)	(29.38%, 42.72%)	(2.0776)	
GRE Quantitative Percentile								0.0861***	
								(0.0226)	
Gender								0.0384	
								(0.1075)	
Demographic 1								-0.2591	
								(0.1750)	
Demographic 2								0.1470	
								(0.1796)	
Advanced Math			0.9310***					0.4198**	
			(0.1981)					(0.1642)	
Committee Score								0.0566	
								(0.0356)	
Year Transition More	0.4302***			0.0282	0.5923	0.7481***	54.06%		
	(0.0810)			(0.0501)	(0.4762, 0.7368)	(0.2432)	(43.09%, 64.65%)		
Inverse Mills		-0.1978	-0.1119						
		(0.4424)	(0.6804)						
$\sqrt{\text{MSE}}$			1.9632***						
			(0.0678)						
Sigma									0.1401***
									(0.0370)
Observations	1000								
LL | Parameters | AIC	-2827.7	20	5695.3						
LR Statistic | DF | P-Value	43.5305	9	0.0000						

Robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Static Estimates (Optimal)

A4: Waitlist Summary Statistics (Suboptimal)

Table 27a: Waitlist Summary Statistics (Suboptimal)

	Mean	SD
Transition	0.3700	0.4828
Waitlist	0.3120	0.4633
Accept	0.1510	0.3580
Matriculate	0.0310	0.1733
Success	0.3100	0.4625
GRE Quantitative Percentile	84.9310	7.5286
Gender	0.3970	0.4893
Demographic 1	0.5500	0.4975
Demographic 2	0.2620	0.4397
Advanced Math	0.5300	0.4991
Committee Score	5.1470	2.0277
Year Transition/Matriculate More	0.5000	0.5000
Observations	1000	

▸▸ Waitlist-Hybrid Unrestricted AMEs (Suboptimal)

A4: Waitlist Summary Statistics (Optimal)

Table 27b: Waitlist Summary Statistics (Optimal)

	Mean	SD
Transition	0.3650	0.4814
Waitlist	0.3240	0.4680
Accept	0.1500	0.3571
Matriculate	0.0390	0.1936
Success	0.3150	0.4645
GRE Quantitative Percentile	84.9310	7.5286
Gender	0.3970	0.4893
Demographic 1	0.5500	0.4975
Demographic 2	0.2620	0.4397
Advanced Math	0.5300	0.4991
Committee Score	5.1470	2.0277
Year Transition/Matriculate More	0.5000	0.5000
Observations	1000	

Waitlist-Hybrid Unrestricted AMEs (Optimal)

A4: Waitlist Correlation Matrix (Suboptimal)

Table 28a: Waitlist Correlation Matrix (Suboptimal)

	Trans	Wait	Acc	Mat	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year
Transition	1											
Waitlist	0.8787	1										
Accept	0.5503	0.2766	1									
Matriculate	0.2334	-0.0084	0.4241	1								
Success	0.0685	0.0760	-0.0049	-0.0700	1							
GRE Quantitative Percentile	0.3041	0.2745	0.2057	0.0675	0.2520	1						
Gender	0.0428	0.0138	0.0288	-0.0390	-0.0533	-0.0398	1					
Demographic 1	-0.0146	0.0017	-0.0340	0.0574	0.0413	0.2483	0.0068	1				
Demographic 2	0.0804	0.0553	0.0472	-0.0016	-0.0208	-0.0918	-0.0605	-0.6587	1			
Advanced Math	0.0660	0.0417	0.0502	0.0413	0.1243	0.2799	-0.0181	0.1953	-0.1042	1		
Committee Score	0.0364	0.0268	0.0493	-0.1154	0.1146	0.2029	0.0783	-0.0960	-0.0869	0.2372	1	
Year Transition/Matriculate More	0.1947	0.2029	-0.1480	0.0058	0.0000	-0.0410	-0.0225	-0.0241	0.0318	0.0521	0.0192	1
Observations	1000											

⏩ Waitlist-Hybrid Unrestricted AMEs (Suboptimal)

A4: Waitlist Correlation Matrix (Optimal)

Table 28b: Waitlist Correlation Matrix (Optimal)

	Trans	Wait	Acc	Mat	Suc	GRE	Gen	Dem1	Dem2	Math	Com	Year
Transition	1											
Waitlist	0.9131	1										
Accept	0.5541	0.3614	1									
Matriculate	0.2657	0.0923	0.4796	1								
Success	0.0448	0.0135	0.0528	0.0302	1							
GRE Quantitative Percentile	0.2503	0.1908	0.2389	0.0801	0.3210	1						
Gender	-0.0038	0.0191	-0.0489	-0.0262	-0.0310	-0.0398	1					
Demographic 1	0.0136	0.0034	0.0478	0.0161	-0.0444	0.2483	0.0068	1				
Demographic 2	0.0065	0.0005	-0.0146	0.0327	0.0513	-0.0918	-0.0605	-0.6587	1			
Advanced Math	0.0647	0.0397	0.0645	0.0034	0.0994	0.2799	-0.0181	0.1953	-0.1042	1		
Committee Score	0.0813	0.0710	0.0704	-0.1063	0.1557	0.2029	0.0783	-0.0960	-0.0869	0.2372	1	
Year Transition/Matriculate More	0.1641	0.1624	-0.1736	0.0052	-0.0624	-0.0410	-0.0225	-0.0241	0.0318	0.0521	0.0192	1
Observations	1000											

Waitlist-Hybrid Unrestricted AMEs (Optimal)

A4: Full Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

Table 29a: Unrestricted Waitlist-Hybrid Estimates (Suboptimal)

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-0.1272	-10.7137***	-2.1064***	1.2719	0.1632	2.9322	0.3823	-7.5200***	-1.6199***	-7.8289***	-1.6641***	-0.0895	-0.0123
	(0.1075)	(1.0455)	(0.1549)	(2.3247)	(0.2970)	(3.3958)	(0.4416)	(0.9558)	(0.1909)	(1.7347)	(0.3277)	(4.9230)	(0.6751)
GRE Quantitative Percentile		0.1111***	0.0218***	-0.0210	-0.0027	0.0300	0.0039	0.0810***	0.0175***	0.0802***	0.0170***	0.0253	0.0035
		(0.0119)	(0.0019)	(0.0253)	(0.0032)	(0.0272)	(0.0035)	(0.0113)	(0.0023)	(0.0206)	(0.0040)	(0.0616)	(0.0083)
Gender		0.3462**	0.0681**	0.3959	0.0508	-0.7582**	-0.0989**	-0.2071	-0.0446	-0.0143	-0.0030	-0.3825	-0.0525
		(0.1473)	(0.0286)	(0.2998)	(0.0387)	(0.3460)	(0.0441)	(0.1473)	(0.0314)	(0.2373)	(0.0504)	(0.7540)	(0.1051)
Demographic 1		-0.2186	-0.0430	-0.1573	-0.0202	-1.1218**	-0.1463**	-0.2002	-0.0431	-0.1192	-0.0253	1.3825	0.1897
		(0.2038)	(0.0400)	(0.4585)	(0.0589)	(0.5566)	(0.0710)	(0.1986)	(0.0427)	(0.3732)	(0.0792)	(1.1353)	(0.1567)
Demographic 2		0.4335**	0.0852**	0.2759	0.0354	-1.0373**	-0.1352**	-0.1484	-0.0320	-0.2127	-0.0452	0.6914	0.0949
		(0.2208)	(0.0431)	(0.4717)	(0.0604)	(0.5093)	(0.0662)	(0.2220)	(0.0477)	(0.3782)	(0.0803)	(1.1838)	(0.1640)
Advanced Math				0.4859	0.0623	0.0426	0.0056			0.1456	0.0310	0.9409	0.1291
				(0.3240)	(0.0414)	(0.3589)	(0.0467)			(0.2555)	(0.0543)	(0.8525)	(0.1095)
Committee Score				0.0074	0.0009	0.1281	0.0167			0.0319	0.0068	-0.9581***	-0.1315***
				(0.0681)	(0.0087)	(0.0789)	(0.0102)			(0.0577)	(0.0122)	(0.2989)	(0.0272)
Year Transition/Matriculate More		1.0063***	0.1979***									1.4252*	0.1955*
		(0.1492)	(0.0265)									(0.8409)	(0.1053)
$\ddot{m}_2$	1.2940***			-4.0419	-0.5186								
	(0.2636)			(2.9946)	(0.3836)								
$\overline{m_{2A}}$						-15.3294**	-1.9986**						
						(7.5904)	(0.9596)						
$\sqrt{\text{MSE}}$	0.0023												
	(0.0062)												
Observations	1000												
LL | Parameters | AIC	58.6	37	-43.2										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

A4: Full Unrestricted Waitlist-Hybrid Estimates (Optimal)

Table 29b: Unrestricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success 1		Success 3		Matriculate	
	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	0.0760	-8.0751***	-1.6888***	-8.1102***	-0.7772***	-2.5555	-0.3393	-10.5564***	-2.1616***	-9.8893***	-1.9466***	1.5968	0.2120
	(0.1019)	(0.9328)	(0.1635)	(2.6400)	(0.2379)	(2.4701)	(0.3250)	(1.0083)	(0.1698)	(1.7557)	(0.2871)	(9.6640)	(1.2760)
GRE Quantitative Percentile		0.0849***	0.0177***	0.0691**	0.0066**	0.0616**	0.0082**	0.1188***	0.0243***	0.0974***	0.0192***	-0.0715	-0.0095
		(0.0109)	(0.0020)	(0.0312)	(0.0029)	(0.0269)	(0.0034)	(0.0119)	(0.0020)	(0.0203)	(0.0035)	(0.1162)	(0.0149)
Gender		0.0593	0.0124	-0.5322	-0.0510	-0.1302	-0.0173	-0.0708	-0.0145	-0.0561	-0.0111	2.9518*	0.3919**
		(0.1415)	(0.0296)	(0.3773)	(0.0363)	(0.3031)	(0.0403)	(0.1494)	(0.0306)	(0.2465)	(0.0486)	(1.5616)	(0.1629)
Demographic 1		-0.2872	-0.0601	0.2441	0.0234	0.4601	0.0611	-0.7174***	-0.1469***	-0.4344	-0.0855	3.4735**	0.4612**
		(0.1955)	(0.0407)	(0.6404)	(0.0616)	(0.4962)	(0.0654)	(0.2061)	(0.0414)	(0.3668)	(0.0718)	(1.6588)	(0.1801)
Demographic 2		-0.0789	-0.0165	0.4959	0.0475	0.1979	0.0263	-0.0743	-0.0152	0.5744	0.1131	5.5471**	0.7364***
		(0.2150)	(0.0450)	(0.6305)	(0.0605)	(0.5096)	(0.0675)	(0.2199)	(0.0450)	(0.3821)	(0.0743)	(2.2266)	(0.2082)
Advanced Math				0.2685	0.0257	-0.0613	-0.0081			-0.1088	-0.0214	4.2756***	0.5676***
				(0.3837)	(0.0367)	(0.3647)	(0.0484)			(0.2639)	(0.0520)	(1.6021)	(0.1723)
Committee Score				-0.0174	-0.0017	0.0656	0.0087			0.1487**	0.0293**	-0.8898***	-0.1181***
				(0.0989)	(0.0095)	(0.0890)	(0.0118)			(0.0684)	(0.0133)	(0.2508)	(0.0236)
Year Transition/Matriculate More		0.8006***	0.1674***									3.3305***	0.4422***
		(0.1410)	(0.0277)									(1.2386)	(0.1208)
$\ddot{m}_2$	0.8325***			-0.6135	-0.0588								
	(0.2037)			(0.8874)	(0.0841)								
$\overline{m_{2A}}$						-9.4220***	-1.2509***						
						(3.5284)	(0.4445)						
$\sqrt{\text{MSE}}$	0.0015												
	(0.0022)												
Observations	1000												
LL | Parameters | AIC	216.3	37	-358.6										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Optimal)

A4: Full Restricted Waitlist-Hybrid Estimates (Suboptimal)

Table 30a: Restricted Waitlist-Hybrid Estimates (Suboptimal)

	$\overline{m_{2A}}$	Cutoff 1	$P(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Intercept	-0.1272	-0.5708***	36.10%	-8.2311*	22.49%	-8.9130***	-4.6536**	-0.0895		
	(0.1075)	(0.1028)	(31.60%, 40.87%)	(4.3333)	(9.89%, 43.40%)	(1.2161)	(1.8940)	(4.9230)		
GRE Quantitative Percentile						0.0954***	0.0461**	0.0253		
						(0.0145)	(0.0215)	(0.0616)		
Gender						0.0556	-0.1249	-0.3825		
						(0.1153)	(0.1735)	(0.7540)		
Demographic 1						-0.1889	-0.4255*	1.3825		
						(0.1565)	(0.2577)	(1.1353)		
Demographic 2						0.1535	-0.3378	0.6914		
						(0.1686)	(0.2665)	(1.1838)		
Advanced Math							0.2123	0.9409		
							(0.1808)	(0.8525)		
Committee Score							0.0564	-0.9581***		
							(0.0427)	(0.2989)		
Year Transition/Matriculate More		-0.4893***	25.73%		98.22%			1.4252*		
		(0.1071)	(23.09%, 28.56%)		(28.45%, 99.99%)			(0.8409)		
$\ddot{m}_2$	1.2940***									
	(0.2636)									
$\overline{m_{2A}}$				25.3524*						
				(14.1439)						
$\sqrt{\text{MSE}}$	0.0023									
	(0.0062)									
Beta									1.0000***	
									(0.0000)	
Sigma										0.2317***
										(0.0530)
Observations	1000									
LL | Parameters | AIC	32.3	21	-22.7							
LR Statistic | DF | P-Value	52.4857	16	0.0000							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Waitlist-Hybrid Estimates (Suboptimal)

A4: Full Restricted Waitlist-Hybrid Estimates (Optimal)

Table 30b: Restricted Waitlist-Hybrid Estimates (Optimal)

	$\overline{m_{2A}}$	Cutoff 1	P(Cutoff)$_1$	Cutoff 3	P(Cutoff)$_3$	Success 1	Success 3	Matriculate	Beta	Sigma
Intercept	0.0760	-0.4805***	38.21%	-17.7954**	6.51%	-10.8672***	-9.9856***	1.5968		
	(0.1019)	(0.0672)	(35.16%, 41.37%)	(8.2222)	(0.67%, 41.91%)	(0.9634)	(1.6344)	(9.6640)		
GRE Quantitative Percentile						0.1217***	0.0996***	-0.0715		
						(0.0112)	(0.0187)	(0.1162)		
Gender						-0.0289	-0.1703	2.9518*		
						(0.1212)	(0.2000)	(1.5616)		
Demographic 1						-0.6107***	-0.2214	3.4735**		
						(0.1657)	(0.3133)	(1.6588)		
Demographic 2						-0.0829	0.4767	5.5471**		
						(0.1796)	(0.2952)	(2.2266)		
Advanced Math							-0.0150	4.2756***		
							(0.2301)	(1.6021)		
Committee Score							0.1114**	-0.8898***		
							(0.0557)	(0.2508)		
Year Transition/Matriculate More		-0.4876***	27.53%		100.00%			3.3305***		
		(0.0944)	(24.44%, 30.84%)		(29.39%, 100.00%)			(1.2386)		
$\ddot{m}_2$	0.8325***									
	(0.2037)									
$\overline{m_{2A}}$				57.4973**						
				(28.4481)						
$\sqrt{\text{MSE}}$	0.0015									
	(0.0022)									
Beta									0.9991***	
									(0.0023)	
Sigma										0.3013***
										(0.0253)
Observations	1000									
LL | Parameters | AIC	209.6	21	-377.2							
LR Statistic | DF | P-Value	13.3895	16	0.6441							

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

⏩ Restricted Waitlist-Hybrid Estimates (Optimal)

A4: Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

Table 31a: Unrestricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math		Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.5431***	-1.4659***	0.9467	0.5784	-0.1272	-10.3436***	-2.0520***	0.4355	0.0559	1.1203	0.1452	-7.9849***	-1.6931***	-0.0895	-0.0123
	(0.8439)	(0.1646)	(0.7367)	(1.1325)	(0.1075)	(1.0179)	(0.1548)	(2.2994)	(0.2949)	(3.3782)	(0.4386)	(1.7499)	(0.3282)	(4.9230)	(0.6751)
GRE Quantative Percentile	0.0746***	0.0167***	0.0567***	-0.0021		0.1071***	0.0213***	-0.0132	-0.0017	0.0494*	0.0064*	0.0825***	0.0175***	0.0253	0.0035
	(0.0100)	(0.0020)	(0.0088)	(0.0144)		(0.0116)	(0.0019)	(0.0254)	(0.0033)	(0.0283)	(0.0035)	(0.0208)	(0.0040)	(0.0616)	(0.0083)
Gender	-0.0329	-0.0074	0.3124***	-0.0158		0.3354**	0.0665**	0.4355	0.0559	-0.7111**	-0.0922**	-0.0038	-0.0008	-0.3825	-0.0525
	(0.1376)	(0.0308)	(0.1188)	(0.0155)		(0.1459)	(0.0286)	(0.2995)	(0.0386)	(0.3476)	(0.0442)	(0.2378)	(0.0504)	(0.7540)	(0.1051)
Demographic 1	0.6096***	0.1366***	-1.5775***	0.0034		-0.2308	-0.0458	-0.1845	-0.0237	-1.1081**	-0.1436**	-0.1696	-0.0360	1.3825	0.1897
	(0.1830)	(0.0401)	(0.1549)	(0.0874)		(0.2012)	(0.0398)	(0.4586)	(0.0590)	(0.5651)	(0.0718)	(0.3723)	(0.0788)	(1.1353)	(0.1567)
Demographic 2	0.0620	0.0139	-1.3593***	-0.0012		0.4034*	0.0800*	0.2709	0.0348	-0.9403*	-0.1219*	-0.2598	-0.0551	0.6914	0.0949
	(0.2042)	(0.0457)	(0.1784)	(0.0535)		(0.2181)	(0.0430)	(0.4721)	(0.0605)	(0.5150)	(0.0668)	(0.3772)	(0.0798)	(1.1838)	(0.1640)
Advanced Math			0.9118***	0.0045				0.4602	0.0591	0.0070	0.0009	0.1485	0.0315	0.9409	0.1291
			(0.1212)	(0.0343)				(0.3230)	(0.0414)	(0.3631)	(0.0470)	(0.2563)	(0.0543)	(0.8525)	(0.1095)
Committee Score				0.0036				0.0044	0.0006	0.1274	0.0165	0.0297	0.0063	-0.9581***	-0.1315***
				(0.0094)				(0.0679)	(0.0087)	(0.0800)	(0.0103)	(0.0578)	(0.0122)	(0.2989)	(0.0272)
Year Transition/Matriculate More						0.9912***	0.1966***							1.4252*	0.1955*
						(0.1473)	(0.0265)							(0.8409)	(0.1053)
$\ddot{m}_2$					1.2940***			-3.6069	-0.4630						
					(0.2636)			(2.6429)	(0.3395)						
$\overline{m_{2A}}$										-15.1650**	-1.9656**				
										(7.4987)	(0.9426)				
$\sqrt{\text{MSE}}$			1.8530***	0.0759*	0.0023										
			(0.0392)	(0.0446)	(0.0062)										
Observations	1000														
LL | Parameters | AIC	-1606.1	52	3316.2												

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

A4: Unrestricted Waitlist-Dynamic Estimates (Optimal)

Table 31b: Unrestricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math		Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.3390***	-1.4261***	1.0066	0.6966	0.0760	-8.0015***	-1.6759***	-5.7584**	-0.5618**	-2.4065	-0.3210	-10.5954***	-2.0568***	1.5968	0.2120
	(0.8340)	(0.1646)	(0.7368)	(0.7448)	(0.1019)	(0.9284)	(0.1635)	(2.3684)	(0.2290)	(2.4585)	(0.3254)	(1.8360)	(0.2883)	(9.6640)	(1.2760)
GRE Quantative Percentile	0.0723***	0.0163***	0.0560***	-0.0031		0.0841***	0.0176***	0.0435	0.0042	0.0600**	0.0080**	0.1061***	0.0206***	-0.0715	-0.0095
	(0.0099)	(0.0020)	(0.0088)	(0.0096)		(0.0108)	(0.0020)	(0.0284)	(0.0028)	(0.0267)	(0.0034)	(0.0212)	(0.0035)	(0.1162)	(0.0149)
Gender	-0.0369	-0.0083	0.3113***	0.0147		0.0576	0.0121	-0.5486	-0.0535	-0.1122	-0.0150	-0.0530	-0.0103	2.9518*	0.3919**
	(0.1370)	(0.0308)	(0.1188)	(0.0224)		(0.1413)	(0.0296)	(0.3736)	(0.0366)	(0.3018)	(0.0403)	(0.2502)	(0.0486)	(1.5616)	(0.1629)
Demographic 1	0.6053***	0.1362***	-1.5742***	0.0189		-0.2889	-0.0605	0.2771	0.0270	0.4360	0.0582	-0.5275	-0.1024	3.4735**	0.4612**
	(0.1822)	(0.0401)	(0.1549)	(0.0592)		(0.1951)	(0.0407)	(0.6083)	(0.0596)	(0.4936)	(0.0654)	(0.3723)	(0.0716)	(1.6588)	(0.1801)
Demographic 2	0.0525	0.0118	-1.3575***	0.0178		-0.0836	-0.0175	0.4928	0.0481	0.1412	0.0188	0.5069	0.0984	5.5471**	0.7364***
	(0.2032)	(0.0457)	(0.1783)	(0.0479)		(0.2146)	(0.0449)	(0.6023)	(0.0589)	(0.5071)	(0.0676)	(0.3865)	(0.0743)	(2.2266)	(0.2082)
Advanced Math			0.9134***	0.0177				0.3699	0.0361	-0.0607	-0.0081	-0.1135	-0.0220	4.2756***	0.5676***
			(0.1212)	(0.0396)				(0.3815)	(0.0371)	(0.3636)	(0.0485)	(0.2680)	(0.0521)	(1.6021)	(0.1723)
Committee Score				0.0010				-0.0240	-0.0023	0.0639	0.0085	0.1486**	0.0288**	-0.8898***	-0.1181***
				(0.0077)				(0.0990)	(0.0097)	(0.0887)	(0.0118)	(0.0694)	(0.0133)	(0.2508)	(0.0236)
Year Transition/Matriculate More						0.7980***	0.1672***							3.3305***	0.4422***
						(0.1408)	(0.0277)							(1.2386)	(0.1208)
$\ddot{m}_2$					0.8325***			-0.8030	-0.0783						
					(0.2037)			(0.9111)	(0.0878)						
$\overline{m_{2A}}$										-9.3770***	-1.2508***				
										(3.5018)	(0.4443)				
$\sqrt{\text{MSE}}$			1.8530***	0.1909***	0.0015										
			(0.0392)	(0.0714)	(0.0022)										
Observations	1000														
LL | Parameters | AIC	-1817.3	52	3738.6												

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Optimal)

A4: Restricted Waitlist-Dynamic Estimates (Suboptimal)

Table 32a: Restricted Waitlist-Dynamic Estimates (Suboptimal)

	Advanced Math	Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-6.4496***	0.7051	0.4766	-0.1272	0.0125	1.0126	-7.6194	21.22%	-10.4830***	-0.0895		
	(0.8614)	(0.7692)	(1.1502)	(0.1074)	(0.1709)	(0.7244, 1.4155)	(16.6895)	(0.06%, 99.16%)	(1.5411)	(4.9230)		
GRE Quantative Percentile	0.0735***	0.0595***	-0.0009						0.1081***	0.0253		
	(0.0102)	(0.0092)	(0.0145)						(0.0194)	(0.0616)		
Gender	-0.0306	0.3225***	-0.0108						0.0622	-0.3825		
	(0.1372)	(0.1189)	(0.0190)						(0.2133)	(0.7540)		
Demographic 1	0.6102***	-1.5739***	0.0001						-0.2810	1.3825		
	(0.1826)	(0.1550)	(0.0820)						(0.3277)	(1.1353)		
Demographic 2	0.0690	-1.3414***	0.0028						0.0016	0.6914		
	(0.2045)	(0.1791)	(0.0456)						(0.3387)	(1.1838)		
Advanced Math		0.9135***	0.0042						0.2552	0.9409		
		(0.1213)	(0.0332)						(0.2883)	(0.8525)		
Committee Score			0.0033						0.0851	-0.9581***		
			(0.0092)						(0.0889)	(0.2989)		
Year Transition/Matriculate More					-0.2633***	0.7782		96.83%		1.4252*		
					(0.0511)	(0.5489, 1.1034)		(0.00%, 100.00%)		(0.8409)		
$\ddot{m}_2$				1.2940***								
				(0.2635)								
$\overline{m_{2A}}$							22.8654					
							(49.3074)					
$\sqrt{\text{MSE}}$		1.8540***	0.0764**	0.0023								
		(0.0397)	(0.0368)	(0.0062)								
Beta											1.0000***	
											(0.0001)	
Sigma												0.2696***
												(0.0581)
Observations	1000											
LL | Parameters | AIC	-1668.9	36	3409.8									
LR Statistic | DF | P-Value	125.6201	16	0.0000									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Waitlist-Hybrid Estimates (Optimal)

A4: Restricted Waitlist-Dynamic Estimates (Optimal)

Table 32b: Restricted Waitlist-Dynamic Estimates (Optimal)

	Advanced Math	Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-6.3235***	0.7422	0.6027	0.0760	0.2802**	1.3234	-19.0703	6.39%	-12.9838***	1.5968		
	(0.8356)	(0.7499)	(0.9450)	(0.1019)	(0.1135)	(1.0595, 1.6531)	(25.5492)	(0.20%, 69.98%)	(1.8174)	(9.6640)		
GRE Quantative Percentile	0.0721***	0.0593***	-0.0020						0.1324***	-0.0715		
	(0.0100)	(0.0090)	(0.0119)						(0.0210)	(0.1162)		
Gender	-0.0378	0.3117***	0.0171						-0.1848	2.9518*		
	(0.1368)	(0.1191)	(0.0250)						(0.2315)	(1.5616)		
Demographic 1	0.6050***	-1.5827***	0.0150						-0.2909	3.4735**		
	(0.1821)	(0.1550)	(0.0678)						(0.3667)	(1.6588)		
Demographic 2	0.0517	-1.3602***	0.0133						0.4564	5.5471**		
	(0.2030)	(0.1789)	(0.0537)						(0.3242)	(2.2266)		
Advanced Math		0.9113***	0.0169						-0.0830	4.2756***		
		(0.1217)	(0.0418)						(0.3430)	(1.6021)		
Committee Score			0.0011						0.1675*	-0.8898***		
			(0.0076)						(0.0895)	(0.2508)		
Year Transition/Matriculate More					-0.1868***	1.0980		100.00%		3.3305***		
					(0.0389)	(0.9042, 1.3334)		(0.00%, 100.00%)				
$\ddot{m}_2$				0.8325***								
				(0.2037)								
$\overline{m_{2A}}$							62.2680					
							(95.2035)					
$\sqrt{\text{MSE}}$		1.8555***	0.1919***	0.0015								
		(0.0393)	(0.0722)	(0.0022)								
Beta											0.9997***	
											(0.0014)	
Sigma												0.3148***
												(0.0580)
Observations	1000											
LL | Parameters | AIC	-1846.5	36	3765.0									
LR Statistic | DF | P-Value	58.3577	16	0.0000									

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Waitlist-Hybrid Estimates (Optimal)

A4: Unrestricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

Table 33a: Unrestricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

	Selection		Advanced Math		Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-6.3791***	-2.0904***	-3.4140*	-1.1948*	2.8988	0.5784	-0.1272	-10.7133***	-2.1064***	1.2903	0.1656	2.9778	0.3880	-7.8155***	-1.6616***	-0.0895	-0.0123
	(0.6110)	(0.1573)	(1.8865)	(0.6482)	(2.6796)	(1.1325)	(0.1075)	(1.0455)	(0.1549)	(2.3256)	(0.2971)	(3.4020)	(0.4419)	(1.7335)	(0.3277)	(4.9230)	(0.6751)
GRE Quantative Percentile	0.0661***	0.0217***	0.0388**	0.0136**	0.0386	-0.0021		0.1111***	0.0218***	-0.0213	-0.0027	0.0297	0.0039	0.0801***	0.0170***	0.0253	0.0035
	(0.0070)	(0.0019)	(0.0184)	(0.0063)	(0.0259)	(0.0144)		(0.0119)	(0.0019)	(0.0253)	(0.0032)	(0.0272)	(0.0035)	(0.0206)	(0.0040)	(0.0616)	(0.0083)
Gender	0.2029**	0.0665**	0.1609	0.0563	0.6097***	-0.0158		0.3463**	0.0681**	0.3972	0.0510	-0.7582**	-0.0988**	-0.0146	-0.0031	-0.3825	-0.0525
	(0.0880)	(0.0286)	(0.1476)	(0.0514)	(0.2189)	(0.0155)		(0.1473)	(0.0286)	(0.2998)	(0.0387)	(0.3461)	(0.0441)	(0.2372)	(0.0504)	(0.7540)	(0.1051)
Demographic 1	-0.1336	-0.0438	0.6288***	0.2201***	-1.2569***	0.0034		-0.2181	-0.0429	-0.1480	-0.0190	-1.1235**	-0.1464**	-0.1191	-0.0253	1.3825	0.1897
	(0.1213)	(0.0397)	(0.2086)	(0.0703)	(0.2773)	(0.0874)		(0.2038)	(0.0400)	(0.4597)	(0.0591)	(0.5571)	(0.0710)	(0.3730)	(0.0792)	(1.1353)	(0.1567)
Demographic 2	0.2567*	0.0841*	0.0479	0.0168	-1.2976***	-0.0012		0.4340**	0.0853**	0.2845	0.0365	-1.0456**	-0.1362**	-0.2130	-0.0453	0.6914	0.0949
	(0.1321)	(0.0430)	(0.2235)	(0.0782)	(0.3262)	(0.0535)		(0.2208)	(0.0431)	(0.4727)	(0.0605)	(0.5099)	(0.0662)	(0.3781)	(0.0803)	(1.1838)	(0.1640)
Advanced Math					0.7518***	0.0045				0.4844	0.0622	0.0382	0.0050	0.1458	0.0310	0.9409	0.1291
					(0.2175)	(0.0343)				(0.3240)	(0.0414)	(0.3591)	(0.0467)	(0.2554)	(0.0543)	(0.8525)	(0.1095)
Committee Score						0.0036				0.0078	0.0010	0.1283	0.0167	0.0319	0.0068	-0.9581***	-0.1315***
						(0.0094)				(0.0681)	(0.0087)	(0.0790)	(0.0102)	(0.0577)	(0.0122)	(0.2989)	(0.0272)
Year Transition/Matriculate More	0.5989***	0.1963***						1.0063***	0.1978***							1.4252*	0.1955*
	(0.0887)	(0.0266)						(0.1492)	(0.0265)							(0.8409)	(0.1053)
$\ddot{m}_2$							1.2940***			-4.0491	-0.5195						
							(0.2636)			(3.0009)	(0.3844)						
$\overline{m_{2A}}$												-15.3487**	-1.9997**				
												(7.6081)	(0.9604)				
Inverse Mills			-0.2428	-0.0850	-0.7438												
			(0.3553)	(0.1244)	(0.5381)												
$\sqrt{\text{MSE}}$					1.8994***	0.0759*	0.0023										
					(0.0681)	(0.0446)	(0.0062)										
Observations	1000																
LL | Parameters | AIC	-493.9	60	1107.9														

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Suboptimal)

A4: Unrestricted Waitlist-Dynamic Estimates, z Missing (Optimal)

Table 33b: Unrestricted Waitlist-Dynamic Estimates, z Missing (Optimal)

	Selection		Advanced Math		Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Transition		Accept 2		Accept 3		Success		Matriculate	
	Coefficient	AME	Coefficient	AME	Coefficient	Coefficient	Coefficient	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME	Coefficient	AME
Intercept	-4.8615***	-1.6782***	-3.3441*	-1.1783*	1.7804	0.6966	0.0760	-8.0675***	-1.6874***	-8.0113***	-0.7675***	-2.6713	-0.3552	-9.8952***	-1.9480***	1.5968	0.2120
	(0.5539)	(0.1650)	(1.8743)	(0.6499)	(2.4321)	(0.7448)	(0.1019)	(0.9324)	(0.1635)	(2.6333)	(0.2376)	(2.4696)	(0.3249)	(1.7561)	(0.2871)	(9.6640)	(1.2760)
GRE Quantative Percentile	0.0510***	0.0176***	0.0418**	0.0147**	0.0425*	-0.0031		0.0848***	0.0177***	0.0679**	0.0065**	0.0626**	0.0083**	0.0975***	0.0192***	-0.0715	-0.0095
	(0.0065)	(0.0020)	(0.0177)	(0.0061)	(0.0231)	(0.0096)		(0.0109)	(0.0020)	(0.0311)	(0.0029)	(0.0269)	(0.0034)	(0.0203)	(0.0035)	(0.1162)	(0.0149)
Gender	0.0357	0.0123	0.0632	0.0223	0.2910	0.0147		0.0588	0.0123	-0.5356	-0.0513	-0.1269	-0.0169	-0.0544	-0.0107	2.9518*	0.3919**
	(0.0858)	(0.0296)	(0.1457)	(0.0513)	(0.1923)	(0.0224)		(0.1415)	(0.0296)	(0.3777)	(0.0363)	(0.3026)	(0.0403)	(0.2465)	(0.0486)	(1.5616)	(0.1629)
Demographic 1	-0.1712	-0.0591	0.4139**	0.1458**	-1.1405***	0.0189		-0.2921	-0.0611	0.2970	0.0285	0.4746	0.0631	-0.4386	-0.0863	3.4735**	0.4612**
	(0.1174)	(0.0404)	(0.1970)	(0.0683)	(0.2555)	(0.0592)		(0.1954)	(0.0407)	(0.6549)	(0.0630)	(0.4958)	(0.0654)	(0.3666)	(0.0718)	(1.6588)	(0.1801)
Demographic 2	-0.0370	-0.0128	-0.0556	-0.0196	-0.8560***	0.0178		-0.0849	-0.0178	0.5432	0.0520	0.2156	0.0287	0.5681	0.1118	5.5471**	0.7364***
	(0.1295)	(0.0447)	(0.2144)	(0.0755)	(0.2866)	(0.0479)		(0.2149)	(0.0449)	(0.6450)	(0.0618)	(0.5092)	(0.0676)	(0.3819)	(0.0743)	(2.2266)	(0.2082)
Advanced Math					1.0447***	0.0177				0.2587	0.0248	-0.0625	-0.0083	-0.1103	-0.0217	4.2756***	0.5676***
					(0.1960)	(0.0396)				(0.3828)	(0.0366)	(0.3641)	(0.0484)	(0.2639)	(0.0520)	(1.6021)	(0.1723)
Committee Score						0.0010				-0.0151	-0.0014	0.0660	0.0088	0.1486**	0.0292**	-0.8898***	-0.1181***
						(0.0077)				(0.0990)	(0.0095)	(0.0888)	(0.0118)	(0.0684)	(0.0133)	(0.2508)	(0.0236)
Year Transition/Matriculate More	0.4860***	0.1678***						0.8005***	0.1674***							3.3305***	0.4422***
	(0.0848)	(0.0277)						(0.1410)	(0.0277)							(1.2386)	(0.1208)
$\ddot{m}_2$							0.8325***			-0.7079	-0.0678						
							(0.2037)			(0.9064)	(0.0857)						
$\overline{m_{2A}}$												-9.4003***	-1.2500***				
												(3.5065)	(0.4439)				
Inverse Mills			-0.3667	-0.1292	0.0196												
			(0.4403)	(0.1548)	(0.5749)												
$\sqrt{\text{MSE}}$					1.7743***	0.1909***	0.0015										
					(0.0631)	(0.0714)	(0.0022)										
Observations	1000																
LL | Parameters | AIC	-697.6	60	1515.2														

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01.

Unrestricted Waitlist-Hybrid AMEs (Optimal)

A4: Restricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

Table 34a: Restricted Waitlist-Dynamic Estimates, z Missing (Suboptimal)

	Selection	Advanced Math	Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-6.3774***	-3.6104*	1.9751	0.4746	-0.1272	0.0067	1.0067	-7.6729	20.96%	-10.5320***	-0.0895		
	(0.6114)	(1.8642)	(2.8711)	(1.1418)	(0.1074)	(0.1835)	(0.7026, 1.4426)	(17.2934)	(0.05%, 99.31%)	(1.5304)	(4.9230)		
GRE Quantative Percentile	0.0661***	0.0409**	0.0487*	-0.0009						0.1085***	0.0253		
	(0.0070)	(0.0181)	(0.0280)	(0.0143)						(0.0199)	(0.0616)		
Gender	0.2030**	0.1684	0.6455***	-0.0112						0.0494	-0.3825		
	(0.0880)	(0.1470)	(0.2186)	(0.0187)						(0.2128)	(0.7540)		
Demographic 1	-0.1339	0.6306***	-1.2537***	0.0006						-0.2576	1.3825		
	(0.1213)	(0.2079)	(0.2757)	(0.0805)						(0.3574)	(1.1353)		
Demographic 2	0.2568*	0.0595	-1.2486***	0.0033						-0.0032	0.6914		
	(0.1322)	(0.2233)	(0.3282)	(0.0442)						(0.3556)	(1.1838)		
Advanced Math			0.7454***	0.0034						0.1449	0.9409		
			(0.2189)	(0.0336)						(0.3600)	(0.8525)		
Committee Score				0.0037						0.0972	-0.9581***		
				(0.0088)						(0.0985)	(0.2989)		
Year Transition/Matriculate More	0.5995***					-0.2599***	0.7763		96.87%		1.4252*		
	(0.0885)					(0.0503)	(0.5370, 1.1223)		(0.00%, 100.00%)		(0.8409)		
$\ddot{m}_2$					1.2940***								
					(0.2635)								
$\overline{m_{2A}}$								23.0017					
								(51.1060)					
Inverse Mills		-0.2362	-0.7093										
		(0.3630)	(0.5634)										
$\sqrt{\text{MSE}}$			1.9039***	0.0765**	0.0023								
			(0.0722)	(0.0360)	(0.0062)								
Beta												0.9998***	
												(0.0000)	
Sigma													0.2696***
													(0.0612)
Observations	1000												
LL | Parameters | AIC	-557.4	44	1202.9										
LR Statistic | DF | P-Value	126.9760	16	0.0000										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

Restricted Waitlist-Hybrid Estimates (Suboptimal)

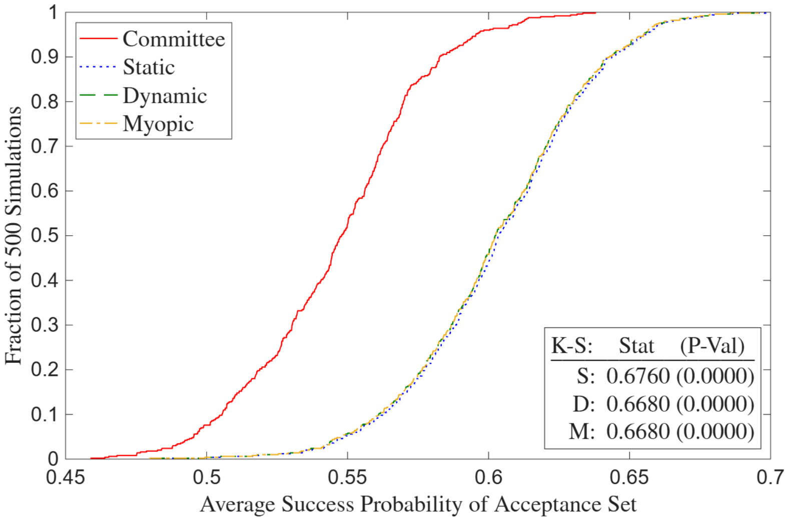
Table 34b: Restricted Waitlist-Dynamic Estimates, z Missing (Optimal)

	Selection	Advanced Math	Committee Score	$\ddot{m}_2$	$\overline{m_{2A}}$	Cutoff 1	$L(\text{Cutoff})_1$	Cutoff 3	$P(\text{Cutoff})_3$	Success	Matriculate	Beta	Sigma
Intercept	-4.8608***	-3.2487*	1.6331	0.6036	0.0760	0.2815**	1.3251	-19.2388	6.37%	-13.0378***	1.5968		
	(0.5558)	(1.8601)	(2.4088)	(0.9529)	(0.1019)	(0.1155)	(1.0567, 1.6616)	(26.4958)	(0.20%, 70.27%)	(1.8451)	(9.6640)		
GRE Quantative Percentile	0.0510***	0.0409**	0.0453**	-0.0020						0.1337***	-0.0715		
	(0.0065)	(0.0175)	(0.0228)	(0.0120)						(0.0215)	(0.1162)		
Gender	0.0356	0.0626	0.2901	0.0170						-0.1820	2.9518*		
	(0.0858)	(0.1457)	(0.1924)	(0.0251)						(0.2325)	(1.5616)		
Demographic 1	-0.1722	0.4234**	-1.1463***	0.0148						-0.2869	3.4735**		
	(0.1173)	(0.1972)	(0.2572)	(0.0685)						(0.3762)	(1.6588)		
Demographic 2	-0.0379	-0.0479	-0.8645***	0.0131						0.4501	5.5471**		
	(0.1294)	(0.2146)	(0.2886)	(0.0544)						(0.3363)	(2.2266)		
Advanced Math			1.0331***	0.0163						-0.1344	4.2756***		
			(0.1988)	(0.0424)						(0.3691)	(1.6021)		
Committee Score				0.0009						0.1628*	-0.8898***		
				(0.0079)						(0.0885)	(0.2508)		
Year Transition/Matriculate More	0.4854***					-0.1868***	1.0993		100.00%		3.3305***		
	(0.0851)					(0.0391)	(0.9017, 1.3402)		(0.00%, 100.00%)		(1.2386)		
$\ddot{m}_2$					0.8325***								
					(0.2037)								
$\overline{m_{2A}}$								62.8952					
								(98.9921)					
Inverse Mills		-0.3929	-0.0454										
		(0.4454)	(0.5823)										
$\sqrt{\text{MSE}}$			1.7803***	0.1919***	0.0015								
			(0.0632)	(0.0724)	(0.0022)								
Beta												1.0000***	
												(0.0000)	
Sigma													0.3158***
													(0.0582)
Observations	1000												
LL | Parameters | AIC	-727.6	44	1543.2										
LR Statistic | DF | P-Value	59.9834	16	0.0000										

Murphy-Topel robust standard errors in parentheses: *p<0.10, **p<0.05, ***p<0.01. Confidence intervals are 95%.

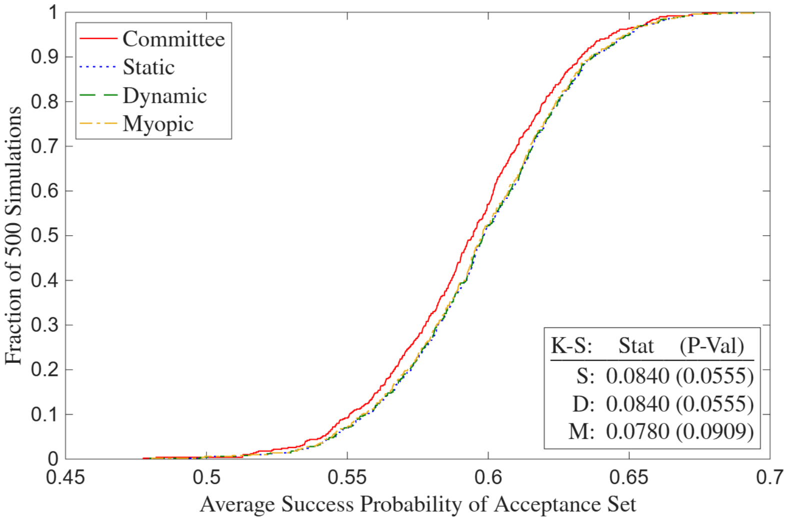
▶▶ Restricted Waitlist-Hybrid Estimates (Optimal)

A4: Full Deviation Gains (Suboptimal)



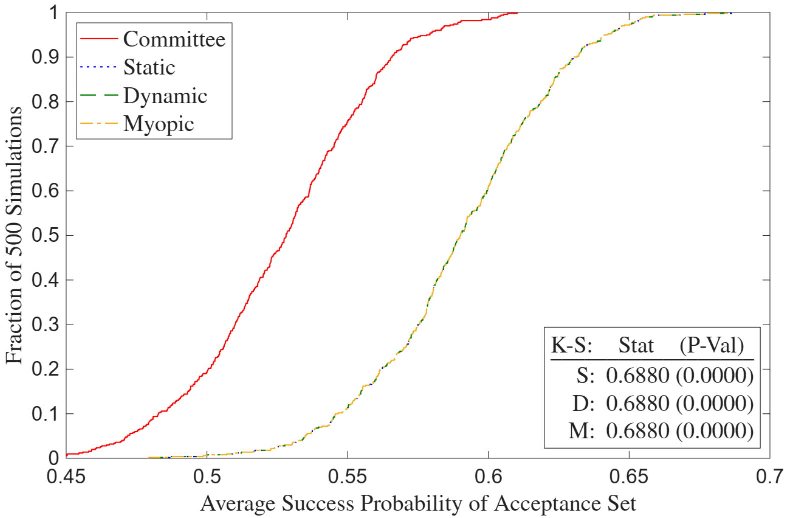
Deviation Gains (Suboptimal)

A4: Full Deviation Gains (Suboptimal)



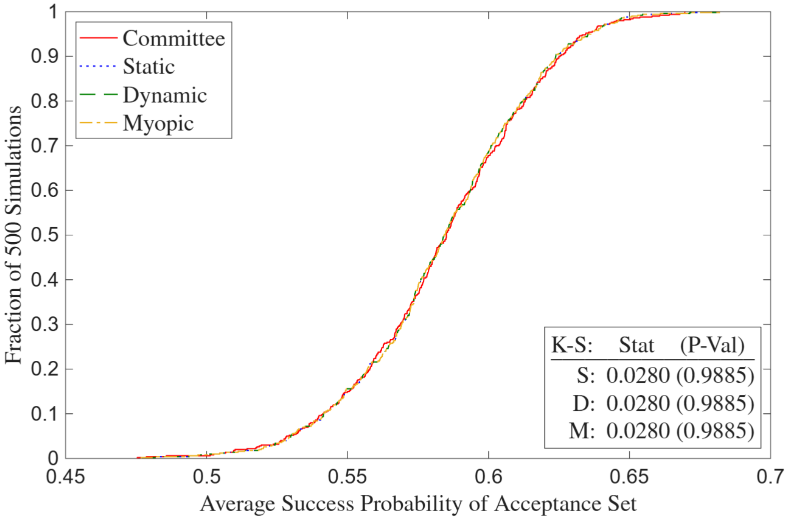
Deviation Gains (Optimal)

A4: Deviation Gains, z Missing (Suboptimal)



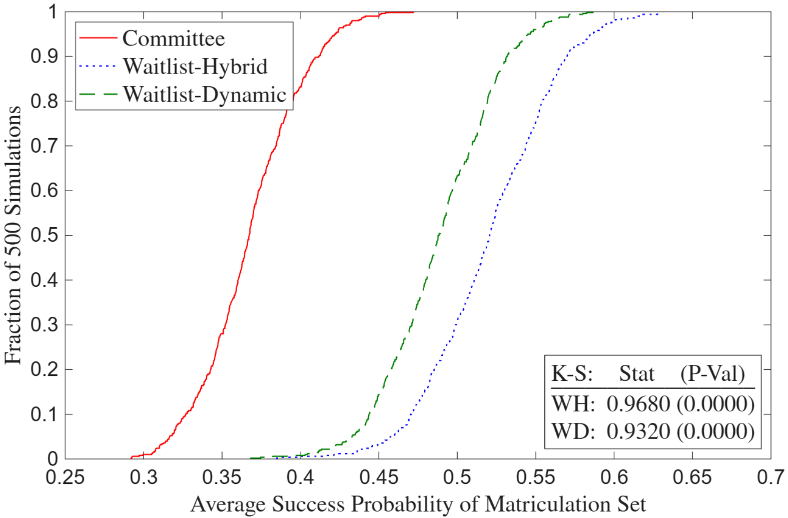
Deviation Gains (Suboptimal)

A4: Deviation Gains, z Missing (Optimal)



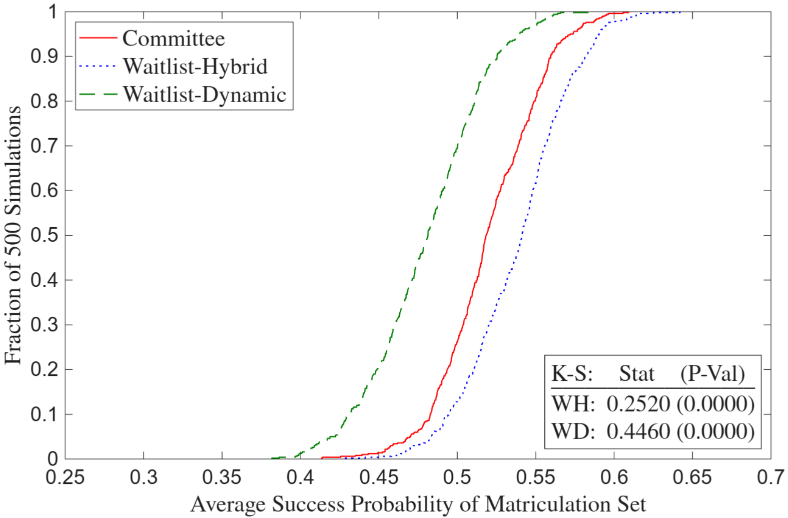
Deviation Gains (Optimal)

A4: Waitlist Deviation Gains, z Missing (Suboptimal)



Waitlist Deviation Gains (Suboptimal)

A4: Waitlist Deviation Gains, z Missing (Optimal)



Waitlist Deviation Gains (Optimal)

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